Check Your Blind Spots: Decision-Guided Perception by Probing the Planner with Unseen Objects

Supplementary Material

Anonymous CVPR submission

This document complements the main paper with complete proofs, the evaluation protocol, and all experiments that the main paper summarizes. It is organized by scientific purpose; the roadmap in Tab. S1 maps each claim of the main paper to the section and the evidence that support it.

Conventions.

- Unprefixed numbers (*e.g.*, Theorem 1, Tab. 2, Sec. 4.3) refer to the main paper; numbers with the prefix S refer to this document.
- G is the decision value of Eq. (5), *i.e.*, the fraction of the decision loss of the cheap pipeline that a pipeline removes.
- Detector-input pixels (px) are all pixels fed to a detector in a frame, padded to multiples of 32 pixels and including the cheap pass and the truncation guard, in % of the native frame unless a column states otherwise; native-resolution inference therefore exceeds 100%.
- Δ denotes a 95% paired bootstrap interval (2,000 resamples over scenes, logs, or sequences) of the difference DGSP $-$ compared pipeline; intervals above zero, *i.e.*, DGSP significantly better, are in bold, and intervals below zero are marked with *.
- In columns of decision values, † / * mark that DGSP is significantly better / worse.
- The *lift* estimates object states: monocular geometry, the learned looming estimate, and the range refiner, which acts once after the loop on the verified support objects. "+ comp." denotes a pipeline with this lift (the "shared components"); the main paper applies it to every row and omits the suffix.
- The reference F of the main paper is whole-frame inference at the verification resolution with the lift (u960 + comp.). The diagnostics of Sec. S1.8 that isolate the conditions of the theorem remove the range refiner from every pipeline and from F; they say so.
- "Cheap detector" and "cheap pipeline" denote low-resolution inference (u320 in the main paper), and "hybrid" denotes the ablation "support + fixed tube" of the main paper.
- Unless stated otherwise, results are on the test splits, *i.e.*, the splits not used for development: nuScenes held-out (228 scenes of its trainval set; not the official test set), Argoverse 2 val (150 logs), and Argoverse-HD val (24 sequences), with the frozen configuration of DGSP.

Contents

S1 Formal guarantees **3**

S2 Implementation, evaluation protocol, and baseline fairness **12**

S3 Decision–pixel trade-offs and component analysis **17**

S4 Unseen-object recovery, field behavior, and coverage **25**

S5 Braking severity, operating points, and robustness **30**

S6 Embedded latency, deadlines, closed loop, and planning **34**

S7 Qualitative results, failure cases, and reproducibility **41**

S8 Reference tables **46**

Table S1. **Roadmap.** Each claim of the main paper, the section of this document that supports it, and the evidence.

Claim in the main paper	Section	Evidence
Method and guarantee (Secs. 3.2 to 3.4 and 4.2)		
Theorem 1 and Corollaries 1–2, including the empty support at KEEP	S1.2–S1.5	complete proofs
Counterexamples without the field, the support, insertion stability, or the iteration	S1.6	six constructions, Tab. S2
Where the deployed system departs from the theorem, and by how much	S1.8	Tab. S3, Fig. S1, Tab. S4
Decision fidelity (Sec. 4.2, Tab. 1)		
Agreement and repair rate of all pipelines, with the lift in every row and in F	S1.8	Tab. S5
The same without the range refiner; repair rate *vs*. pixels	S1.8	Tab. S6, Fig. S2
Setup (Sec. 4.1)		
Planner, loss, and decision value G	S2.1	Eqs. (S23) and (S24)
One configuration, selected on development data and frozen	S2.2	Tabs. S7 and S8
Detector-input pixels and the Argoverse 2 ego mask	S2.3–S2.4	Tab. S9
Every baseline uses the detector and the shared components and is tuned	S2.5–S2.6	Tabs. S10 and S11
Decision quality versus pixel cost (Sec. 4.3)		
No published, learned, or object-level method under the per-frame cap is significantly better; all rows and variants that Tab. 2 omits; multiple comparisons	S3.1	Fig. S3, Tab. S43
Whole frame with the components needs 1.9–14.5$\times$ the pixels	S3.2	Figs. S4 and S5, Tabs. S44 and S45
The advantage stems from the selection of the regions; strict cap including the guard	S3.3	Tabs. S12 and S13, Fig. S6
Mean pixel budget for the frame-level methods; crops along anchored trajectories	S3.4–S3.5	Tabs. S14 and S15
The gain removes excessive braking; insufficient braking is that of native inference	S3.6	Fig. S7, Tabs. S17 and S46
The result holds for another YOLO detector and fine-tuned detectors	S3.7	Tabs. S18 to S20
What the field contributes (Sec. 4.4, Tab. 4, Tab. 5)		
The field recovers objects that the draft misses; natural misses	S4.1	Tab. S21
The verified and guard crops contain the decisive object on 92–96% of the frames	S4.2	Fig. S8, Tab. S48
The field is an action-adaptive corridor; it adapts to the action, the ego speed, the path, and the planner	S4.3	Fig. S9, Tabs. S22 to S25
No other probe or grid setting improves on the field on all datasets	S4.4	Tab. S26
Braking, operating points, and robustness (Sec. 4.3, Fig. 3, Limitations)		
Missed and false braking episodes, by severity	S5.1	Tabs. S27 and S28
At constrained operating points, DGSP misses no significantly more episodes than the cheap detector	S5.2	Fig. S10, Tabs. S30 and S49
Held-out sequences, two evaluations under protocols fixed in advance; strata; weightings of the loss	S5.3–S5.5	Tabs. S31 and S32
The gain shrinks when insufficient braking weighs more (Limitations)	S5.6	Tab. S33
Embedded and closed-loop results (Secs. 4.5 and 4.6)		
Latency and energy on the Jetson; latency of all methods	S6.1	Tab. S34, Fig. S11, Tab. S50
Decision value under per-frame deadlines; latency–decision frontier; all methods under deadlines	S6.2	Fig. S12, Tabs. S35, S36 and S51
Controlled closed loop and planning benchmarks; PDMS, anchored crops, and interactions on NAVSIM	S6.3–S6.5	Tabs. S37 to S41
Frames of Fig. 4 are chosen by a fixed rule; failure cases	S7	Figs. S13 to S15, Tab. S42

S1. Formal guarantees

This section proves Theorem 1 and Corollaries 1 and 2 (Secs. S1.2 to S1.4), shows that termination is guaranteed (Sec. S1.5) and what fails when a condition is omitted (Sec. S1.6), and relates the conditions to the deployed system and its measured exactness (Secs. S1.7 and S1.8).

S1.1. Setting and notation

A *scene* is a finite set of objects; each object o has a box $b(o)$ in image coordinates and a state $x(o)$ in the ego frame (range, lateral extent, and time-to-collision). Objects with equal states are distinct, and a *probe* o_x is a hypothetical object with state x. A *planner* π maps a scene to an action through the states of its objects; the ego state is fixed within a frame and omitted.

- *Field.* The field of a scene S contains the states of a single inserted object that changes the action,

$$Z(S) = \{\, x \,:\, \pi(S \cup \{o_x\}) \neq \pi(S) \,\}. \tag{S1}$$

- *Insertion stability.* π is insertion-stable if, for every scene S and every finite set X of objects whose states lie outside $Z(S)$,

$$\pi(S \cup X) = \pi(S). \tag{S2}$$

- *Certifying support.* Let $\mathcal{Y}(S)$ be the family that contains the empty set and every singleton $\{o\}$ with $x(o) \notin Z(S)$. A support $\mathrm{Sup}(S) \subseteq S$ is certifying if

$$\pi\big((S \setminus D) \cup Y\big) = \pi(S) \qquad \text{for all } D \subseteq S \setminus \mathrm{Sup}(S) \text{ and } Y \in \mathcal{Y}(S). \tag{S3}$$

- *Verified region and merged scene.* A crop is an axis-aligned image rectangle, and a verified region R is a finite set of crops; $S|_R$ denotes the objects of a scene S whose box center lies in a crop of R. $V(R)$ denotes what verification returns: the reference detector runs on each crop of R, in the order in which Algorithm 1 adds them, the boxes are mapped to image coordinates, and duplicates of overlapping crops are merged. V is a function of the crops, not only of their union, since a detection depends on the crop that contains it. The merged scene is

$$M(R) = (C \setminus C|_R) \cup V(R). \tag{S4}$$

- *Requested crops.* The crops requested for a scene S are its field footprint $R_Z(S)$ and the set $\mathrm{crops}(\mathrm{Sup}(S))$, one crop per support object, which contains the box center of that object. Algorithm 1 stops when every crop requested for the current merged scene lies inside the union of the crops of R.
- *Locality.* $V(R) = F|_R$, with identical boxes and states. This is a condition on verification, not a definition of V: $F|_R$ restricts the hypothetical whole-frame reference scene, whereas $V(R)$ is computed on crops.
- *Coverage.* Every object of F with a state in $Z(M)$ has its box center in R.

Definition 1 (Max-composable planner)**.** Let the actions $a \in \{0, \dots, K\}$ be ordered by intensity, with thresholds $0 = \tau_0 < \tau_1 < \cdots < \tau_K < \tau_{K+1} = \infty$. A planner π is max-composable if, for a per-object demand $d \geq 0$,

$$\pi(S) = q\Big(\max_{o \in S} d\big(x(o)\big)\Big), \qquad q(u) = \max\{a \,:\, u \geq \tau_a\}, \qquad \max \emptyset = 0. \tag{S5}$$

The braking controller of the experiments is max-composable with $K = 2$, the actions KEEP (0), DECELERATE (1), and HARD-BRAKE (2), and $\tau_1 = 1.0$ and $\tau_2 = 3.5\,\mathrm{m/s^2}$ (Sec. S2.1).

S1.2. Decision exactness (Theorem 1)

Theorem 1 (Decision exactness)**.** *Let Algorithm 1 stop with verified region R and merged scene M. It returns $\pi(F)$ if (i) π is insertion-stable; (ii)* Sup *is certifying; (iii)* locality*: $V(R) = F|_R$, with identical boxes and states; and (iv)* coverage*: every object of F whose state lies in $Z(M)$ has its box center in R.*

Proof strategy. The proof first removes the unverified draft objects using the certification by the support (Step 3) and then inserts the remaining reference objects using insertion stability (Step 4).

Proof. Step 0: the state at termination. Let R be the verified region at termination and $M = M(R)$. Since the loop stopped, every crop requested for M lies inside R:

$$R \supseteq R_Z(M) \cup \mathrm{crops}\big(\mathrm{Sup}(M)\big). \tag{S6}$$

Step 1: the support has been verified. By locality, $V(R) = F|_R$, so that Eq. (S4) gives $M = (C \setminus C|_R) \cup F|_R$, and the objects of M whose box center lies in R are exactly those of $F|_R$. Every support object has its box center in its crop and hence, by Eq. (S6), in R, so that $\mathrm{Sup}(M) \subseteq F|_R$. The unverified part of the merged scene therefore lies outside the support:

$$D \coloneqq C \setminus C|_R = M \setminus F|_R \subseteq M \setminus \mathrm{Sup}(M). \tag{S7}$$

Step 2: the unverified reference objects lie outside the field of M. Let $\bar{F} = F \setminus F|_R$. Every $o \in \bar{F}$ has its box center outside R, and by coverage,

$$x(o) \notin Z(M) \qquad \text{for all } o \in \bar{F}. \tag{S8}$$

Step 3: certification transfers the action to $F|_R$. We apply Eq. (S3) to $S = M$ with the set D of Eq. (S7), for which $M \setminus D = F|_R$. With $Y = \emptyset$, and with $Y = \{o\}$ for $o \in \bar{F}$, which belongs to $\mathcal{Y}(M)$ by Eq. (S8),

$$\pi(F|_R) = \pi(M), \tag{S9}$$

$$\pi\big(F|_R \cup \{o\}\big) = \pi(M) = \pi(F|_R) \qquad \text{for all } o \in \bar{F}. \tag{S10}$$

Since a planner sees objects only through their states, Eqs. (S1) and (S10) give $x(o) \notin Z(F|_R)$ for every $o \in \bar{F}$.

Step 4: insertion stability transfers the action to F. Since $\bar{F}$ is finite and all its states lie outside $Z(F|_R)$, insertion stability Eq. (S2) with $S = F|_R$ and $X = \bar{F}$ gives, together with Eq. (S9),

$$\pi(F) = \pi(F|_R \cup \bar{F}) = \pi(F|_R) = \pi(M). \tag{S11}$$

The algorithm returns $\pi(M)$, which proves the claim. If $\bar{F} = \emptyset$, Step 4 is vacuous and Eq. (S9) suffices. □

Remark S1 (Where the conditions enter). Locality is used in Step 1 to identify the verified objects with $F|_R$, coverage in Step 2, certification in Step 3, and insertion stability in Step 4. Certification is needed because the merged scene contains unverified draft objects, which may be spurious or mislocalized; insertion stability is needed because $\bar{F}$ may contain arbitrarily many objects that the procedure never observes. Coverage is a condition on the reference scene, not on the probe grid: a finite grid satisfies it if every reference object whose state lies in the field has its box center in the verified region, and Sec. S4.2 measures how often this holds for the decisive object. The proof uses the footprint only through coverage: the footprint is the means of reaching it.

S1.3. Max-composable planners (Corollary 1)

Corollary 1 (Max-composable planners)**.** *A max-composable planner is insertion-stable with $Z(S) = \{x : d(x) \geq \tau_{a+1}\}$, where $a = \pi(S)$, and $\mathrm{Sup}(S) = \{o \in S : d(x(o)) \geq \tau_a\}$ for $a \geq 1$ and $\mathrm{Sup}(S) = \emptyset$ for $a = 0$ is certifying.*

Proof strategy. The action depends on the scene only through the maximum demand m. An inserted object can only raise m, and it changes the action exactly if it raises m to τ_{a+1} or above; removing objects can only lower m, and the action is retained as long as one object with demand at least τ_a remains. For $a = 0$, the lower bound $\tau_0 = 0$ holds for every scene, so that no object needs to remain.

Proof. Let $a = \pi(S)$ and $m = \max_{o \in S} d(x(o))$, with $m = 0$ if $S = \emptyset$. By Eq. (S5),

$$\tau_a \leq m < \tau_{a+1}. \tag{S12}$$

(i) Field. For a state x, $\pi(S \cup \{o_x\}) = q\big(\max(m, d(x))\big)$. Since $\max(m, d(x)) \geq m \geq \tau_a$, the action cannot decrease, and since $m < \tau_{a+1}$,

$$\pi(S \cup \{o_x\}) \neq a \iff \max(m, d(x)) \geq \tau_{a+1} \iff d(x) \geq \tau_{a+1}. \tag{S13}$$

Hence $Z(S) = \{x : d(x) \geq \tau_{a+1}\}$; for $a = K$, $\tau_{K+1} = \infty$ and the field is empty.

(ii) Insertion stability. If $d(x) < \tau_{a+1}$ for every state x of a finite set X, then

$$\tau_a \leq m \leq \max\Big(m, \max_{x \in X} d(x)\Big) < \tau_{a+1}, \tag{S14}$$

so that $\pi(S \cup X) = a$.

(iii) Certification. Let $T = \{o \in S : d(x(o)) \geq \tau_a\}$ if $a \geq 1$ and $T = \emptyset$ if $a = 0$. Let $D \subseteq S \setminus T$, $Y \in \mathcal{Y}(S)$, and $W = (S \setminus D) \cup Y$; we show $\tau_a \leq \max_{o \in W} d(x(o)) < \tau_{a+1}$.

- *Upper bound.* Every object of S has a demand of at most $m < \tau_{a+1}$, and the object of Y, if any, has a state outside $Z(S)$ and hence, by (i), a demand below τ_{a+1}.
- *Lower bound,* $a \geq 1$. Since $m \geq \tau_a > 0$, S is nonempty, and since it is finite, an object o^* with $d(x(o^*)) = m$ exists. It belongs to T and, since $D \cap T = \emptyset$, to W.
- *Lower bound,* $a = 0$. The bound $\tau_0 = 0 \leq \max_{o \in W} d(x(o))$ holds for every scene, including $W = \emptyset$.

Hence $\pi(W) = a$ for all such D and Y, *i.e.*, $\mathrm{Sup}(S) = T$ satisfies Eq. (S3). □

Remark S2 (The case KEEP). For $a = 0$, the formula $\{o \in S : d(x(o)) \geq \tau_0\}$ would contain every object of the scene, since $\tau_0 = 0$. This set is also certifying but needlessly large: removing objects cannot raise the action, and an inserted object outside the field does not raise it either, so that the empty support suffices. At KEEP, the procedure therefore requests no support crops, and its verified region consists of the footprint of the field only.

Remark S3 (Smaller supports and the deployed choice). For $a \geq 1$, the certification uses a single object of T with demand at least τ_a, so that every subset of S that contains such an object is certifying (this also follows from Lemma S3 below). The deployed system verifies all of T, which also provides the objects to which the range refiner is applied.

S1.4. Bottleneck selectors (Corollary 2)

Let $\mathcal{T}$ be a fixed finite set of candidates with a total order $\prec$, let $\beta : \mathcal{T} \to \bar{\mathbb{R}}$ be a preference and $h : \mathcal{T} \times \mathcal{X} \to \bar{\mathbb{R}}$ a per-object score, and let

$$s(c, S) = \min\Big(\beta(c), \min_{o \in S} h\big(c, x(o)\big)\Big), \qquad \min \emptyset = +\infty. \tag{S15}$$

The planner selects the first maximizer with respect to $\prec$, *i.e.*, $\pi(S) = c^*$ if and only if

$$s(c^*, S) \geq s(c, S) \;\text{ for all } c \in \mathcal{T}, \qquad s(c^*, S) > s(c, S) \;\text{ for all } c \prec c^*. \tag{S16}$$

Two properties of Eq. (S15) are used throughout: scores compose by the minimum, and removing objects cannot decrease a score,

$$s(c, S \cup S') = \min\big(s(c, S), s(c, S')\big), \qquad S' \subseteq S \;\Rightarrow\; s(c, S') \geq s(c, S). \tag{S17}$$

Corollary 2 (Bottleneck selection)**.** *A bottleneck selector is insertion-stable, and* $\mathrm{Sup}(S) = \{o \in S : h(c, x(o)) \leq s(\pi(S), S)$ *for some* $c \neq \pi(S)\}$ *is certifying.*

The corollary follows from Theorem 1 and the following two lemmas.

Lemma S1 (Insertion stability)**.** *If* $\pi(S \cup \{o\}) = \pi(S)$ *for every object* o *of a finite nonempty set* X*, then* $\pi(S \cup X) = \pi(S)$.

Proof strategy. By Eq. (S17), the scores of $S \cup X$ are the minima of the scores of the single insertions. The single insertion that lowers the score of c^* most bounds every other candidate from above.

Proof. Let $c^* = \pi(S)$. By Eq. (S17), $s(c, S \cup X) = \min_{o \in X} s(c, S \cup \{o\})$ for every c. Let $o_0 \in X$ minimize $s(c^*, S \cup \{o\})$, so that $s(c^*, S \cup X) = s(c^*, S \cup \{o_0\})$. For every c,

$$s(c, S \cup X) \;\leq\; s(c, S \cup \{o_0\}) \;\leq\; s(c^*, S \cup \{o_0\}) \;=\; s(c^*, S \cup X), \tag{S18}$$

where the second inequality holds since $c^* = \pi(S \cup \{o_0\})$. If equality holds throughout for some $c \neq c^*$, then c ties with c^* in $S \cup \{o_0\}$, and by Eq. (S16) for $S \cup \{o_0\}$, $c^* \prec c$. Hence c^* satisfies Eq. (S16) for $S \cup X$. □

Lemma S2 (Certification by the binding objects)**.** *Let* $c^* = \pi(S)$, $\theta = s(c^*, S)$, *and* $\mathrm{Sup}(S) = \{o \in S : h(c, x(o)) \leq \theta$ *for some* $c \neq c^*\}$. *Then* $\pi\big((S \setminus D) \cup Y\big) = c^*$ *for every* $D \subseteq S \setminus \mathrm{Sup}(S)$ *and* $Y \in \mathcal{Y}(S)$.

Proof strategy. An object outside the support has a score above θ for every competing candidate, so that removing it cannot raise any competing score above the score of c^*; removing objects can only raise the score of c^*.

Proof. Let $S^+ = S \cup Y$ and $W = S^+ \setminus D$; $\pi(S^+) = c^*$ since $Y \in \mathcal{Y}(S)$. For $c \neq c^*$, $s(c, S^+) \leq s(c^*, S^+) \leq \theta$, whereas every object $o \in D$ satisfies $h(c, x(o)) > \theta$; removing D therefore leaves the minimum unchanged,

$$s(c, W) = s(c, S^+) \qquad \text{for all } c \neq c^*. \tag{S19}$$

Moreover, $s(c^*, W) \geq s(c^*, S^+)$ by Eq. (S17). Hence $s(c, W) \leq s(c^*, W)$ for every $c \neq c^*$, and equality implies $s(c, S^+) = s(c^*, S^+)$, a tie in S^+ that Eq. (S16) resolves in favor of c^*, *i.e.*, $c^* \prec c$. □

Computing the field. The field requires only the per-candidate scores of the probe. With $c^* = \pi(S)$ and $\theta = s(c^*, S)$,

$$x \in Z(S) \iff \exists\, c \neq c^* : \min\big(s(c, S), h(c, x)\big) > h(c^*, x), \text{ or equality holds and } c \prec c^*. \tag{S20}$$

If $h(c^*, x) < \theta$, the score of c^* after the insertion is $h(c^*, x)$, and Eq. (S20) is the condition that another candidate overtakes it. If $h(c^*, x) \geq \theta$, the insertion leaves the score of c^* at θ, every competitor stays at most θ, and strictly below it if $c \prec c^*$; neither branch of Eq. (S20) can then hold, in agreement with $x \notin Z(S)$. Each cell thus costs $|\mathcal{T}|$ score evaluations and no re-planning, and single-object probes suffice by Lemma S1. The support, in contrast, requires a planner that exposes the scores h of all candidates.

Max-composable planners as bottleneck selectors. With $\mathcal{T} = \{0, \ldots, K\}$ ordered by $0 \prec 1 \prec \cdots \prec K$, $b(a) = -a$, and

$$h(a, x) = \begin{cases} -(K{+}1) & \text{if } d(x) \geq \tau_{a+1}, \\ +\infty & \text{otherwise,} \end{cases} \tag{S21}$$

so that

$$s(a, S) = \begin{cases} -a & \text{if } \max_{o \in S} d(x(o)) < \tau_{a+1}, \\ -(K{+}1) & \text{otherwise,} \end{cases} \tag{S22}$$

and the unique maximizer is $q(\max_{o \in S} d(x(o)))$, independently of the order. Lemma S1 therefore contains the insertion stability of Corollary 1. The binding set of Lemma S2 is $\{o : d(x(o)) \geq \tau_1\}$ for $a \geq 1$ and empty for $a = 0$; for $a \geq 1$, it is larger than the support of Corollary 1. The smaller support suffices because of the following property.

Lemma S3 (Witness supports)**.** *Let $c^* = \pi(S)$ and $\theta = s(c^*, S)$, and suppose that $s(c^*, S \cup Y) = \theta$ for every $Y \in \mathcal{Y}(S)$. Then every $T \subseteq S$ with $s(c, T) \leq \theta$ for all $c \neq c^*$, with strict inequality if $c \prec c^*$, is a certifying support.*

Proof. Let $D \subseteq S \setminus T$, $Y \in \mathcal{Y}(S)$, and $W = (S \setminus D) \cup Y$, so that $T \subseteq W \subseteq S \cup Y$. By Eq. (S17) and the assumption, $s(c^*, W) \geq s(c^*, S \cup Y) = \theta$, and $s(c, W) \leq s(c, T) \leq \theta$ for $c \neq c^*$, strictly if $c \prec c^*$. Hence c^* satisfies Eq. (S16). □

For a max-composable planner, an object outside the field has $h(a, x) = +\infty$ at the selected action a, so that the selected score is never lowered and the assumption of Lemma S3 holds. Every object with demand at least τ_a bounds all lower actions at $-(K{+}1) < -a$, whereas every higher action $a' > a$ is bounded by its preference $-a' < -a$; the support of Corollary 1 is therefore a witness, and for $a = 0$, the empty set is one, since no action precedes 0 and every higher action is bounded by its preference. For a general bottleneck selector, an insertion outside the field can lower the selected score, and a witness is not certifying (construction (f) in Sec. S1.6).

S1.5. Termination

Each round that does not stop adds at least one crop that does not lie inside the union of the previous crops. The union therefore grows strictly, by at least one image pixel, and the loop stops after finitely many rounds; this argument needs neither locality nor a bound on the number of rounds. Under locality, the requested crops are moreover drawn from a small finite set, namely the footprints of subsets of the finite ground grid and the crops of the finitely many objects of C and F. The experiments cap the loop at three rounds, and a frame that reaches the cap returns the action of its last merged scene; Theorem 1 covers only frames on which the loop reaches its stopping condition. More than 99.9% of the frames reach it within the cap, and raising the cap to ten rounds resolves no further disagreement with $\pi(F)$ (Fig. S1). The deployed stopping test approximates containment: a requested crop counts as verified once at least 95% of its area lies inside one earlier crop, and for a support crop, the guard crops also count. A rectangle with more than half of its area inside another rectangle has its center there, so that a support crop not clipped by the image boundary still has the box center of its object in that earlier crop; a footprint, however, can leave up to 5% of its area unverified, which can affect coverage (Sec. S1.8).

S1.6. Counterexamples for omitted or weakened conditions

The following constructions show failures when the coverage of the field, the certification by the support, insertion stability, or the iteration to a fixed point is omitted or weakened; Tab. S2 summarizes them. In (a)–(d), π is max-composable, and all constructions retain locality, so they do not test whether locality can be dropped.

Table S2. **Counterexamples for omitted or weakened conditions.** For each construction, the procedure that drops or weakens the condition returns an action that differs from $\pi(F)$; DGSP returns $\pi(F)$ in (a)–(d) and (f).

	Procedure or planner	Condition violated	Returned *vs.* $\pi(F)$
(a)	field only	support (empty support not certifying)	$a \geq 1$ *vs.* 0
(b)	support only	coverage of the field	0 *vs.* ≥ 1
(c)	pivots instead of the support	certification (two objects justify the action)	a *vs.* 0
(d)	a single round	fixed point	0 *vs.* 1
(e)	additive demands	insertion stability	0 *vs.* 1
(f)	witness instead of the binding set	certification (bottleneck selector)	c_{A} *vs.* c_{B}

(a) Field only. Let, for some $a \geq 1$,

$$C = \{p\} \text{ with } \tau_a \leq d(p) < \tau_{a+1}, \qquad F = \emptyset,$$

so that $\pi(C) = a$ and $\pi(F) = 0$. The field $Z(C) = \{x : d(x) \geq \tau_{a+1}\}$ does not contain the state of p, so that a procedure that verifies only the footprint of the field returns $a \neq \pi(F)$ whenever the image region of p lies outside this footprint; the empty support is not certifying. Verifying the support removes p.

(b) Support only. Let

$$C = \emptyset, \qquad F = \{o\} \text{ with } d(o) \geq \tau_1.$$

Then $\pi(C) = 0$, and the support is empty; a support-only procedure verifies no region and returns $0 \neq \pi(F)$. The field $Z(\emptyset) = \{x : d(x) \geq \tau_1\}$ contains the state of o, so that its footprint is verified and o is detected.

(c) Pivots do not replace the support. Let

$$C = \{p_1, p_2\} \text{ with } \tau_a \leq d(p_j) < \tau_{a+1} \;\; (j = 1, 2), \qquad F = \emptyset.$$

Removing either object leaves the action at a; there is no pivot, the empty set of pivots is not certifying, and a pivot-based procedure returns a, whereas $\pi(F) = 0$. The support contains both objects.

(d) A single round is insufficient. Let $K = 2$ and

$$C = \{p\} \text{ with } d(p) \geq \tau_2, \qquad F = \{o\} \text{ with } \tau_1 \leq d(o) < \tau_2,$$

where o lies outside the crop of p. Since $\pi(C) = K$, the field of the first round is empty. The first round removes p, the merged scene is empty, and the action becomes 0. The state of o is contained only in the field for action 0, which is computed in the second round.

(e) Joint effects. Let the demands add up, $\pi(S) = q\big(\sum_{o \in S} d(x(o))\big)$, and let

$$C = \emptyset, \qquad F = \{o_1, o_2\} \text{ with } d(x(o_1)) = d(x(o_2)) = 0.6\,\tau_1, \quad 1.2\,\tau_1 < \tau_2.$$

The field $Z(\emptyset) = \{x : d(x) \geq \tau_1\}$ contains neither state, and the empty support of $\emptyset$ is certifying. If the image regions of o_1 and o_2 lie outside the footprint, the procedure terminates after one round and returns 0, whereas $\pi(F) = 1$; only insertion stability is violated.

(f) Witnesses are not certifying for bottleneck selectors. Let $\mathcal{T} = \{c_{\mathrm{A}}, c_{\mathrm{B}}\}$ with $c_{\mathrm{A}} \prec c_{\mathrm{B}}$ and $\beta \equiv 5$. The draft contains o_1 and o_2, which affect only c_{B}; the reference scene contains o_1 and an object o_x with state x that the draft misses:

$$h(c_{\mathrm{B}}, o_1) = 4, \quad h(c_{\mathrm{B}}, o_2) = 2, \qquad h(c_{\mathrm{A}}, x) = 3, \quad h(c_{\mathrm{B}}, x) = +\infty.$$

The scores $(s(c_{\mathrm{A}}, \cdot), s(c_{\mathrm{B}}, \cdot))$ are

$$C : \; (5, 2) \;\Rightarrow\; \pi(C) = c_{\mathrm{A}}, \qquad F : \; (3, 4) \;\Rightarrow\; \pi(F) = c_{\mathrm{B}}, \qquad C \cup \{o_x\} : \; (3, 2),$$

so that the state of x lies outside $Z(C)$. The set $\{o_1\}$ is a witness, since $s(c_\mathrm{B}, \{o_1\}) = 4 \leq 5$, but the assumption of Lemma S3 fails, since inserting o_x lowers the selected score from 5 to 3. Verifying only o_1 leaves the merged scene equal to the draft, and the procedure returns c_A. The binding set $\{o_1, o_2\}$ removes the spurious o_2; the field of $\{o_1\}$ then contains x (scores $(3, 4)$), and the procedure returns c_B. In driving terms, o_2 is a spurious obstacle close to an alternative trajectory, o_1 a real obstacle farther from it, and o_x an obstacle on the current trajectory that the draft misses.

S1.7. Planners outside the class

PDM-Closed is not a bottleneck selector: its collision and time-to-collision terms are minima over objects, but they are combined multiplicatively and additively with other terms, and its candidate set depends on the objects. Its support is obtained by removal queries, *i.e.*, as the set of pivots, which is not guaranteed to be certifying; we therefore measure violations of the certification condition Eq. (3) on 1,000 navtest frames: of 9,449 pairs of objects outside the support, 6 change the plan only when removed together (D of size two, $Y = \emptyset$), and of 139,675 combinations of a probe outside the field and the removal of an object outside the support, 11 do ($|D| = |Y| = 1$); adding the corresponding objects to the support changes the score of none of the 1,000 frames.

Empirical check of Corollary 2. We instantiate a bottleneck selector on NAVSIM navtest with 15 candidates, namely the route centerline offset by 0 and ± 1.5 m combined with constant accelerations of $+2$, $+1$, 0, -1.5, and -3 m/s^2; the preference β is 5 s times the normalized progress, and $h(c, o)$ is the time of the first overlap of the ego footprint along c with object o under constant velocity, capped at 5 s. The field is probed with a static vehicle on the 7×7 grid used for PDM-Closed (Tab. S7), and the support is the binding set of Lemma S2. With the whole-frame detections inside the crops, *i.e.*, with locality enforced, the procedure agrees with $\pi(F)$ on 99.71% [99.58, 99.83] of the 12,146 scenes and on 95.2% of the 724 scenes in which the cheap draft does not. In each of the 35 remaining scenes, an object outside the crops changes the selection when it is inserted alone, *i.e.*, the finite grid of static probes misses a sensitive state; no disagreement is due to a joint effect. With real crop detections, the agreement is 96.8%, compared with 95.9% for whole-frame inference at the native resolution. On 1,000 scenes, none of the 769,561 pairs of probes whose single insertions leave the selection unchanged changes it jointly, consistent with Lemma S1; this checks the implementation on the sampled pairs, whereas the guarantee itself follows from the lemma.

S1.8. From the theorem to the deployed system

Table S3 lists, for each condition of Theorem 1, how the deployed system realizes it and where its effect is measured. The reference scene F of all exactness measurements is whole-frame inference at the verification resolution with the same detector and the same learned looming estimate, so that $\pi(F)$ is the action that the theorem promises.

Attribution of the disagreement. Fig. S1 removes the approximations of the practical system one at a time, starting from the deployed procedure without the refiner.

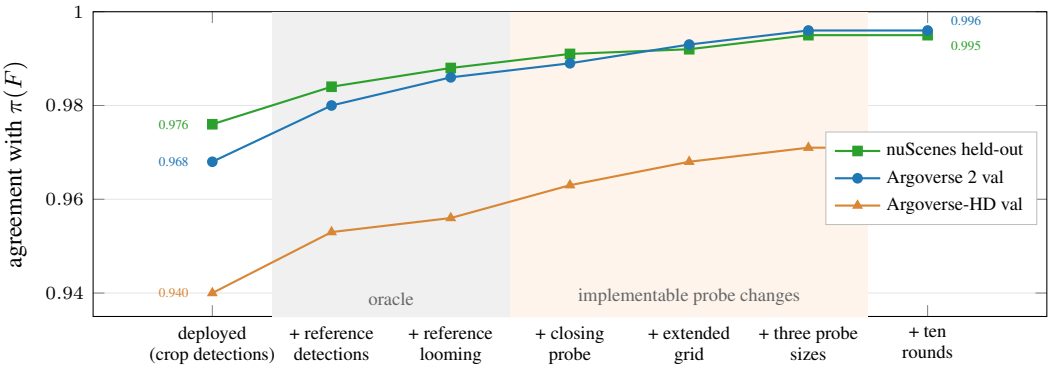


Figure S1. **Which approximation causes the remaining disagreement with $\pi(F)$?** Agreement of DGSP (without refiner) with whole-frame inference at the verification resolution, as the approximations are removed cumulatively from left to right (test splits). Grey: oracle diagnostics, not deployable, that replace the crop detections and their looming estimate by those of the reference; orange: implementable changes of the probe; the last step raises the limit of three rounds to ten. Values in Tab. S47.

Removed cumulatively, the oracle interventions account for 50%, 56%, and 26% of the disagreeing frames and the probe changes for most of the remainder, which raises the agreement to 99.5%, 99.6%, and 97.1%; the loop terminates

Table S3. **Conditions of Theorem 1 in the deployed system.** "Exact": the condition holds by construction; "approx.": the deployed system approximates it; "bounded": the iteration is capped; the last column states the measured effect.

Condition	Deployed system	Status	Measured effect
Insertion stability	Braking controller and time-to-collision warning are max-composable (Secs. S2.1 and S4.3)	exact	Corollary 1; for PDM-Closed, outside the class, joint effects are rare (Sec. S1.7)
Certifying support	Closed form of Corollary 1: $\{o : d(x(o)) \geq \tau_a\}$ for $a \geq 1$, $\emptyset$ at KEEP; each support object contributes a crop of twice its box	exact	superset of a minimal certifying support (Sec. S1.3)
Coverage	Closed-form field evaluated on a finite ground grid with one probe size and a static probe; footprint: bounding rectangle of the sensitive probes; stopping test with a 95% overlap instead of containment (Sec. S1.5)	approx.	implementable probe changes resolve 28%, 33%, and 26% of the disagreements (Fig. S1); coverage of the decisive object 92–96% (Sec. S4.2)
Locality	Crop detection differs from whole-frame detection by truncation and temporal cues; truncation guard and same-scale history. The guard replaces a truncated detection by the matching detection of an enlarged crop (at most two enlargements, after which a detection is merged as it is; dropped if no detection of the enlarged crop covers half of it). The kept detection can have its center outside R, and its draft twin (IoU ≥ 0.5) is removed; guard crops count as pixels but are not part of R	approx.	reference detections and looming in the crops resolve 50%, 56%, and 26% of the disagreements (Fig. S1)
Fixed point	At most three rounds	bounded	ten rounds resolve no disagreement (Fig. S1)
(none)	Range refiner, the last step of the lift: applied once after the loop to the verified support objects of the final scene, and to the support of the whole-frame scene for F	outside the theorem	agreement with the refiner in DGSP and in F: 98.0%, 96.9%, and 94.2% (Tab. S5); without it in both: 97.6%, 96.8%, and 94.0%

after at most one round in 96%, 89%, and 86% of the frames. Applied alone instead of cumulatively, substituting the reference detections inside the crops resolves 42.7%, 46.9%, and 30.0% of the 218, 382, and 897 disagreeing frames, taking the lift and looming estimate of the matching reference detection 30.3%, 54.5%, and 30.8%, the closing probe 20.2%, 19.4%, and 19.3% (identical at 10 and 20 m/s), the extended grid 23.9%, 27.7%, and 15.8%, the three probe templates 22.9%, 21.5%, and 12.8%, and ten rounds none. Each change of the probe also creates 4 to 111 new disagreements, and 26.6%, 19.9%, and 42.4% of the frames are resolved by no single intervention; the causes therefore overlap, most on Argoverse-HD.

Crops outside the verified region. As a random-crop diagnostic of whether computation outside the field and the support changes the action (it cannot show that no region outside does), random crops with the sizes of the draft boxes are added strictly outside every verified crop and the field footprint, at 25% and 50% more crop pixels than DGSP on each frame (five seeds), and processed like the crops of DGSP without a looming history. They change the action on 0.009%, 0.015–0.024%, and 0.044–0.072% of the frames (mean over seeds), compared with the disagreement of 2.4%, 3.2%, and 6.0% between the deployed procedure and $\pi(F)$, and the decision value by -0.0002, -0.0005 to -0.0007, and -0.0012 to -0.0018; most changes on Argoverse-HD stem from annotated objects outside the field. In the frames that remain, the annotated scene supports the action of DGSP in 44, 34, and 398 frames and the action of whole-frame inference in 5, 5, and 2.

Disagreements of the deployed system. Without any refiner, the agreement with $\pi(F)$ is 97.6%, 96.8%, and 94.0% for DGSP, 93.0%, 84.0%, and 77.0% for the cheap detector, and 97.4%, 95.1%, and 92.2% for native-resolution inference with the learned looming estimate; with the refiner in both DGSP and F, it is 98.0%, 96.9%, and 94.2% for DGSP and 97.7%, 95.8%, and 92.6% for native-resolution inference. Table S4 compares the actions of the deployed system with those of F, without any intervention. In the disagreeing frames, the action of DGSP equals the required action in 70%, 81%, and 73% of the cases, and native-resolution inference with the same components behaves alike. Without any refiner, in the frames in which DGSP and $\pi(F)$ disagree, the annotated scene requires the action of

Table S4. **Disagreements with whole-frame inference** at the verification resolution with the learned looming estimate and without the refiner (F); each pipeline uses the refiner if it has one. "Disagr.": share of frames; "Pipeline" / "F": share of the disagreeing frames in which the action equals the one required by the annotated scene; "Crit.": share of frames in which braking is required and the pipeline takes a lower action than F.

Pipeline	Disagr.	Pipeline	F	Crit.
nuScenes held-out				
Cheap (u320)	7.0%	31%	64%	1.1%
Native + comp.	3.8%	70%	26%	0.7%
DGSP	3.7%	70%	27%	0.9%
Argoverse 2 val				
Cheap (u320)	16.0%	28%	63%	1.8%
Native + comp.	13.1%	78%	17%	3.4%
DGSP	12.5%	81%	14%	3.5%
Argoverse-HD val				
Cheap (u320)	23.0%	21%	62%	4.6%
Native + comp.	17.2%	66%	22%	5.2%
DGSP	16.0%	73%	21%	5.1%

DGSP in 128, 219, and 722 frames, that of $\pi(F)$ in 81, 126, and 130, and neither in 9, 37, and 45. Among the 60 costliest frames in which DGSP brakes less than required, the decisive object is detected but its time-to-collision, lateral position, or range is estimated inaccurately in 46, it lies in a crop but is missed in 11, and it lies outside every crop in 3; the residual safety-critical errors thus stem from the monocular lift rather than from the selection of the regions.

Table S5. **Fidelity to the reference action $\pi(F)$ with the shared lift**, *i.e.*, with the range refiner in every pipeline and in F (the convention of the main paper; test splits, braking controller, own history, computed from stored actions without the decision loss). A: share of frames with the action of $\pi(F)$; Rep.: the same on the frames in which the cheap detector differs from $\pi(F)$ (671, 2,245, and 4,006 frames); Trans.: the same on the frames in which $\pi(F)$ changes from the previous frame; px: detector-input pixels in % of the native frame, including the guard. Capped: nominal per-frame cap at the pixels of DGSP. † / *: DGSP significantly higher / lower (paired unit bootstrap, 95%).

	nuScenes held-out				Argoverse 2 val				Argoverse-HD val			
Pipeline	A	Rep.	Trans.	px	A	Rep.	Trans.	px	A	Rep.	Trans.	px
Always KEEP	0.911	0.580	0.429	0.0	0.745	0.625	0.366	0.0	0.679	0.495	0.385	0.0
u320	0.927†	0.000†	0.653†	4.3	0.810†	0.000†	0.555†	2.6	0.733†	0.000†	0.497†	3.1
Whole frame u512	0.978	0.800	0.763	10.2	0.963†	0.875	0.766	6.7	0.950	0.868	0.608	7.1
Whole frame u640	0.983	0.861*	0.827*	17.1	0.969	0.893	0.798	10.3	0.963	0.902	0.671	11.6
Whole frame u1280	0.984*	0.887*	0.845*	65.4	0.976*	0.931*	0.843*	40.0	0.965	0.930*	0.664	44.4
Whole frame native	0.977	0.821	0.782	103.1	0.958†	0.887	0.778	101.2	0.925†	0.869*	0.633	101.3
AnA-style, iterated	0.985*	0.845*	0.838*	10.5	0.973*	0.904	0.796	7.9	0.967	0.913	0.683	20.6
AdaScale	0.985*	0.879*	0.850*	30.6	0.990*	0.967*	0.927*	26.8	0.985*	0.965*	0.866*	29.9
SAHI	0.974†	0.861*	0.822*	187.7	0.928†	0.799†	0.737†	165.2	0.921	0.855	0.637	153.8
FOVEA, instance saliency	0.980	0.848*	0.834*	69.7	0.958†	0.893	0.791	42.6	0.960	0.907	0.633	47.6
UFP	0.980	0.829	0.803	18.1	0.972	0.904	0.820*	9.0	0.957	0.884	0.683	12.8
AnA-style, iterated, capped	0.979	0.770	0.780	5.8	0.964†	0.867†	0.758†	5.0	0.936†	0.814†	0.649	7.4
Counterfactual, iterated, capped	0.973†	0.697†	0.742†	5.2	0.961†	0.841†	0.739†	4.2	0.906†	0.693†	0.608	5.9
AdaScale, capped	0.972†	0.693†	0.749	5.5	0.950†	0.804†	0.716†	3.7	0.916	0.730	0.603	5.1
SAHI, capped	0.945†	0.697†	0.686†	6.4	0.879†	0.689†	0.624†	4.2	0.867†	0.685†	0.558†	5.5
FOVEA, instance saliency, capped	0.962†	0.745	0.726	7.5	0.934†	0.816†	0.686†	5.1	0.858†	0.655	0.556	6.8
UFP, capped	0.980	0.803	0.782	7.2	0.958†	0.855†	0.757	4.8	0.945	0.850	0.658	6.3
Chanakya-style, capped	0.933†	0.088†	0.691†	4.6	0.819†	0.052†	0.574†	2.9	0.748†	0.059†	0.517†	3.8
Support + fixed tube	0.978	0.773	0.770	10.8	0.969	0.891	0.784	7.1	0.942	0.838	0.644	10.8
Support + fixed tube, strict cap	0.973†	0.736†	0.754	7.5	0.963†	0.872†	0.773	5.1	0.918†	0.774†	0.608†	6.8
Support only	0.972†	0.681†	0.726†	5.0	0.958†	0.828†	0.724†	4.1	0.905†	0.691†	0.599†	5.6
Field only	0.943†	0.244†	0.700†	6.8	0.839†	0.173†	0.615†	3.7	0.780†	0.192†	0.533†	4.3
DGSP	0.980	0.788	0.773	7.8	0.969	0.892	0.790	5.4	0.942	0.835	0.637	7.0

Table S6. **Diagnostic: fidelity without the range refiner** in any pipeline or in F, which isolates the loop that Theorem 1 covers. Columns and marks as in Tab. S5; the cheap detector differs from $\pi(F)$ on 639, 1,891, and 3,447 frames.

	nuScenes held-out				Argoverse 2 val				Argoverse-HD val			
Pipeline	A	Rep.	Trans.	px	A	Rep.	Trans.	px	A	Rep.	Trans.	px
Always KEEP	0.899	0.534	0.441	0.0	0.662	0.417	0.372	0.0	0.592	0.293	0.363	0.0
u320	0.930†	0.000†	0.657†	4.3	0.840†	0.000†	0.642†	2.6	0.770†	0.000†	0.498†	3.1
Whole frame u512	0.971†	0.729	0.761	10.2	0.953†	0.801†	0.781†	6.7	0.942	0.831	0.609	7.1
Whole frame u640	0.979	0.817*	0.822*	17.1	0.964	0.850	0.826	10.3	0.958	0.884	0.662	11.6
Whole frame u1280	0.982*	0.873*	0.856*	65.4	0.974*	0.922*	0.863*	40.0	0.964*	0.919*	0.676	44.4
Whole frame native	0.974	0.806*	0.795	103.1	0.951†	0.882	0.800	101.2	0.922†	0.849*	0.625	101.3
AnA-style, iterated	0.981*	0.786*	0.822*	10.5	0.963	0.828†	0.815	7.9	0.960	0.870	0.672	20.6
AdaScale	0.981*	0.829*	0.820*	30.6	0.986*	0.944*	0.923*	26.8	0.985*	0.963*	0.850*	29.9
SAHI	0.967†	0.822*	0.816*	187.7	0.918†	0.824†	0.783†	165.2	0.920	0.890	0.631	153.8
FOVEA, instance saliency	0.976	0.825*	0.824*	69.7	0.951†	0.854	0.818	42.6	0.956	0.885	0.648	47.6
UFP	0.975	0.765	0.786	18.1	0.957†	0.807†	0.816	9.0	0.953	0.856	0.668	12.8
AnA-style, iterated, capped	0.974	0.704†	0.771	5.8	0.955†	0.786†	0.787†	5.0	0.932†	0.766†	0.633	7.4
Counterfactual, iterated, capped	0.966†	0.585†	0.712†	5.2	0.934†	0.644†	0.753†	4.2	0.890†	0.573†	0.596	5.9
AdaScale, capped	0.967†	0.585†	0.731	5.5	0.928†	0.595†	0.732†	3.7	0.908	0.623	0.592	5.1
SAHI, capped	0.936†	0.604†	0.665†	6.4	0.840†	0.529†	0.598†	4.2	0.851†	0.616†	0.549†	5.5
FOVEA, instance saliency, capped	0.955†	0.673	0.720†	7.5	0.911†	0.675†	0.729†	5.1	0.842†	0.602	0.537†	6.8
UFP, capped	0.973	0.725	0.756	7.2	0.937†	0.738†	0.766†	4.8	0.936	0.801	0.641	6.3
Chanakya-style, capped	0.936†	0.085†	0.688†	4.6	0.847†	0.045†	0.652†	2.9	0.787†	0.076†	0.512†	3.8
Support + fixed tube	0.974†	0.718	0.767	10.8	0.967	0.868	0.821	7.1	0.939	0.803	0.645	10.8
Support + fixed tube, strict cap	0.969†	0.681†	0.750	7.5	0.962†	0.850†	0.813	5.1	0.924†	0.753†	0.633	6.8
Support only	0.964†	0.563†	0.701†	5.0	0.929†	0.617†	0.732†	4.1	0.888†	0.562†	0.578†	5.6
Field only	0.944†	0.225†	0.720†	6.8	0.880†	0.283†	0.733†	3.7	0.822†	0.247†	0.568†	4.3
DGSP	0.976	0.734	0.769	7.8	0.968	0.870	0.822	5.4	0.940	0.803	0.643	7.0

Fidelity with the shared lift. Table S5 states the fidelity in the convention of the main paper: every pipeline and the reference F use the lift, including the range refiner. DGSP agrees with $\pi(F)$ on 98.0%, 96.9%, and 94.2% of the frames and repairs 78.8%, 89.2%, and 83.5% of the disagreements of the cheap detector. Under the nominal cap, it repairs significantly more than the published, learned, and object-level methods in 27 of the 36 comparisons and significantly less in none (the nine others: AnA-style refinement, FOVEA, and UFP on nuScenes, and AdaScale, FOVEA, and UFP on Argoverse-HD); under the strict cap, it repairs significantly more than the hybrid and the primary crop heuristics in all 12 comparisons. Whole-frame inference at 512 pixels, with 1.0–1.3× the pixels of DGSP, repairs 80.0%, 87.5%, and 86.8%, which does not differ significantly on any dataset; its raw agreement is significantly lower only on Argoverse 2. At similar pixels, a lower whole-frame resolution is therefore as faithful to the reference as DGSP; the two differ in decision value (Tab. S44) and in latency (Tab. S34). At changes of $\pi(F)$, DGSP agrees on 77.3%, 79.0%, and 63.7% of the frames, and on the frames in which it differs from $\pi(F)$, the annotated scene requires its action in 58%, 55%, and 81% and that of $\pi(F)$ in 38%, 37%, and 15%.

Fidelity without the range refiner. The raw agreement is dominated by KEEP: without the refiner, always keeping the speed agrees with $\pi(F)$ on 89.9%, 66.2%, and 59.2% of the frames. Table S6 therefore also reports the repair rate, *i.e.*, the agreement on the frames in which the draft differs from $\pi(F)$, and the agreement at changes of $\pi(F)$; both are computed from stored actions, without any detector run and without the decision loss. DGSP repairs 73.4%, 87.0%, and 80.3% of the disagreements of the draft. Under the nominal cap, it repairs significantly more than the published, learned, and object-level methods in 29 of the 36 comparisons and significantly less in none; under the strict cap, it repairs significantly more than the hybrid and the primary crop heuristics in all 12 comparisons. Under the nominal cap of Tab. 5, with 1.3–1.5× the pixels, the hybrid ties DGSP in repair rate on all three datasets. Every pipeline that repairs significantly more uses more pixels: iterated AnA-style refinement on nuScenes (1.3×), whole-frame inference from 640 pixels on nuScenes (2.2×) and at 1280 pixels elsewhere (6–8×), AdaScale (4–5×), and SAHI and FOVEA on nuScenes (9–24×). Fig. S2 plots the repair rate against the pixels. At the same pixels on Argoverse-HD, whole-frame inference at 512 pixels repairs 83.1% (not significant), and DGSP is significantly more faithful only when braking is required (95.4% *vs*. 89.7%). The agreement at changes of $\pi(F)$ is the weakest metric of DGSP (76.9%, 82.2%, and 64.3%): when the set of crops changes, the history of an object detected in a crop starts anew. On the frames in which DGSP differs from $\pi(F)$, the annotated scene requires its action in 59%, 57%, and 80% and that of $\pi(F)$ in 37%, 33%, and 14%; for whole-frame inference at 1280 pixels and for AdaScale, the two shares lie within 11 points of each other.

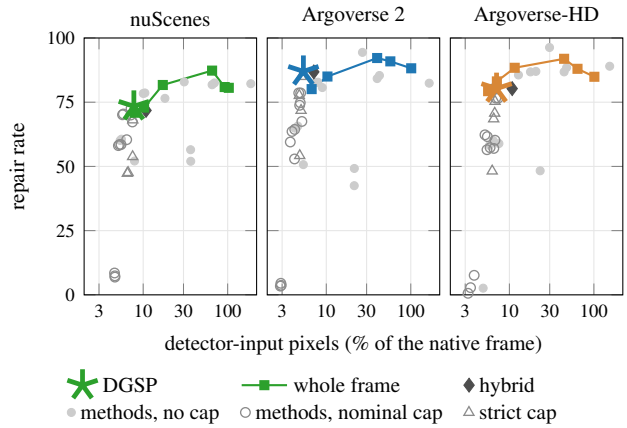


Figure S2. **Repair rate *vs.* pixels**, without the refiner. Share of the cheap detector's disagreements with $\pi(F)$ that a pipeline resolves (u960, the reference, omitted; values in Tab. S6). DGSP lies on the upper-left frontier of the tested pipelines: every pipeline above it uses more pixels.

Fidelity and decision value. Fidelity does not fix the decision value: G scores the actions against the annotated scene, whereas $\pi(F)$ is itself the action of a detector. The reference attains $G = 0.085$ $[-0.203, +0.347]$, 0.290 $[0.234, 0.350]$, and 0.296 $[0.097, 0.462]$ on nuScenes, Argoverse 2, and Argoverse-HD, and DGSP without the refiner 0.181, 0.241, and 0.422. The differences, $+0.096$ $[-0.085, +0.305]$, -0.049 $[-0.116, +0.007]$, and $+0.126$ **[+0.015, +0.332]**, arise entirely on the 218, 382, and 897 frames on which the two actions differ: since the previous action in the loss is that of the cheap pipeline for all pipelines (Sec. S2.1), every loss term is equal on the other frames, and their contribution is exactly zero. On the differing frames, DGSP has less excessive-braking loss on all three datasets ($+0.037$ $[+0.020, +0.067]$, $+0.029$ $[+0.018, +0.041]$, and $+0.131$ $[+0.021, +0.336]$, in units of the loss of the cheap pipeline) and more insufficient-braking loss on Argoverse 2 and Argoverse-HD (-0.076 $[-0.143, -0.023]$ and -0.007 $[-0.011, -0.003]$). Its higher point estimates on nuScenes and Argoverse-HD therefore come from fewer excessive braking actions on the few frames on which its crops and object histories yield a different scene than whole-frame inference. With the refiner, the differences to the reference are $+0.125$, $+0.003$, and $+0.122$, none significant.

S2. Implementation, evaluation protocol, and baseline fairness

This section specifies the planner and the decision value (Sec. S2.1), the benchmarks, the configuration, and the selection protocol (Sec. S2.2), the accounting of detector-input pixels (Sec. S2.3), the ego mask on Argoverse 2 (Sec. S2.4), and the baselines and the checks that make their comparison fair (Secs. S2.5 and S2.6).

S2.1. Planner and decision value

For completeness, this section restates the benchmark components of [9] that our experiments adopt: the monocular lift, the braking controller, and the loss behind the decision value G. The decision-exactness objective and its measurement by the action agreement with $\pi(F)$ (Sec. S1.8) use the controller but not the loss.

Monocular lifting. Each 2D detection is lifted to an object state in the ego frame with a pinhole camera model. The range is a combination of a ground-plane cue, computed from the bottom edge of the box, and a height cue, computed from the box height and a class-specific height prior. The weight of the ground-plane cue increases with the distance of the contact point from the horizon. The lateral extent is obtained from the horizontal box edges at the estimated range, and the time-to-collision from the change of the box between consecutive frames. The "oracle" geometry replaces the range of matched boxes by the annotated range; it is used only in diagnostic analyzes, and the deployed pipeline estimates ranges from the image detections.

Braking controller. For an object within the ego corridor (half-width $1.2\,\mathrm{m}$) and within $80\,\mathrm{m}$, at range r and with ego speed v, the demand is

$$d = \min\Big(\max\Big(\tfrac{v^2}{2\,(r - r_0 - v\,t_r)},\ \tfrac{v_{\mathrm{rel}}^2}{2\,(r - r_0)} \Big),\ 9 \Big)\ \mathrm{m/s^2}, \tag{S23}$$

with stand-off distance $r_0 = 2\,\mathrm{m}$, reaction time $t_r = 0.4\,\mathrm{s}$, and closing speed $v_{\mathrm{rel}} = r/\mathrm{ttc}$, where the denominators are bounded from below by $0.1\,\mathrm{m}$. Objects outside the corridor have $d = 0$. The action is HARD-BRAKE if the maximum

demand is at least $\tau_2 = 3.5\,\mathrm{m/s^2}$, DECELERATE if it is at least $\tau_1 = 1.0\,\mathrm{m/s^2}$, and KEEP otherwise. The commanded decelerations are $u = 6.0$, 2.5, and $0\,\mathrm{m/s^2}$, respectively.

Loss. Let $d^\star$ denote the maximum demand over the annotated objects; the object that attains it is the *decisive object* of the frame, and an annotated object is *decision-relevant* if its own demand reaches τ_1. The loss of action a with command $u(a)$ is

$$L = \lambda_r \left[d^\star - u(a)\right]_+^2 + \lambda_b \left[u(a) - d^\star\right]_+^2 + \lambda_j \left|u(a) - u(a_{\mathrm{prev}})\right| + \lambda_c \mathbb{1}\left[d^\star - u(a) \geq 8\right], \tag{S24}$$

with $(\lambda_r, \lambda_b, \lambda_j, \lambda_c) = (1,\ 0.12,\ 0.02,\ 6)$. The previous action a_{prev} is that of the cheap pipeline for all methods, such that all pipelines are evaluated against the same history. The loss is asymmetric: insufficient braking is penalized as a safety shortfall, whereas excessive braking costs comfort and progress.

Perception in the loop for planners. For NAVSIM and nuPlan, an annotated object that projects into the front camera is retained only if it is matched to a detection (Hungarian matching, compatible class, IoU ≥ 0.3). Unmatched annotated objects within the camera frustum are removed, unmatched detections are converted to static objects on the ground plane with the class-median size, and objects outside the camera frustum remain privileged.

S2.2. Benchmarks, configuration, and selection protocol

Table S7. **Benchmarks and configurations.** Configurations are selected on the development split, and the final configuration is frozen before its evaluation on the test split. The nuScenes development and test splits (the latter is not the official test set) are disjoint sets of 152 and 228 scenes of its trainval set, also disjoint from the scenes used for training. Front-camera resolutions: nuScenes 1600×900, Argoverse 2 1550×2048, Argoverse-HD 1920×1200 at 30 Hz. One configuration is used for the three megapixel datasets, KITTI, and Bench2Drive; NAVSIM uses the listed variant for its trajectory planner.

Benchmark	Development	Test	Metric	Configuration	Selected on
nuScenes Argoverse 2 Argoverse-HD	152 scenes train, 700 logs –	228 scenes, 9,181 fr. val, 150 logs, 11,818 fr. val, 24 seq., 15,006 fr.	G	Braking field with a static probe ($v_{\mathrm{close}} = 0$), ≤ 3 rounds, crops at 960/long side ($s = 0.6$, 0.469, and 0.5), truncation guard, scene history with the learned looming estimate, range refiner	nuScenes development split and Argoverse 2 training logs
KITTI	15 sequences	6 seq., 3,536 fr.	G	Configuration of the megapixel datasets	Not tuned
NAVSIM	OpenScene mini	navtest, 12,146 scenes	PDMS	Field and support obtained by probing PDM-Closed (7×7 grid, 8–55 m longitudinal, ± 10.5 m lateral; a cell is sensitive if the 4 s trajectory moves by more than 0.5 m), crop scale 0.5, one round, guard	OpenScene mini
Bench2Drive	–	219 routes	DS	Configuration of the megapixel datasets without the refiner, since matched objects retain their true state	Not tuned

The grid of the braking field comprises the ranges $\{12.5, 17.5, 22.5, 27.5, 35, 45, 55, 70\}$ m and lateral offsets from -6 to 6 m in steps of 1.5 m, with a static probe of 1.8×1.6 m; the loop runs at most three rounds, crops at the scale of the frame resized to a long side of 960 pixels, the truncation guard enlarges a crop at most twice, and each support object contributes a crop of twice its box size around its center. *Merge rule:* a draft detection is replaced if its box center lies in a verified crop; detections of overlapping crops with IoU ≥ 0.5 keep the more confident one, and a draft detection with IoU ≥ 0.5 to a kept detection is also removed once guard crops were used. A static probe ($v_{\mathrm{close}} = 0$) carries no closing rate, so that its demand reduces to the first term of Eq. (S23), that of a stationary obstacle. The sensitivity analyzes also use $v_{\mathrm{close}} > 0$, whose meaning follows the planner: for the braking controller, the probe has the time-to-collision r/v_{close}, *i.e.*, v_{close} is the closing speed of Eq. (S23); for the time-to-collision warning, it has the time-to-collision $r/(v + v_{\mathrm{close}})$. Both coincide with a stationary obstacle at $v_{\mathrm{close}} = 0$.

Range refiner. The refiner is a ResNet-18 [8] that receives a 64×64 crop and box features and regresses a correction of the log-range. It is trained on detections from KITTI train (14,605), nuScenes training scenes (62,527), and 340

Table S8. **Configurations evaluated on the test splits** (decision value G; detector-input pixels in % in parentheses). M and L use a larger detector on the crops.

Configuration	nuScenes	Argoverse 2	Argov.-HD
Earlier configuration	0.170 (10.0)	0.122 (5.3)	0.091 (7.0)
Hand-crafted looming	0.199 (7.7)	0.262 (5.3)	0.338 (7.0)
DGSP	0.211 (7.8)	0.294 (5.4)	0.418 (7.0)
M: YOLOv8m on the crops	0.298 (6.5)	0.281 (4.4)	0.428 (5.7)
L: YOLO11m on the crops	0.315 (7.7)	0.292 (5.4)	0.431 (7.0)

Argoverse 2 training logs (96,918), with model selection on KITTI val and 60 further Argoverse 2 training logs. No scene or log of a development or test split and no Argoverse-HD data are used. On the held-out detections, it reduces the mean absolute log-range error of the lift from 0.263 to 0.058.

Looming estimate. The looming estimate applies a ResNet-18 with shared weights to the 64×64 crops of an object in the last three frames and combines the features with the box features, the ratios of consecutive box heights, and the frame interval. It regresses the relative closing rate, *i.e.*, the inverse time-to-collision, and an object is assigned a time-to-collision only if the estimated rate exceeds a small threshold. All three observations must stem from detections at the same scale; an object with fewer observations carries no time-to-collision. The estimate is trained on detections from the Argoverse 2 training logs of the refiner and from the nuScenes training scenes, with model selection on held-out logs and scenes of both. It applies to crop detections and, for the baselines that are given the shared components, to whole-frame detections alike. The hand-crafted alternative in Tabs. S8 and S16 takes the median of the pairwise growth estimates over the same three observations.

Protocol. The configuration was selected on development data only, namely the nuScenes development split (152 scenes) and 300 Argoverse 2 training logs that are disjoint from the logs used for the refiner and the looming estimate. A component or parameter was adopted if the interval of its effect was above zero on one development set and its point estimate was not negative on the other, and shared values were preferred over dataset-specific ones. The configuration, its code, the weights of the refiner and of the looming estimate, and the decision criterion were then frozen before the evaluation on the three test splits; Argoverse-HD was not used for selection. Every component that distinguishes the final configuration from those in Tab. S8 was selected on the development data. The parameter grids of the crop heuristics were defined and tuned on the same development data. The analyzes in Secs. 4.2 and 4.4 were conducted on the test splits with the final configuration; no configuration was changed on their basis. The ego mask on Argoverse 2 is a fixed image band (rows $\geq$ 1830 of 2048) that covers the hood of the ego vehicle; it is applied identically to all pipelines by discarding the detections whose box lies in the band by at least one half (Sec. S2.4).

Configurations. Table S8 lists the configurations that were evaluated on the test splits. The learned looming estimate improves on the hand-crafted one on all three datasets ($[+0.004, +0.026]$, $[+0.013, +0.050]$, and $[+0.054, +0.107]$), and DGSP improves on the earlier configuration by 0.041, 0.171, and 0.327 (intervals $[-0.005, +0.087]$, $[+0.115, +0.229]$, and $[+0.225, +0.437]$). Across three training seeds of the refiner, the standard deviation of G is 0.002, 0.005, and 0.006. A larger detector on the crops raises the decision value on nuScenes to 0.298 and 0.315.

S2.3. Pixel accounting

The detector-input pixels of DGSP are measured on the detector input tensors, including the rounding of each crop to a multiple of 32 pixels and the re-detections of the truncation guard. DGSP processes no crop in 42%, 34%, and 27% of the frames on nuScenes, Argoverse 2, and Argoverse-HD and 0.7, 0.9, and 1.0 crops per frame on average; the 95th percentile of the detector-input pixels per frame is 17.6%, 12.9%, and 17.5%. In Tab. 2, the detector-input pixels of a method include the cheap pass if the method uses it, the padded mosaics of UFP, and the canvases of FOVEA. A crop heuristic adds crops in its own order until its budget is reached, and the re-detections of the truncation guard are added on top. Each heuristic is evaluated at three budgets; at the smallest one, which is used in Tab. S12, the heuristics receive 1.2 to 2.4× the detector-input pixels of DGSP, since they issue a larger number of small crops and more guard re-detections. The 64×64 inputs of the range refiner, which amount to less than 0.3% of a frame per support object, and those of the looming estimate are not counted as detector-input pixels; their runtime is included in all latency measurements.

Table S9. **Argoverse 2 val without the ego mask.** "Δ" is the 95% interval of (DGSP − method). All baselines use the refiner; "+ looming": and the learned looming estimate. Crop heuristics: best variant at the smallest budget, with both components.

Method	$G \uparrow$	px (%)	Δ
u640	0.118	10.3	**[+0.437, +0.560]**
u960	0.496	23.2	**[+0.089, +0.152]**
u1280	0.534	40.0	**[+0.055, +0.109]**
u1536	0.609	57.3	[-0.022, +0.037]
Native	0.639	101.2	[-0.065, +0.021]
u960 + looming	0.546	23.2	**[+0.036, +0.104]**
u1280 + looming	0.587	40.0	[-0.004, +0.061]
u1536 + looming	0.669	57.3	[-0.088, -0.020]*
Native + looming	0.693	101.2	[-0.120, -0.033]*
Uncertainty crops	0.478	10.8	**[+0.108, +0.173]**
Temporal-criticality crops	0.290	10.7	**[+0.294, +0.361]**
Ego-corridor crops	0.036	7.9	**[+0.528, +0.636]**
Support + ego-corridor crops	0.619	8.9	[-0.008, +0.003]
Random crops	0.117	12.6	**[+0.454, +0.546]**
DGSP w/o refiner	0.610	7.4	**[+0.002, +0.016]**
DGSP	0.617	7.4	–

S2.4. Ego mask on Argoverse 2

The front camera of Argoverse 2 images the hood of the ego vehicle at the bottom of the frame. At low resolution, the COCO-pretrained detector reports the hood as a vehicle in the ego corridor: it determines the action of the cheap detector in 51% of the frames and accounts for 81% of its decision loss, whereas it rarely affects native-resolution inference (0.3% of the frames). Since this spurious object can be removed for every pipeline by a static mask, the main paper reports the results with the mask, and Tab. S9 reports the results without it. In this setting, the hood detection belongs to the support of the draft action, and DGSP verifies and removes it without any mask; it attains $G = 0.617$ with 7.4% of the native pixels and significantly outperforms whole-frame inference with the refiner up to a long side of 1280 pixels, with both components up to 960 pixels, and the crop heuristics based on uncertainty, temporal criticality, and the ego corridor. On nuScenes and Argoverse-HD, no part of the ego vehicle is visible, and the corresponding mask changes none of the results.

S2.5. Baselines

Table S10 lists, for each baseline, whether it runs official code, a re-implementation, or an adaptation ("-style"), and the check, where one exists, that it reproduces its reported behavior before its decision value is measured; all rows of Tab. 2 additionally share the shared components, so that none of them is the original system as published.

Crop heuristics. All crop heuristics use the detector, the crop scale, the truncation guard, the merge rule, the scene history with the learned looming estimate, and the range refiner of DGSP, and select crops until the pixel budget is reached (Sec. S2.3). *Uncertainty crops* select the draft detections in the order of increasing confidence, with a context of twice the box size. *Temporal-criticality crops* select the objects of the merged scene of the previous frame in the order of decreasing braking demand, with boxes enlarged by 25%. *Ego-corridor crops* select a tube along the ego corridor from 10 to 70 m, which is widened until the budget is exhausted; the variant *support + ego corridor* additionally selects the support of the current draft. *Random crops* place boxes with the sizes of the draft detections at random positions in the lower 65% of the image. Each heuristic has a small grid of variants, *e.g.*, the context of uncertainty crops and the number of objects of temporal-criticality crops; Tab. S12 reports the best variant at the smallest budget, and Fig. S6 the best variant at each budget.

Shared components. Every row of Tab. 2 runs the cheap pass at a long side of 320 pixels and YOLOv8s at the scale that the method chooses, with the learned looming estimate on the row's own same-scale history, the range refiner on the support of the braking controller in the row's scene, and, for the methods that crop, the truncation guard and the merge rule of DGSP.

Selection on development data. Each family receives 12 candidates on the development data, including its official or previously reported form, which is run if it is the best on development data; Argoverse-HD uses the selections of

Table S10. **Baselines, implementations, and fidelity checks.** All baselines use YOLOv8s as the detector unless the official implementation prescribes otherwise. "Official": the released code; "re-implementation": our implementation of the published method; "-style": the objective or selection rule of the cited work adapted to our planner and pipeline, not a reproduction of the original system. –: no quantitative fidelity check exists for this row.

Baseline	Implementation	Fidelity check
AdaScale [2]	Re-implementation (original: Caffe, R-FCN); ResNet-18 scale classifier over $\{320, \ldots, 960\}$	–
Budget-Adaptive Routing [5]	Official estimators, offloader, and arbiter; the decision-value targets are our addition	Official VOC results reproduced within ± 0.01
FOVEA [13]	Official grid generator (KDE from the draft detections or from a dataset prior)	Fine-tuned AP_{50} on par with the uniform control (0.524 *vs.* 0.522)
LZU [14]	Official zoom and unzoom operators around YOLOv8s	Fine-tuned AP_{50} above the uniform control (0.500 *vs.* 0.497)
SAHI [1]	Official library	Recall of small objects increases, as reported
UFP [10]	Official foreground packing module; YOLOv8s on the mosaics in Tab. 2, detector fine-tuned on mosaics in the fine-tuned setting	Fine-tuned form requires a detector trained on mosaics, as in the original work
ESOD [12]	Official implementation (YOLOv5s, input size 1536)	Released weights; ranking on VisDrone reproduced
ZoomDet [15]	Official implementation (736×736 input)	–
Chanakya-style policy [6]	Objective of [6] implemented directly, since the official controller adapts detector settings that do not map to our actions	Pixel budget matched to DGSP on development data within 5%
AnA-style refinement [3]	Ours, within our pipeline	–
Counterfactual importance [7]	Ours, with our planner	–
Crop heuristics	Ours (uncertainty, temporal criticality, ego corridor, random)	–

Argoverse 2. The selected settings are:

- AdaScale: scale network shifted by one ladder step, ladder up to u1280;
- SAHI: slices of 640 pixels, overlap 0.1; FOVEA: u1280 canvas; UFP: merge scale 2.0, mosaics of 512 pixels;
- both routers: u1280 as the strong mode, half of the frames routed; Chanakya-style policy: cost-sensitive learner;
- AnA-style refinement: draft threshold 0.05, no time-to-collision threshold; counterfactual importance: perturbations of 40% in range and 2 s in time-to-collision.

The official form of each family was a candidate and was not selected on the development data; this does not imply that it is weaker on every dataset.

Budget enforcement.

Under the per-frame cap, crop methods take crops in their own order until the pixels of DGSP are reached, whole-frame and warping methods take the largest scale or canvas within the cap and otherwise only the cheap pass, SAHI reduces its slice input, and UFP its mosaic. The AnA-style rows re-detect every detection of a draft with a confidence threshold t that lies in the ego corridor within 80 m and below a time-to-collision threshold, together with the support of the current action, in one pass as in [3] or iterated; the row with $t = 0.25$, the threshold of the other pipelines, is reported regardless of the selection. The counterfactual rows re-detect the draft objects whose removal or whose perturbation in range or time-to-collision changes the action and do not search for unseen objects.

Routers. The oracle router runs the cheap detector on every frame and escalates the fraction ρ of the frames with the largest true decision value, *i.e.*, the difference between the decision loss of the cheap pipeline and that of u960, to u960 with the shared components; ρ is set such that the mean detector-input pixels equal those of DGSP. It requires the decision loss of both pipelines on the frame and is therefore not realizable; Fig. S6 includes it as a reference for frame-level escalation. In Tab. 2, both learned routers escalate the selected frames to u1280 with the shared components; a frame without a recent escalated frame carries no looming history, as in deployment. The learned routers do not realize

Table S11. **Published methods in their official form**, *i.e.*, with the monocular lift of [9] and without the shared components. Δ: 95% interval of (DGSP − method); intervals above zero in bold. *: best estimator or setting, at the mean pixels of DGSP.

	Argoverse 2 val			nuScenes held-out		
Method	px (%)	G	Δ	px (%)	G	Δ
AdaScale [2]	17.0	−0.057	**[+0.268, +0.432]**	17.3	0.083	**[+0.068, +0.198]**
FOVEA [13]*	25.8	−0.023	**[+0.233, +0.403]**	42.7	0.074	**[+0.040, +0.253]**
SAHI [1]	165.2	−0.248	**[+0.425, +0.670]**	187.7	0.130	[−0.084, +0.274]
BAR [5]*	5.3	0.021	**[+0.208, +0.334]**	7.9	−0.012	**[+0.125, +0.348]**
DGSP	5.4	0.294	–	7.8	0.211	–

the headroom of the oracle, as the decision value of a frame is difficult to predict before escalation. The routers with decision-value targets use the estimators of [5], trained to predict the decision value instead of the detection-based target. On the nuScenes training frames, the detection-based target is positive for 62% of the frames, whereas the decision value is positive for 4.9%, with a rank correlation of 0.03; the detection-based target is thus a poor proxy for the decision.

Temporal escalation. Periodic escalation runs u960 with the components on every k-th frame and the cheap pass otherwise, with $k = 10$, 7, and 6 such that the mean detector-input pixels equal those of DGSP; random escalation escalates each frame with probability $1/k$ (five seeds). An escalated frame has a looming history only if the two previous frames were escalated, as in deployment. Periodic escalation attains $G = -0.022$, 0.018, and 0.065 (DGSP minus it: $[+0.119, +0.349]$, $[+0.218, +0.337]$, and $[+0.265, +0.435]$), and every phase and seed of both schedules lies significantly below DGSP, also with a continuous looming history. The schedules miss fewer required braking episodes than DGSP (0.35, 0.14, and 0.20 *vs*. 0.41, 0.19, and 0.38) at about five times its false braking episodes (22, 35, and 28 *vs*. 4.0, 6.5, and 6.3 per 1,000 frames).

S2.6. Fairness checks

With the shared components, every published method attains a higher decision value than in its official form (Tab. S11), which is why Tab. 2 compares all methods with them.

Further settings of the official forms. SAHI with slices of 384–640 pixels, FOVEA with a dataset prior, and the frozen LZU and UFP pipelines attain G between −0.861 and 0.130 on nuScenes, and SAHI attains at most −0.248 on Argoverse 2.

S3. Decision–pixel trade-offs and component analysis

This section asks whether the decision value of DGSP is bought by pixels or by the selection of the regions. It lists all comparisons of Tab. 2 with their intervals (Sec. S3.1), separates the contribution of the components from that of the resolution for whole-frame inference (Sec. S3.2), compares the crop heuristics at matched budgets (Sec. S3.3), decomposes the improvement into the terms of the loss (Sec. S3.6), and uses other detectors (Sec. S3.7).

S3.1. Published, learned, and object-level baselines

Fig. S3 and Tab. S43 report the paired comparisons with whole-frame inference and with the published, learned, and object-level baselines of Tab. 2, without a cap and under the per-frame cap at the pixels of DGSP, including the rows and variants that Tab. 2 omits; the crop heuristics, which the main paper only summarizes, are compared in Sec. S3.3.

Multiple comparisons. From the same paired resamples, we compute two-sided bootstrap p-values, $p = 2\min(P^*(\Delta \le 0), P^*(\Delta \ge 0))$. Of the 36 capped comparisons with published, learned, and object-level methods, 29 are significant without correction, 25 after the Holm–Bonferroni correction, and 28 after the Benjamini–Hochberg correction; over all 14 capped rows, these counts are 35, 30, and 34 of 42. The comparisons that remain significant only under Benjamini–Hochberg are one-pass AnA-style refinement on nuScenes, FOVEA with instance saliency on Argoverse 2, and the one-pass counterfactual on Argoverse-HD; iterated AnA-style refinement on nuScenes ($p = 0.046$) is significant under neither correction. The 7 significant comparisons with the support-only, field-only, and pivot variants of Tab. 3 all remain significant under both corrections, and no comparison in either family favors a baseline.

Under the per-frame cap, DGSP is significantly better in 35 of the 42 comparisons with these 14 methods and variants (29 of the 36 with the 12 primary methods), and no method is significantly better. The strongest object-level

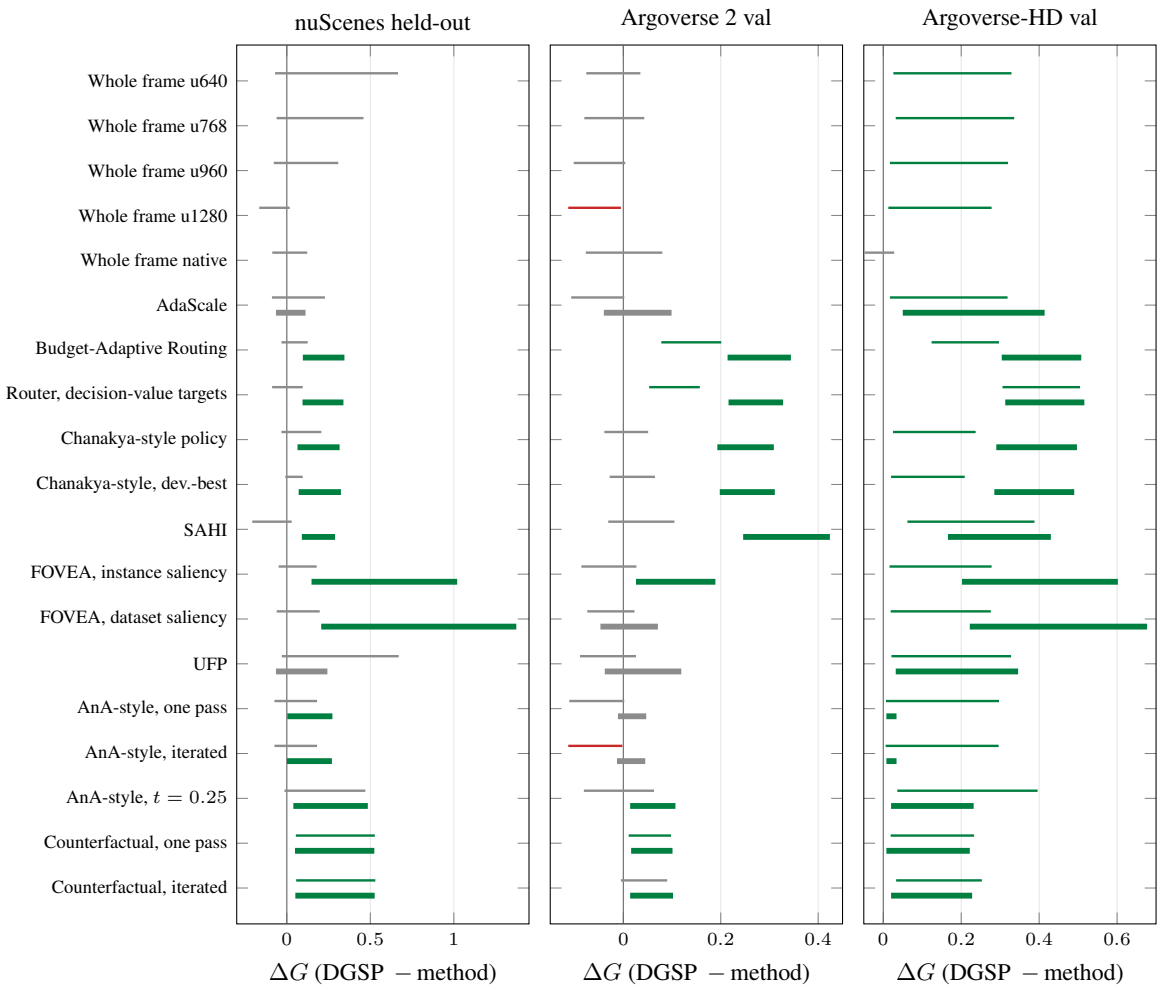


Figure S3. **Is any compared method significantly better than DGSP?** 95% paired intervals of DGSP − method without a cap (thin) and under the per-frame cap at the pixels of DGSP (thick). Green: DGSP significantly better; red: significantly worse; grey: not significant. Under the cap, no interval lies below zero. Whole-frame rows have no cap. Values in Tab. S43.

alternative on Argoverse 2, iterated AnA-style refinement, finds the decisive object as often as DGSP (96.1% *vs*. 96.5% of the frames that require braking) and, where both find it, with the same estimated range; it verifies every corridor detection on every frame with 1.5× the pixels of DGSP, which gives the learned looming estimate a continuous history, and under the cap, it falls below DGSP. DGSP produces 4.2, 5.6, and 15.0 spurious braking detections per 1,000 frames on the three datasets, the fewest of all methods with a positive decision value on nuScenes.

S3.2. Whole-frame inference with and without the components

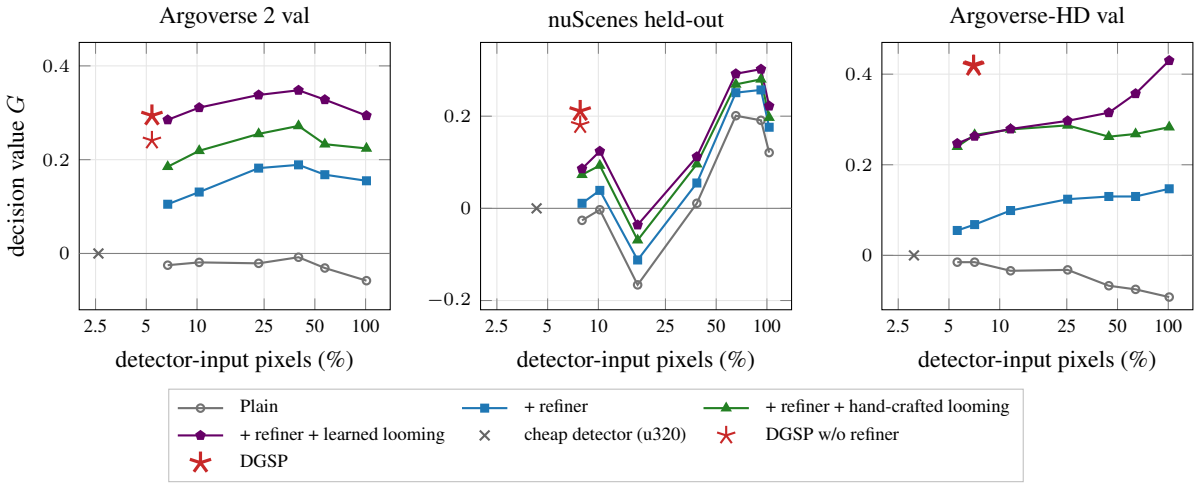


Figure S4. **How much of the gain is due to the components?** Decision value of whole-frame inference at each long side, without and with the shared components, *vs*. detector-input pixels (log scale). The components raise the decision value at every resolution. DGSP lies above the curve with the refiner on Argoverse 2 and Argoverse-HD, and whole-frame inference with both components exceeds it only with 1.9×, 8.4×, and 14.5× its pixels on Argoverse 2, nuScenes, and Argoverse-HD. Values in Tab. S44; intervals in Tab. S45.

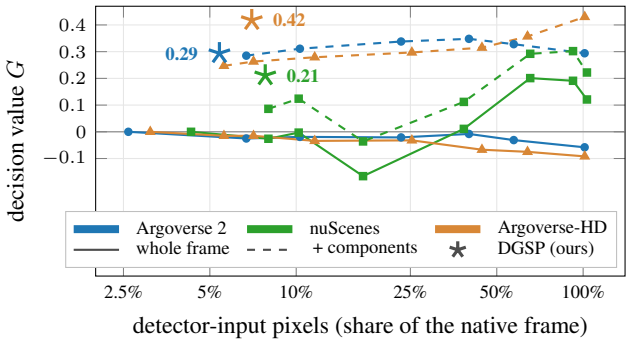


Figure S5. **Decision value *vs*. detector-input pixels.** Whole-frame YOLOv8s from a long side of 320 pixels (the cheap pipeline, $G = 0$) to the native resolution, without (solid) and with (dashed) the shared components; stars: DGSP.

Fig. S5 plots the decision value of whole-frame inference against its pixels, and Tab. S44 lists the plotted values, together with whole-frame inference that is equipped with the shared components, and Tab. S45 the intervals with the refiner in both pipelines. DGSP is significantly better than whole-frame inference with the refiner at every resolution on Argoverse 2 and Argoverse-HD and at the three lowest resolutions on nuScenes. With the learned looming estimate in addition, DGSP remains significantly better at every long side up to 1280 pixels on Argoverse-HD, with lower interval bounds between 0.013 and 0.028.

S3.3. Crop heuristics under nominal and strict pixel caps

At the smallest budget of their tuned grid, the heuristics receive 1.2 to 2.4× the detector-input pixels of DGSP; DGSP significantly outperforms four of the five on all three datasets and matches the heuristic that combines its own support

Table S12. **Crop heuristics with identical components at matched budgets.** No heuristic is significantly better than DGSP in either setting. Left: smallest budget of the tuned grid of each heuristic; "Single": with the refiner and the looming cue of a single pair of boxes; "Learned": with the refiner and the learned looming estimate; Δ against the heuristic with the learned estimate. Right: capped at the pixels of DGSP on every frame, with the learned estimate; "px": realized detector-input pixels relative to DGSP (the truncation guard lets the heuristics exceed the cap). Bold intervals: above zero.

	Smallest budget of the tuned grid				Per-frame cap at the pixels of DGSP		
Heuristic	px (%)	G, single	G, learned	Δ	G	px	Δ
nuScenes held-out							
Uncertainty	10.2	−0.067	−0.059	**[+0.032, +0.483]**	−0.058	1.32×	**[+0.030, +0.481]**
Temporal criticality	9.4	0.142	0.173	**[+0.008, +0.076]**	0.167	1.22×	**[+0.014, +0.082]**
Ego corridor	10.1	0.026	0.052	**[+0.021, +0.314]**	0.053	1.30×	**[+0.020, +0.312]**
Support + ego corridor	10.8	0.148	0.174	[-0.012, +0.096]	0.174	1.40×	[-0.012, +0.096]
Random	12.2	−0.034	−0.037	**[+0.100, +0.420]**	–	–	–
DGSP	7.8	**0.211**		–	**0.211**	1.00×	–
Argoverse 2 val							
Uncertainty	8.0	0.163	0.219	**[+0.020, +0.126]**	0.221	1.51×	**[+0.018, +0.123]**
Temporal criticality	8.3	0.157	0.242	**[+0.006, +0.093]**	0.246	1.55×	**[+0.002, +0.090]**
Ego corridor	6.6	0.174	0.260	**[+0.011, +0.062]**	0.262	1.24×	**[+0.010, +0.060]**
Support + ego corridor	7.1	0.194	0.296	[-0.013, +0.007]	0.296	1.33×	[-0.014, +0.008]
Random	8.9	0.061	0.018	**[+0.212, +0.347]**	–	–	–
DGSP	5.4	**0.294**		–	0.294	1.00×	–
Argoverse-HD val							
Uncertainty	12.3	0.141	0.282	**[+0.049, +0.252]**	0.285	1.76×	**[+0.046, +0.249]**
Temporal criticality	11.6	0.185	0.378	**[+0.007, +0.079]**	0.377	1.66×	**[+0.009, +0.080]**
Ego corridor	10.9	0.183	0.389	**[+0.001, +0.063]**	0.393	1.55×	[-0.004, +0.062]
Support + ego corridor	10.8	0.180	0.408	[-0.001, +0.023]	0.409	1.54×	[-0.002, +0.022]
Random	16.6	0.039	0.139	**[+0.192, +0.362]**	–	–	–
DGSP	7.0	**0.418**		–	**0.418**	1.00×	–

with an ego-corridor tube. The learned looming estimate raises the decision value of every informed heuristic on all three datasets, and DGSP retains its advantage when the heuristics are given it; the advantage therefore stems from the selection of the regions.

In Tab. S12, every heuristic is capped at the pixels of DGSP on the same frame. DGSP significantly outperforms the uncertainty and temporal-criticality heuristics on all three datasets and the ego-corridor heuristic on nuScenes and Argoverse 2, and it matches the remaining ones, although the guard lets the heuristics exceed the cap by 22–76%.

Table S13. **Strict per-frame cap including the truncation guard.** Each crop or guard re-detection that would raise the detector-input pixels of a frame above those of DGSP on that frame is skipped, and a pending truncated detection is then kept as detected; the realized pixels never exceed those of DGSP. All rows use the refiner and the learned looming estimate. The primary variant of each heuristic was fixed by a rule on the development data before the run; Δ: DGSP minus the row; px: realized pixels in % of the native frame.

	nuScenes held-out		Argoverse 2 val		Argoverse-HD val	
Primary row	G (px)	Δ	G (px)	Δ	G (px)	Δ
Support + ego corridor (hybrid)	0.179 (7.5)	**[+0.012, +0.051]**	0.274 (5.1)	**[+0.010, +0.031]**	0.392 (6.8)	**[+0.011, +0.046]**
Uncertainty	−0.060 (6.5)	**[+0.058, +0.435]**	0.016 (4.8)	**[+0.186, +0.370]**	0.280 (6.3)	**[+0.017, +0.263]**
Temporal criticality	0.153 (6.6)	**[+0.022, +0.105]**	0.207 (5.0)	**[+0.052, +0.127]**	0.383 (6.7)	**[+0.004, +0.073]**
Ego corridor	0.077 (7.5)	**[+0.033, +0.233]**	0.114 (4.8)	**[+0.103, +0.265]**	0.377 (6.5)	[-0.006, +0.099]
DGSP	**0.211** (7.8)	–	**0.294** (5.4)	–	**0.418** (7.0)	–

Strict cap. Under the nominal cap, the guard lets the hybrid and the heuristics use 1.10–1.47× the pixels of DGSP on average and up to 5–13× on single frames. Under the strict cap of Tab. S13, which includes the guard, DGSP is

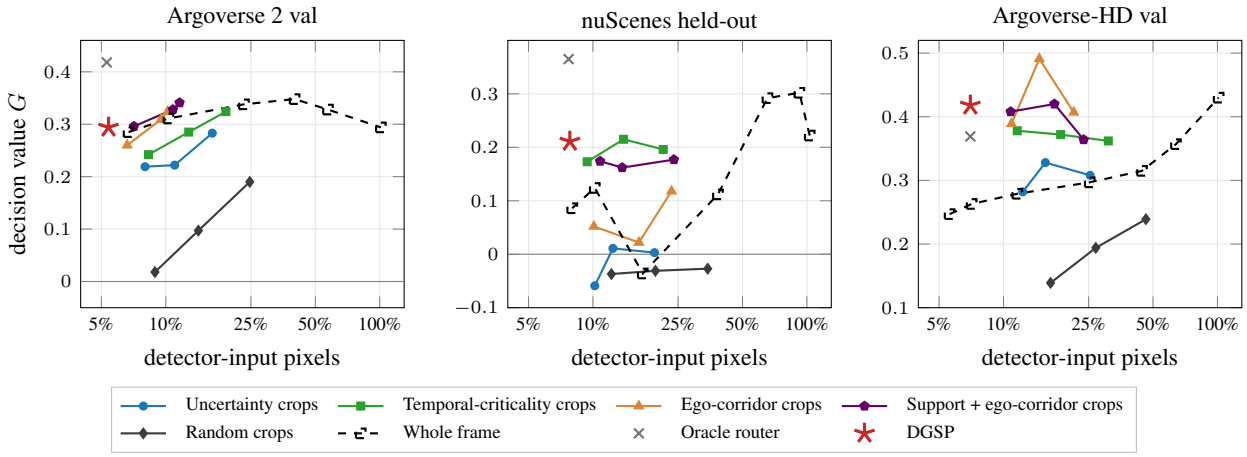


Figure S6. **Decision value *vs*. detector-input pixels with identical components.** All baselines use the range refiner and the learned looming estimate of DGSP, and the crop heuristics also its truncation guard and scene history; each heuristic is shown with its best tuned variant at three budgets. The oracle router escalates the frames with the highest true decision value and is not realizable.

significantly better than the hybrid on all three datasets, which it only matches under the nominal cap, and than the primary uncertainty, temporal-criticality, and ego-corridor heuristics in 8 of 9 comparisons; the exception is the ego corridor on Argoverse-HD ($+0.041$, $[-0.006, +0.099]$). Against the secondary variant of each heuristic, DGSP is significantly better in 7 of 9 comparisons; on Argoverse-HD, it does not differ significantly from the other temporal-criticality variant ($+0.034$) and from the other ego-corridor variant, whose point estimate is higher (0.458, $\Delta = -0.040$, $[-0.144, +0.061]$). Against the other hybrid variant, it is significantly better on nuScenes and Argoverse-HD and not on Argoverse 2 ($+0.013$, $[-0.003, +0.029]$). With an equal pixel budget that includes the guard, the selection of the regions therefore matters, and the ties of the hybrid under the nominal cap stem from its additional guard pixels.

On the Jetson (Tab. S34, same frames), the capped hybrid attains the decision value of DGSP (0.373 *vs*. 0.380 on nuScenes and 0.417 *vs*. 0.419 on Argoverse 2) at a higher mean and 95th-percentile latency (24.0 and 75.0 ms *vs*. 21.5 and 57.7 ms; 44.5 and 109.0 ms *vs*. 39.9 and 89.5 ms). Along the pixel axis of DGSP, the decision value is insensitive to the crop parameters: crops with a long side of up to 1280 pixels and a support context of up to three box sizes change G by -0.037 to $+0.020$ on nuScenes and add at most 0.017 on Argoverse 2 and Argoverse-HD at 1.4–$1.8\times$ the pixels.

S3.4. Mean pixel budget for the frame-level methods

Table S14. **Frame-level methods under a mean pixel budget** instead of the per-frame cap of Tab. 2, each run with its own routed history and the refiner. Thresholds are set on the development data of nuScenes and Argoverse 2 to meet the mean pixels of DGSP; Argoverse-HD uses those of Argoverse 2. px: realized mean pixels relative to DGSP on the test split; AdaScale cannot meet the budget with its scale ladder and uses its closest policy. Δ: DGSP minus the row.

	nuScenes held-out		Argoverse 2 val		Argoverse-HD val	
Method	G (px)	Δ	G (px)	Δ	G (px)	Δ
AdaScale	0.197 (1.77×)	[-0.050, +0.076]	0.120 (2.59×)	**[+0.075, +0.292]**	0.319 (2.23×)	**[+0.027, +0.220]**
Budget-Adaptive Routing	0.009 (0.99×)	**[+0.095, +0.339]**	0.026 (1.22×)	**[+0.206, +0.329]**	0.026 (0.78×)	**[+0.291, +0.489]**
Router, decision-value targets	0.126 (0.95×)	[-0.010, +0.188]	0.014 (1.23×)	**[+0.220, +0.341]**	0.001 (0.46×)	**[+0.313, +0.517]**
Chanakya-style policy	0.115 (1.03×)	[-0.018, +0.216]	0.195 (0.95×)	**[+0.039, +0.175]**	0.321 (1.04×)	**[+0.016, +0.193]**
DGSP	**0.211** (1×)	–	**0.294** (1×)	–	**0.418** (1×)	–

A per-frame cap forces a frame-level method to the pixels of DGSP on every frame, although these methods are designed to spend a budget unevenly over a sequence. Table S14 therefore gives the four frame-level methods a mean budget instead; each routes its own history, and its thresholds are fixed on the development data, so the realized test budgets drift from the target by up to a factor of two. With these four rows replacing their capped counterparts, DGSP is significantly better in 28 of the 36 primary comparisons (23 after Holm correction, compared with 29 and 25 under the per-frame cap), and no method is significantly better. Compared with the per-frame cap, the comparisons with

the router and the Chanakya-style policy on nuScenes are no longer significant, that with AdaScale on Argoverse 2 becomes significant, and AdaScale on nuScenes, with $1.77\times$ the pixels of DGSP, remains not significantly different. DGSP also repairs more of the disagreements of the draft than every mean-budget row except AdaScale on Argoverse-HD (-0.028, $[-0.128, +0.046]$).

S3.5. Region selection along anchored trajectories

Table S15. **Crops along anchored ego trajectories in the style of VeteranAD [16]**, with the detector, crop scale, guard, merge rule, and shared components. (a) All 20 anchors; (b) anchors selected by the planner on the current merged scene, re-selected after each verification; (c) (b) with the support of DGSP. Nominal and strict caps as in Sec. S3.3. px: pixels relative to DGSP. † / *: DGSP significantly better / worse (95% paired interval); ‡: also after Holm correction over the nine rows of a dataset. **Bold**: highest in each panel.

Row (cap)	nuScenes	Argoverse 2	Argoverse-HD	px
Natural draft (the evaluation of Tab. 2)				
(a) all anchors, nominal	$0.024^{\ddagger}$	$0.028^{\ddagger}$	$0.031^{\ddagger}$	0.58–1.12×
(b) selected, nominal	0.137	$0.265^{\dagger}$	$0.390^{\ddagger}$	1.36–1.50×
(b) selected, strict	0.182	$0.217^{\dagger}$	$0.361^{\ddagger}$	0.92–0.94×
(c) selected + support, nominal	0.190	0.289	$0.401^{\ddagger}$	1.37–1.45×
(c) selected + support, strict	**0.216**	$0.267^{\ddagger}$	0.408	0.95–0.96×
DGSP	0.211	**0.294**	**0.418**	1×
Draft detections of all relevant objects deleted (Tab. 4)				
(a) all anchors, nominal	$0.122^{\ddagger}$	$0.159^{\ddagger}$	$0.091^{\ddagger}$	0.59–1.12×
(b) selected, nominal	0.650	0.593	0.498	1.32–1.49×
(b) selected, strict	**0.664**	$0.568^{\dagger}$	0.476	0.92–0.94×
(c) selected + support, nominal	0.648	0.597	0.500	1.34–1.45×
(c) selected + support, strict	0.659	$0.567^{\ddagger}$	0.486	0.95–0.96×
DGSP	0.630	**0.599**	**0.508**	1×

VeteranAD [16] gathers perception along anchored candidate trajectories of the ego vehicle and plans autoregressively; since it is an end-to-end model without a module that selects pixels, we implement its selection principle within our pipeline, as for AnA-style refinement. The 20 anchors are k-means centers of the future ego trajectories of the training splits (eight waypoints up to 4 s; Argoverse-HD uses those of Argoverse 2), and each anchor is projected as a corridor, as for the ego-corridor tube. Table S15 shows that verifying all anchors fails, since the boxes of the segments near the ego vehicle exhaust the budget. Anchors selected by the planner score lower than DGSP under both caps on all three datasets of the natural evaluation, significantly on Argoverse 2 and Argoverse-HD, even with 1.4–$1.5\times$ its pixels under the nominal cap, and they repair fewer of the disagreements of the draft with $\pi(F)$ (0.65–0.70, 0.77–0.80, and 0.72–0.75 *vs*. 0.73, 0.87, and 0.80). Adding the support of DGSP closes most of this gap: under the strict cap, the difference is then -0.005 $[-0.053, +0.051]$ on nuScenes and $+0.010$ $[-0.003, +0.027]$ on Argoverse-HD, and only on Argoverse 2 is DGSP still significantly better ($+0.026$, $[+0.009, +0.045]$). After the deletion of the draft detections, the selected anchors attain a slightly higher point estimate than DGSP on nuScenes (0.664 *vs*. 0.630, $[-0.073, +0.012]$), as they cover the corridor ahead regardless of the draft. The search along anchored trajectories therefore needs the support of DGSP to approach it and does not exceed it at equal pixels; on NAVSIM, the same comparison is in Sec. S6.5.

S3.6. Composition of the improvement and detection metrics

Replacing the learned looming estimate by the hand-crafted one or by a single pair of observations significantly lowers G on all three datasets (Tab. S16); removing the truncation guard changes G by at most 0.011, not significantly.

Table S46 decomposes the decision loss. Relative to the cheap detector, DGSP removes most of the loss of excessive braking, and its loss of insufficient braking is close to that of native-resolution inference with the same components on all three datasets; the paired intervals of the difference per frame, $[-0.055, +0.049]$, $[-0.074, +0.052]$, and $[-0.013, +0.015]$, include zero. DGSP also takes fewer unnecessary braking actions than whole-frame inference at any resolution on the three datasets, also when the latter is given its components (15, 53, and 131 per 1,000 frames, compared

Table S16. **Ablation of the implementation components** (decision value G with the refiner; †: significantly worse than the full procedure). The mechanisms, the round count, and the refiner are ablated in Tab. 3.

Variant	nuScenes	Argoverse 2	Argoverse-HD
DGSP (full)	0.211	0.294	0.418
Hand-crafted looming estimate	0.199†	0.262†	0.338†
Single-pair looming	0.176†	0.208†	0.185†
W/o truncation guard	0.222	0.288	0.416

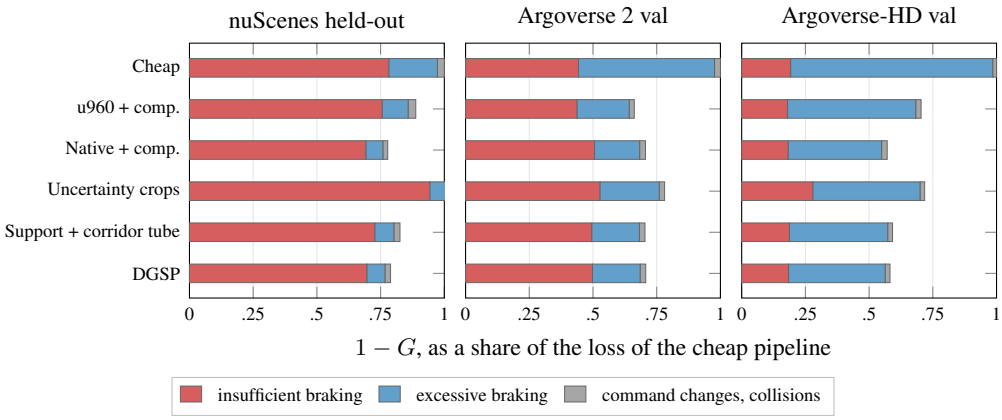


Figure S7. **Which part of the decision loss does DGSP remove?** Decision loss split into insufficient braking, excessive braking, and command changes and collisions, as a share of the total loss of the cheap pipeline; the length of a bar is $1 - G$. All pipelines except the cheap one use the shared components. Values in Tab. S46.

with 42, 126, and 265 for u960 and at least 17, 56, and 142 with the components), and on nuScenes, it delays the onset of hard braking on 1.7 frames per 1,000, compared with 7.8 for the cheap detector ($[-12.65, -1.74]$).

Detection metrics. On the decision-relevant objects, DGSP attains the highest precision on all three datasets, and it produces 50%, 64%, and 34% of the spurious braking detections of whole-frame inference at the verification resolution. The whole-image AP is not the objective of the method, since objects that cannot affect the action are not re-detected.

S3.7. Other detectors

Table S18 repeats the comparison with YOLO11s [11] as the cheap and the reference detector; the configurations are unchanged, and the refiner and the looming estimate, which were trained on YOLOv8s boxes, are applied without retraining. Against whole-frame inference with the refiner only, DGSP is significantly better at every resolution on Argoverse 2 and Argoverse-HD. With the full lift on both sides, this holds only on Argoverse-HD (up to 1280 pixels; native ties): whole-frame inference is significantly better than DGSP at 1280 pixels on nuScenes and at 640 and 960 pixels on Argoverse 2, with 1.9–8.5× its pixels, and no other difference on these two datasets is significant. The pattern is that of YOLOv8s (Tab. S44).

Fine-tuned detectors. In Tab. S19, DGSP uses the fine-tuned detector for verification, and the refiner and the looming estimate are applied without retraining. It attains the highest decision value for both seeds, also compared with fine-tuned whole-frame inference that is given its components at any resolution (at most 0.328 and 0.356); the differences to the latter are significant at long sides of 448, 512, 1280, and 1536 pixels for seed 0 and of 448, 1536, and 1600 pixels for seed 1.

On Argoverse 2 (Tab. S20), DGSP with the fine-tuned verifier attains 0.224 and significantly outperforms every fine-tuned detector in every form, with lower interval bounds of at least 0.14, whereas all of them attain lower decision values than the cheap COCO-pretrained detector, also with the shared components. The fine-tuned detectors learn the projected cuboid boxes of the annotations, which are larger than the tight boxes of COCO; the estimated range of the decisive object is shorter (median ratio 0.96–0.97 relative to u960 with the COCO-pretrained detector), and with the components, u960, FOVEA, LZU, UFP, and ZoomDet brake hard on 8.1–13.8% of the frames that require no braking, compared with 5.0% for DGSP; ESOD brakes less often but misses the decisive object more often. DGSP is affected

Table S17. **Detection metrics.** DGSP attains the highest precision on the decision-relevant objects on all three datasets. "Relevant": the decision-relevant objects of Sec. S2.1; "Spurious": detections per 1,000 frames that trigger braking without a corresponding object. Whole frame with the shared components.

		Relevant		
Pipeline	AP_{50} ↑	Recall ↑	Prec. ↑	Spurious ↓
nuScenes held-out				
Cheap (u320)	0.148	0.759	0.505	16.6
u960	0.246	0.859	0.665	7.6
Native	0.245	0.848	0.731	4.2
Support + ego corridor	0.170	0.821	0.780	**3.8**
DGSP	0.162	0.824	**0.799**	**3.8**
Argoverse 2 val				
Cheap (u320)	0.107	0.804	0.466	16.3
u960	0.178	0.909	0.677	8.7
Native	0.184	0.919	0.665	14.8
Support + ego corridor	0.119	0.900	0.714	5.9
DGSP	0.115	0.901	**0.719**	**5.6**
Argoverse-HD val				
Cheap (u320)	0.117	0.664	0.312	27.7
u960	0.199	0.909	0.410	44.3
Native	0.202	0.923	0.456	16.2
Support + ego corridor	0.118	0.883	0.491	**14.7**
DGSP	0.130	0.890	**0.511**	15.0

Table S18. **YOLO11s as the cheap and the reference detector** (unchanged configurations). Upper rows: whole frame with the refiner only; "+ looming": whole frame with the full lift, as DGSP. † / *: DGSP is significantly better / worse.

	nuScenes		Argoverse 2		Argoverse-HD	
Pipeline	G ↑	px (%)	G ↑	px (%)	G ↑	px (%)
u640	$0.319^{\dagger}$	17.1	$0.145^{\dagger}$	10.3	$0.079^{\dagger}$	11.6
u960	0.408	38.4	$0.158^{\dagger}$	23.2	$0.111^{\dagger}$	25.3
u1280	0.434	65.4	$0.145^{\dagger}$	40.0	$0.101^{\dagger}$	44.4
Native	0.410	103.1	$0.085^{\dagger}$	101.2	$0.119^{\dagger}$	101.3
u512 + looming	0.309	10.2	0.310	6.7	$0.277^{\dagger}$	7.1
u640 + looming	0.379	17.1	0.316^{*}	10.3	$0.300^{\dagger}$	11.6
u960 + looming	0.446	38.4	0.323^{*}	23.2	$0.300^{\dagger}$	25.3
u1280 + looming	0.483^{*}	65.4	0.311	40.0	$0.316^{\dagger}$	44.4
Native + looming	0.443	103.1	0.260	101.2	0.412	101.3
Hand-crafted looming	0.410	7.7	0.246	5.4	0.362	6.9
DGSP	0.418	7.7	**0.261**	5.4	**0.422**	6.9

Table S19. **Decision value on nuScenes held-out with detectors fine-tuned on the training split** (two training seeds). "+ comp.": with the refiner and the learned looming estimate of DGSP.

Method (fine-tuned)	Seed 0	Seed 1
u960	0.241	0.272
u960 + comp.	0.328	0.356
FOVEA [13]	0.235	0.244
LZU [14]	0.219	0.184
UFP [10]	0.202	0.180
ESOD [12]	0.221	0.229
ZoomDet [15]	0.259	0.235
DGSP w/o refiner	0.336	0.336
DGSP	**0.359**	**0.364**

Table S20. **Decision value on Argoverse 2 val with detectors fine-tuned on Argoverse 2 train** (one seed). "Official": monocular lift of [9]; "+ looming": with the learned looming estimate; "+ looming + ref.": also with the range refiner. †: DGSP significantly better.

Method (fine-tuned)	Official	+ looming	+ looming + ref.
u960	$-0.473^{\dagger}$	$-0.126^{\dagger}$	$-0.063^{\dagger}$
FOVEA [13]	$-0.503^{\dagger}$	$-0.126^{\dagger}$	$-0.080^{\dagger}$
LZU [14]	$-0.526^{\dagger}$	$-0.125^{\dagger}$	$-0.080^{\dagger}$
UFP [10]	–	$-0.087^{\dagger}$	$-0.009^{\dagger}$
ESOD [12]	$-0.736^{\dagger}$	$-0.175^{\dagger}$	$-0.120^{\dagger}$
ZoomDet [15]	$-0.384^{\dagger}$	$-0.084^{\dagger}$	$-0.035^{\dagger}$
DGSP	–	$0.144^{\dagger}$	**0.224**

Table S21. **Decision-relevant objects that are missing from the draft.** "Nat.": recovery of the objects that the cheap detector misses and u960 detects. "Del.": recovery after the draft detections of all relevant objects are deleted, with the detector-input pixels in this setting. "Reg.": decision value on the frames in which braking is required but the draft contains no supporting object. All policies use the refiner and the learned looming estimate.

	nuScenes				Argoverse 2				Argoverse-HD			
Policy	Nat.	Del.	px (%)	Reg.	Nat.	Del.	px (%)	Reg.	Nat.	Del.	px (%)	Reg.
Support only	0.197	0.149	5.6	0.000	0.379	0.192	4.1	0.000	0.201	0.237	6.8	0.000
Uncertainty crops	0.256	0.399	16.5	0.061	0.510	0.566	12.8	0.151	0.473	0.637	16.6	0.167
Field only	0.453	0.792	11.7	0.329	0.540	0.754	6.0	0.224	0.738	0.654	5.8	0.678
AnA-style, one pass	**0.718**	0.447	–	–	0.801	0.534	–	–	0.866	0.558	–	–
AnA-style, iterated	**0.718**	0.451	–	–	0.808	0.559	–	–	0.865	0.574	–	–
Counterfactual, one pass	0.205	0.191	–	–	0.419	0.237	–	–	0.202	0.251	–	–
Counterfactual, iterated	0.214	0.197	–	–	0.427	0.251	–	–	0.206	0.268	–	–
DGSP	0.607	**0.847**	14.1	**0.331**	**0.848**	**0.876**	8.6	**0.289**	**0.873**	**0.846**	10.3	**0.684**

least, since it verifies only the regions the decision requires and retains the geometry of the COCO-pretrained draft elsewhere.

S4. Unseen-object recovery, field behavior, and coverage

This section examines the first mechanism of DGSP, the search for objects that the draft misses: how many such objects it recovers (Sec. S4.1), how often the verified region contains the object that decides the action (Sec. S4.2), and how the field differs from a fixed region that has the same pixels (Sec. S4.3).

S4.1. Recovery of objects missing from the draft

On the frames in which braking is required but the draft contains no object that supports braking (3.0–5.7% of the frames), DGSP attains $G = 0.33$, 0.29, and 0.68 on nuScenes, Argoverse 2, and Argoverse-HD, compared with 0 for the support alone and at most 0.17 for uncertainty crops.

The field accounts for most of the recovered objects, and the support adds the objects that lie within the crops of draft detections; their combination recovers more objects than uncertainty crops with fewer pixels on all three datasets. After the deletion, AnA-style refinement recovers at most 57% and counterfactual importance at most 27% of these objects, since neither selects regions by probing hypothetical unseen objects; their recovery depends on whether a missed object falls within crops selected from the remaining draft detections.

Natural misses on nuScenes. Of the natural misses of Tab. S21 on nuScenes, 78 of 117 have a draft box with a confidence between 0.05 and 0.25, which AnA-style refinement verifies and DGSP, with the draft threshold of 0.25 of all other pipelines, does not; on these objects, AnA-style refinement recovers 0.85 and DGSP 0.60, whereas on the others, DGSP recovers more (0.62 *vs*. 0.46). On Argoverse 2 and Argoverse-HD, the field footprint of DGSP recovers more of the natural misses inside it (0.93 *vs*. 0.77 and 0.86) and recovers more overall; the two signals are

complementary.

Low-confidence draft boxes. A post-hoc variant, evaluated only on the nuScenes held-out split and not used elsewhere, tests whether this gap is due to the draft threshold. It runs one additional cheap pass at a confidence of 0.05 and adds the boxes with a confidence between 0.05 and 0.25 that lie in the ego corridor and do not duplicate a draft box to the requests of the first round, leaving the loop unchanged; this seeds 7,377 crops on 42% of the frames. The variant raises the decision value from 0.211 to 0.231 ($\Delta = -0.020$, $[-0.038, -0.002]$), the repair rate from 0.734 to 0.781, and the share of the 270 annotated decision-relevant objects that the cheap detector misses and the pipeline finds from 0.31 to 0.43, at 1.9× the pixels (1.4× without the additional pass). Part of the gap to AnA-style refinement on natural misses is therefore a property of the draft threshold rather than of the field; since the variant was formed after the test results were known, we report it as a direction, not as a result of DGSP.

S4.2. Coverage of the decisive object

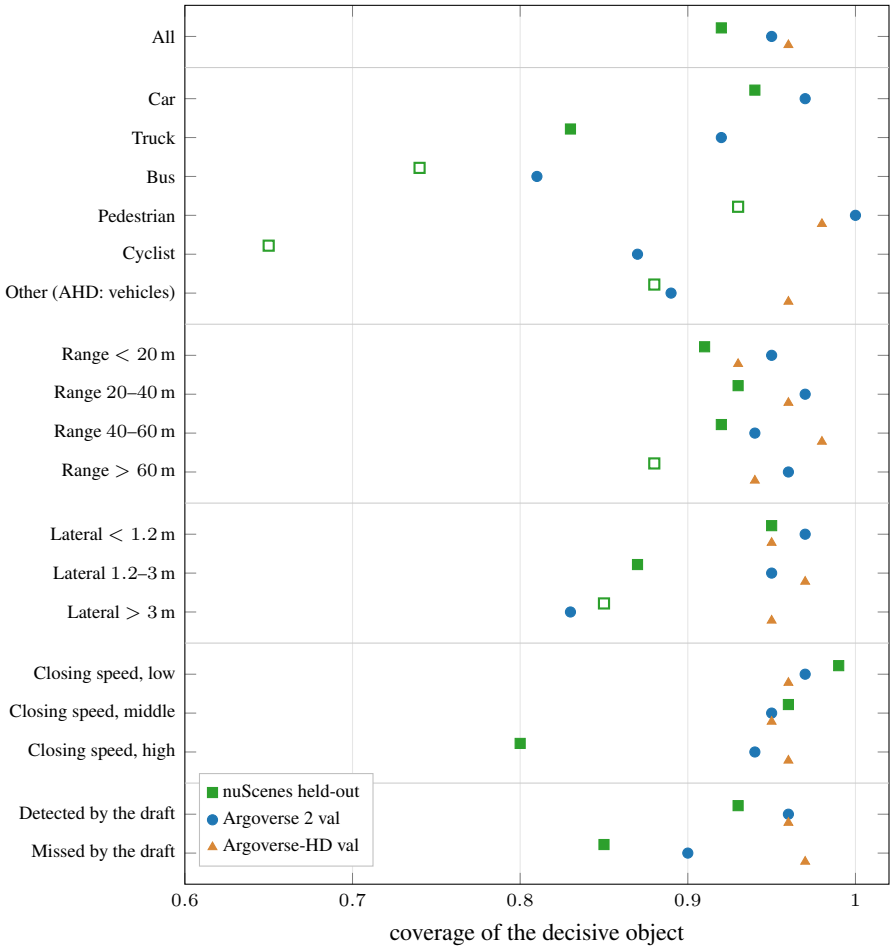


Figure S8. **Where does the verified region miss the decisive object?** Share of the frames that require braking on which the decisive object lies in the verified region of DGSP (field footprint, support and guard crops accumulated up to the final round), by stratum. Coverage is box-center geometry, not detection recall. Hollow markers: strata with fewer than 30 frames. On Argoverse-HD, the annotations do not separate vehicle classes. Values and frame counts in Tab. S48.

Table S48 breaks down whether the decisive object of a frame that requires braking lies in the verified region. It is 0.92 [0.88, 0.95], 0.95 [0.94, 0.97], and 0.96 [0.92, 0.98] overall (0.90 [0.85, 0.93], 0.92 [0.90, 0.95], and 0.91 [0.81, 0.97] with the field and support crops alone, *i.e.*, on the verified region R without the guard crops) and lowest for objects more than 3 m from the ego lane (0.85 and 0.83 on nuScenes and Argoverse 2), for objects that the draft misses (0.85 and 0.90), and, on nuScenes, for the third of the objects with the highest closing speed (0.80 [0.73, 0.87]), which the static probe ($v_{\text{close}} = 0$) of the deployed configuration does not anticipate. The deployable probe

changes of Tab. S47 (closing probe, extended grid, three sizes) with real crop detections raise the coverage of this tercile to 0.997 and the overall coverage to 0.999, 0.976, and 0.962, at 2.0–2.4× the pixels; the decision value changes by $+0.031$ $[-0.007, +0.074]$, $+0.031$ $[-0.007, +0.081]$, and -0.092 $[-0.278, -0.003]$, so that the deployed probe is a deliberate trade-off between coverage and compute. Classes with fewer than 30 frames (buses and cyclists on nuScenes) have wide intervals.

S4.3. Field and fixed tube

Table S22. **The field adapts to the ego speed** (nuScenes held-out; the braking demand is recomputed with the ego speed scaled). Range: farthest sensitive range of the field (mean over frames with a non-empty field); Field: mean image area of the field footprint (% of the frame); px and G: DGSP/ hybrid (support + ego-corridor tube). Δ: 95% interval of (DGSP − hybrid).

Ego speed	Range	Field (%)	px (%)	G	Δ
×0.5	14 m	1.7	5.5 / 6.4	0.241 / 0.237	[-0.001, +0.011]
×1.0	28 m	5.6	7.8 / 10.8	0.211 / 0.174	[-0.012, +0.096]
×1.5	43 m	6.5	8.6 / 12.2	0.196 / 0.153	**[+0.003, +0.087]**

Table S23. **Braking controller with a curved corridor** that follows the path of constant curvature given by the ego yaw rate (decision value under this planner). † / *: DGSP significantly better / worse. The hybrid with a curved support and tube knows the path and serves as a path-aware hand-crafted comparator.

Pipeline	nuScenes	Argoverse 2	Argov.-HD
u960 + comp.	0.101	0.276*	0.282†
Native + comp.	0.235	0.225	0.422
Uncertainty crops	−0.083†	0.168	0.255†
Support + straight tube	0.154	0.209	0.352†
Curved support + curved tube	0.186	0.200	0.379
DGSP, straight controller	0.227	0.254	0.334†
DGSP, curved controller	0.207	0.224	0.394

The field as an action-adaptive corridor. For the braking controller and a static probe, a cell is sensitive exactly when the demand of an object there reaches the threshold τ_{a+1} of the next stronger action and its lateral extent overlaps the corridor, which reduces to a range below $2 + 0.4v + v^2/(2\tau_{a+1})$, the stopping distance at that deceleration with the reaction time and margin of the planner, and to the empty set at HARD-BRAKE. With the same cells, projection, and scheduling, the field and this action-adaptive tube request identical regions on all 36,005 test frames of the three datasets; only the floating-point evaluation of the inequality must follow that of the demand, since 4 of 48 constructed boundary cases differ otherwise. For this planner, the shape of the field is therefore a stopping corridor that changes with the action; what the construction contributes is that it follows from the planner, also for other planners and paths (Tabs. S23 and S24), and that it is combined with the certifying support and the iteration. Table S25 isolates the dependence on the action: tubes that depend only on the ego speed score significantly lower than DGSP on nuScenes under every cap and on Argoverse 2 in three of the four primary settings (five of six with the diagnostic cap), whereas on Argoverse-HD the differences are at most 0.011 in the primary settings (0.016 with the diagnostic cap) and not

Table S24. **Time-to-collision warning as a second max-composable planner** (decision value; detector-input pixels in % in parentheses). All pipelines use the refiner and the learned looming estimate.

Pipeline	nuScenes	Argoverse 2	Argov.-HD
u960	0.157 (38.4)	0.448 (23.2)	0.326 (25.3)
Native	0.222 (103.1)	0.499 (101.2)	0.629 (101.3)
Uncertainty crops	0.187 (11.6)	0.418 (9.2)	0.672 (14.3)
Support + ego-corridor tube	0.146 (10.9)	0.427 (6.9)	0.616 (11.0)
Support + tube, capped	0.157 (12.4)	0.473 (7.7)	0.724 (12.8)
DGSP	0.191 (8.6)	0.496 (6.0)	**0.750** (7.9)

Table S25. **Field *vs*. action-independent stopping-distance tubes.** The tube covers the ego corridor up to the range $2 + 0.4v + v^2/(2\tau)$ at the ego speed v for a fixed τ: the threshold of DECELERATE (τ_1), *i.e*., the field at KEEP applied at every action, or that of HARD-BRAKE (τ_2), *i.e*., the field at DECELERATE. Support, components, crop scale, guard, and merge rule are those of DGSP. Caps as in Sec. S3.3; "strict, diag.": a diagnostic variant of the strict cap that also removes regions whose verification was denied from the observed region. px: pixels relative to DGSP; Δ: DGSP minus the row.

	nuScenes held-out		Argoverse 2 val		Argoverse-HD val	
Tube (cap)	G (px)	Δ	G (px)	Δ	G (px)	Δ
τ_1, nominal	0.154 (1.03)	**[+0.004, +0.117]**	0.285 (1.10)	[-0.008, +0.026]	0.411 (1.06)	**[+0.002, +0.013]**
τ_1, strict	0.148 (0.98)	**[+0.006, +0.123]**	0.248 (0.96)	**[+0.012, +0.080]**	0.429 (0.92)	[-0.039, +0.016]
τ_1, strict, diag.	0.148 (0.98)	**[+0.006, +0.124]**	0.256 (0.96)	**[+0.011, +0.068]**	0.402 (0.92)	**[+0.009, +0.022]**
τ_2, nominal	0.140 (0.80)	**[+0.016, +0.131]**	0.271 (0.93)	**[+0.003, +0.042]**	0.408 (0.90)	**[+0.003, +0.015]**
τ_2, strict	0.141 (0.79)	**[+0.016, +0.131]**	0.272 (0.90)	**[+0.001, +0.042]**	0.417 (0.88)	[-0.011, +0.011]
τ_2, strict, diag.	0.141 (0.79)	**[+0.016, +0.131]**	0.267 (0.90)	**[+0.007, +0.048]**	0.407 (0.88)	**[+0.004, +0.018]**
DGSP	**0.211** (1)	–	**0.294** (1)	–	0.418 (1)	–

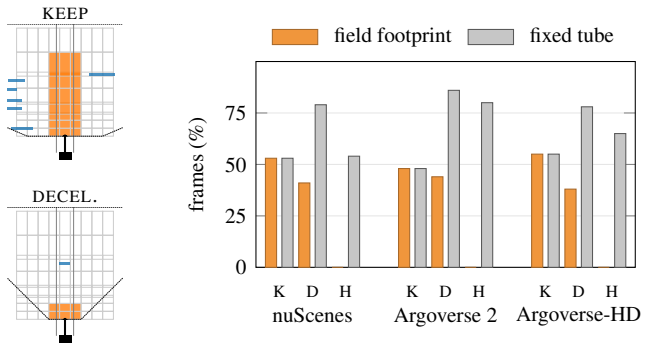


Figure S9. **The field depends on the action, the tube does not.** Left: field (orange) and tube of the hybrid (dashed) at the draft actions KEEP and DECELERATE (nuScenes). Right: share of the evaluated frames with draft action KEEP (K), DECELERATE (D), or HARD-BRAKE (H) on which each requests pixels.

significant under the strict cap. The τ_1 tube recovers more natural misses on Argoverse 2 and Argoverse-HD (0.53 and 0.74 *vs*. 0.51 and 0.69 of the annotated objects that the cheap detector misses), as it requests the long corridor also at DECELERATE, but this does not raise its decision value.

The hybrid matches DGSP with the braking controller because 95%, 90%, and 79% of the footprint of the field lie within its tube on nuScenes, Argoverse 2, and Argoverse-HD. The field adapts where the tube is fixed. Its farthest sensitive range grows with the ego speed (Tab. S22), and at 1.5× the speed, DGSP significantly outperforms the hybrid with fewer pixels. On the 3,470 Argoverse-HD frames whose decisive object lies outside the tube, DGSP attains 0.455, compared with 0.428 for the hybrid ([+0.008, +0.052]). For the time-to-collision warning, the hybrid attains 0.146, 0.427, and 0.616 when it is given the support of this planner and 0.157, 0.473, and 0.724 when it is in addition capped at the pixels of the procedure on every frame, compared with 0.191, 0.496, and 0.750 for the procedure with the field of this planner (Tab. S24); the differences to the capped hybrid are significant on all three datasets ([+0.006, +0.069], [+0.005, +0.045], and [+0.010, +0.044]).

With a braking controller whose corridor follows the path of constant curvature given by the ego yaw rate (Tab. S23), the field and the support follow the path without any change of the procedure. DGSP then attains 0.207, 0.224, and 0.394, significantly above the hybrid with a straight tube on Argoverse-HD ([+0.022, +0.064]), and on the 3,514 Argoverse-HD frames whose decisive object lies outside the straight tube, also above the hybrid whose support and tube follow the same curved path ([+0.007, +0.076]).

Second planner. The time-to-collision warning takes the action *warn* if the smallest time-to-collision of an object in the ego corridor is at most 4 s and *brake* if it is at most 2 s, *i.e*., its demand is the inverse time-to-collision. Its loss is a cost matrix over the required and the taken action, in which a missed brake has the highest cost. A probe at range r has the time-to-collision $r/(v+v_{\text{close}})$ and is sensitive if this reaches the threshold of the next action. Table S24 shows that the procedure, instantiated with the field and the support of this planner, significantly outperforms uncertainty crops

on Argoverse 2 and Argoverse-HD ($[+0.048, +0.109]$ and $[+0.045, +0.117]$), u960 on both datasets, and native-resolution inference on Argoverse-HD, with 6–8% of the pixels; it also attains a higher decision value than the hybrid of the support of this planner with an ego-corridor tube on all three datasets.

S4.4. Sensitivity to the parameters of the field

Table S26. **Decision value over the grid of field parameters** (probe class, closing speed v_{close} of the probe in m/s, and lateral spacing of the ground grid relative to the deployed one), with the refiner. The range spacing, varied by 0.5, 1, and 2, does not change any request and is omitted. px: pixels relative to the deployed field (shaded). * / †: significantly higher / lower than the deployed field (95% paired interval); ‡: lower also after Holm correction over the 81 registered points of a dataset; no point is higher after this correction.

			nuScenes		Argoverse 2		Argoverse-HD	
Probe	v_{close}	Lateral	G	px	G	px	G	px
car	0	×0.5	0.211	1.00	0.294	1.00	0.418	1.00
car	0	×1	0.211	1.00	0.294	1.00	0.418	1.00
car	0	×2	−0.024†	0.78	0.297	0.84	0.418	0.88
car	5	×0.5	0.235	1.25	0.298	1.28	0.384	1.16
car	5	×1	0.235	1.25	0.298	1.28	0.384	1.16
car	5	×2	−0.003	0.89	0.312*	1.02	0.420	0.94
car	10	×0.5	0.237*	1.30	0.295	1.41	0.383	1.26
car	10	×1	0.237*	1.30	0.295	1.41	0.383	1.26
car	10	×2	0.003	0.92	0.312*	1.10	0.419	0.98
truck	0	×0.5	0.246	1.31	0.286	1.22	0.413	1.21
truck	0	×1	0.205	1.14	0.293	1.13	0.409†	1.11
truck	0	×2	0.178	0.87	0.300	0.92	0.420	0.95
truck	5	×0.5	0.266*	1.71	0.301	1.65	0.337†	1.50
truck	5	×1	0.228	1.47	0.304	1.50	0.363†	1.33
truck	5	×2	0.206	1.05	0.315*	1.16	0.396	1.07
truck	10	×0.5	0.268*	1.77	0.299	1.82	0.334†	1.66
truck	10	×1	0.229	1.51	0.299	1.65	0.363†	1.46
truck	10	×2	0.209	1.08	0.314*	1.25	0.398	1.13
pedestrian	0	×0.5	0.115	0.79	0.298	0.85	0.406	0.89
pedestrian	0	×1	−0.075‡	0.71	0.254†	0.80	0.337†	0.84
pedestrian	0	×2	−0.075‡	0.71	0.254†	0.80	0.337†	0.84
pedestrian	5	×0.5	0.127	0.92	0.312*	1.03	0.406	0.96
pedestrian	5	×1	−0.073‡	0.76	0.261	0.89	0.338†	0.86
pedestrian	5	×2	−0.073‡	0.76	0.261	0.89	0.338†	0.86
pedestrian	10	×0.5	0.129	0.94	0.312*	1.11	0.406	1.01
pedestrian	10	×1	−0.073‡	0.78	0.261	0.94	0.337†	0.89
pedestrian	10	×2	−0.073‡	0.78	0.261	0.94	0.337†	0.89

The deployed field uses a static car-sized probe on a fixed ground grid. Table S26 varies the probe class, its closing speed, and the lateral and range spacing of the grid, 81 combinations in all, each run with the detector on all three datasets. The range spacing provably does not change the request: the demand of a probe decreases with its range, so the sensitive cells always form the cells up to some range, every tested grid contains the nearest cell at 12.5 m, and that cell bounds the projected crop; the executed crops are identical on all 1,775 compared frames, and a finer spacing only costs field time. For the car probe, the lateral spacings ×0.5 and ×1 give the same requests. No combination improves on the deployed field on all three datasets, and each that is significantly better on one dataset scores lower on another. The gains on nuScenes need 1.25–1.8× the pixels, and every point above the deployed field on Argoverse-HD lies within 0.003 of it. On Argoverse 2, a truck-sized probe with twice the lateral spacing gains about 0.02 at 1.16–1.25× the pixels; without the refiner, these two points are significantly better also after Holm correction over the 81 points ($\Delta = -0.024$ and -0.023), and with the refiner after correction over the 27 distinct points, since the 81-point family repeats each test three times through the range spacing. The same probe lowers the decision value on nuScenes and Argoverse-HD. A pedestrian-sized probe at the deployed or a coarser lateral spacing fails on nuScenes ($G = -0.07$); we did not analyze the mechanism. Larger probes and closing speeds above zero place the box center of the decisive object inside the executed crops on more frames but do not raise the decision value; we did not analyze why. The deployed field is therefore not a peak selected on the test data, but one dataset prefers a larger probe at more pixels.

Table S27. **Braking episodes and temporal consistency** (episodes that require at least DECELERATE). False: braking episodes without a requirement per 1,000 frames; Rev.: reversals A→B→A within two frames per 100 frames; Missed: share of required episodes without braking; Delay: mean onset delay in seconds, capped at the episode length; lower is better for all four. "+ comp.": with the refiner and the learned looming estimate.

	nuScenes held-out (147 episodes)				Argoverse 2 val (190)				Argoverse-HD val (71)			
Pipeline	False	Rev.	Missed	Delay	False	Rev.	Missed	Delay	False	Rev.	Missed	Delay
Cheap (u320)	22.9	4.71	0.35	0.61	34.8	9.57	0.11	0.17	25.1	10.92	0.17	0.11
u960 + comp.	6.4	1.83	0.35	0.54	7.9	2.10	0.16	0.21	5.5	0.80	0.37	0.15
Native + comp.	4.9	1.63	0.36	0.61	8.6	2.41	0.15	0.19	7.0	1.31	0.38	0.16
Support + corridor tube	4.2	1.55	0.39	0.73	6.1	1.90	0.18	0.24	6.7	1.44	0.38	0.13
Uncertainty crops	7.2	1.92	0.43	0.80	14.9	3.93	0.16	0.26	15.9	3.93	0.39	0.19
DGSP	4.0	1.38	0.41	0.74	6.5	2.06	0.19	0.22	6.3	1.20	0.38	0.13

Table S28. **Required braking episodes by peak demand** (episodes of Tab. S27). Missed: share of the episodes with a peak demand below ($< \tau_2$) or at least ($\geq \tau_2$) the hard-braking threshold that are missed; Loss: loss of insufficient braking within all required episodes, as a share of the total loss of the cheap pipeline. † / *: DGSP significantly better / worse (95% paired bootstrap intervals). "+ comp.": with the refiner and the learned looming estimate.

	nuScenes held-out (109 / 38)			Argoverse 2 val (135 / 55)			Argoverse-HD val (61 / 10)		
Pipeline	Miss. $< \tau_2$	Miss. $\geq \tau_2$	Loss	Miss. $< \tau_2$	Miss. $\geq \tau_2$	Loss	Miss. $< \tau_2$	Miss. $\geq \tau_2$	Loss
Cheap (u320)	0.40	0.18	0.68	0.13*	0.07	0.37	0.20*	0.00	0.15
u960 + comp.	0.40*	0.18	0.64	0.19	0.09	0.33*	0.39	0.20	0.11
Native + comp.	0.42	0.18	0.58	0.18*	0.09	0.40	0.41	0.20	0.11
Support + corridor tube	0.47	0.18	0.61	0.21	0.11	0.38	0.41	0.20	0.11
Uncertainty crops	0.50	0.24	0.83†	0.19	0.09	0.42	0.43	0.20	0.21†
DGSP w/o refiner	0.46	0.18	0.62†	0.16*	0.09	0.36	0.23*	0.10	0.06*
DGSP	0.48	0.21	0.58	0.22	0.11	0.39	0.41	0.20	0.11

S5. Braking severity, operating points, and robustness

The decision value aggregates insufficient and excessive braking. This section reports the braking episodes behind it (Sec. S5.1), asks whether the advantage of DGSP depends on the operating point of the planner (Sec. S5.2), and tests the result on held-out sequences, across strata, and under other weightings of the loss (Secs. S5.3 to S5.5).

S5.1. Braking episodes and their severity

Table S27 evaluates the braking episodes that the annotated scene requires, *i.e.*, maximal runs of frames that require at least DECELERATE, with gaps of one frame merged. DGSP produces the fewest false braking episodes on nuScenes and significantly fewer than u960 with the components on nuScenes and Argoverse 2 and than uncertainty crops on all three datasets, and its action reverses least often on nuScenes. At the HARD-BRAKE level, no significant difference in the share of missed episodes between DGSP and any whole-frame pipeline is detected; with few such episodes (ten on Argoverse-HD), this is weak evidence and does not establish equivalence. Table S28 separates the required episodes by their peak demand. Of the additional episodes that DGSP misses relative to the cheap detector, 8 of 9, 13 of 15, and 13 of 15 have a peak demand below the hard-braking threshold, which still requires braking, and the loss of insufficient braking within the required episodes does not differ significantly from that of the cheap detector or of native-resolution inference with the same components on any dataset. Within the episodes whose peak demand reaches the hard-braking threshold, this loss is lower than that of the cheap detector on nuScenes, by 0.11 of the total loss of the cheap pipeline ($[-0.24, -0.01]$), and higher on the ten such episodes of Argoverse-HD, by 0.02 ($[+0.003, +0.052]$). The refiner, which all pipelines with the components share, raises the missed share of DGSP by 0.020, 0.047, and 0.169 and that of native-resolution inference by 0.041, 0.047, and 0.211, and without the refiner, DGSP misses 0.39, 0.14, and 0.21 of the episodes, which does not differ significantly from the cheap detector on any dataset. In the controlled closed loop, where the refiner is not used, DGSP misses significantly fewer required braking episodes than the cheap detector (Tab. S37). Of the ten longest onset delays of DGSP on each dataset, 8, 8, and 10 stem from the lift or the detection of an object inside a verified region and 2, 2, and 0 from an object outside every crop.

Table S29 extends Tab. S27 to every method of Tab. 2. DGSP produces significantly fewer false braking episodes

Table S29. **Missed and false braking episodes of all methods** at the deployed threshold (definitions of Tab. S27; all rows with the shared lift and their own history). † / *: DGSP significantly lower (better) / higher (worse).

	nuScenes held-out		Argoverse 2 val		Argoverse-HD val	
Pipeline	Missed	False	Missed	False	Missed	False
Cheap (u320)	0.35	22.9†	0.11*	34.8†	0.17*	25.1†
u512 + comp.	0.36	7.0†	0.15	9.0†	0.41	5.6
u640 + comp.	0.35*	7.3†	0.15*	9.3†	0.45	5.7
u960 + comp.	0.35*	6.4†	0.16	7.9†	0.37	5.5
u1280 + comp.	0.33*	6.4†	0.15*	7.8†	0.34	6.3
Native + comp.	0.36*	4.9	0.15*	8.6†	0.38	7.0
AdaScale	0.35*	6.4†	0.17	8.1†	0.34	6.2
Budget-Adaptive Routing	0.36	13.7†	0.12*	26.1†	0.30	15.3†
Router, decision-value targets	0.31*	14.1†	0.14*	26.8†	0.17*	25.2†
Chanakya-style policy	0.40	13.4†	0.14	27.5†	0.35	17.9†
SAHI	0.33*	11.9†	0.13*	16.2†	0.30*	11.2†
FOVEA, instance saliency	0.34*	6.6†	0.14*	9.8†	0.34	6.9
UFP	0.40	5.8†	0.15*	8.9†	0.32	6.3
Counterfactual, iterated	0.49†	4.0	0.21	6.2	0.46	6.1
AnA-style, iterated	0.39	4.9	0.16*	7.0	0.35	5.9
Support + fixed tube	0.39	4.2	0.18	6.1	0.38	6.7
Uncertainty crops	0.43	7.2†	0.16	14.9†	0.39	15.9†
DGSP w/o refiner	0.39	6.8†	0.14*	11.5†	0.21*	6.7
DGSP	0.41	4.0	0.19	6.5	0.38	6.3

than every whole-frame, published, and routed method on nuScenes and Argoverse 2, except native-resolution inference on nuScenes and the object-level methods, but it misses significantly more required episodes than most of them; on Argoverse-HD, its missed share is significantly higher only than that of the cheap detector, the decision-value router, and SAHI.

S5.2. Operating curves and development-selected thresholds

Table S49 sweeps the braking threshold of the planner for DGSP, the hybrid, and whole-frame inference with the same components; the required episodes and the loss of G remain those of the deployed planner, and the deployed threshold reproduces the stored results exactly. Lowering the threshold moves every pipeline with the components along the same trade-off: the missed episodes of DGSP fall from 60, 36, and 27 to 36, 19, and 9 between 1.0 and 0.6, its decision value rises, and its false braking episodes rise from 4.0, 6.5, and 6.3 to 12.0, 10.6, and 7.1 per 1,000 frames; at every threshold, DGSP uses significantly fewer pixels than the hybrid, whose decision value is never significantly higher (on Argoverse 2, it is higher by 0.002–0.006, not significant; on Argoverse-HD, it is significantly lower at 0.6 and 1.2). Table S30 evaluates once, on the test splits and the held-out sequences, the threshold of each pipeline that a rule fixed in advance selected on the development data. At this point, DGSP does not miss significantly more episodes than the cheap detector on any split, brakes falsely 16–27 fewer times per 1,000 frames, and attains a decision value that no pipeline with the same components exceeds significantly; whole-frame inference with the components misses fewer episodes on nuScenes and Argoverse 2 with 3–18$\times$ the pixels. The deployed threshold was not changed in response to these results.

S5.3. Held-out sequences

The frozen configuration and the baselines of Tab. S31 were run once on the 65 training sequences of Argoverse-HD, previously unused for training, configuration selection, and evaluation, with the code, the rows, the metrics, and the decision rule fixed in a protocol before the run; the sequences were used once more, for the constrained operating points of Tab. S30 under a second protocol, and for no other evaluation or choice. The sequences share the sensor, the cities, and the annotation process of the Argoverse-HD test split and contain only 17 episodes that require hard braking. DGSP attains $G = 0.407$ with 7.6% of the native pixels; no baseline with the same components differs significantly from it, although they use 1.6–13$\times$ its pixels, and uncertainty crops are significantly lower. All pipelines with the same components miss 42–44 of the 154 required episodes, compared with 18 for the cheap detector at 34.5 false braking episodes per 1,000 frames. The protocol also contained one candidate extension, fixed on development

Table S30. **Thresholds selected on development data under a missed-braking criterion.** For each pipeline, the braking threshold $\text{thr}_{\text{decel}}$ (m/s^2) is the largest value of $\{0.6, 0.8, 1.0, 1.2, 1.5\}$ at which it misses no more required episodes than the cheap detector on the development data (chosen once, before the evaluation; Argoverse-HD uses the Argoverse 2 choice). [§]: no grid value meets the constraint on the development data (DGSP misses 31 *vs*. 30 of 366 episodes); its lowest-miss value is used. Missed: required episodes without braking; False: false braking episodes per 1,000 frames; ΔG: 95% paired interval of (DGSP − row), bold if above zero; px in % of the native frame. The required episodes and the loss of G remain those of the deployed planner.

Pipeline	$\text{thr}_{\text{decel}}$	Missed	False	G	ΔG	px
nuScenes held-out						
Cheap (u320)	1.0	51/147	22.9	0.000	**[+0.145, +0.396]**	4.3
u960 + comp.	0.8	36/147	10.0	0.172	[−0.087, +0.301]	38.4
Native + comp.	0.8	37/147	8.3	0.276	[−0.091, +0.115]	103.1
Support + corridor tube	0.8	45/147	7.3	0.223	[−0.009, +0.097]	11.3
DGSP	0.8	47/147	7.0	0.261	–	8.0
Argoverse 2 val						
Cheap (u320)	1.0	21/190	34.8	0.000	**[+0.316, +0.435]**	2.6
u960 + comp.	0.6	13/190	12.7	0.415	[−0.088, +0.002]	23.2
Native + comp.	0.6	14/190	14.6	0.365	[−0.068, +0.095]	101.2
Support + corridor tube	0.6	18/190	10.5	0.378	[−0.013, +0.009]	7.8
DGSP	0.6[§]	19/190	10.6	0.376	–	5.7
Argoverse-HD val						
Cheap (u320)	1.0	12/71	25.1	0.000	**[+0.375, +0.584]**	3.1
u960 + comp.	0.6	9/71	6.7	0.366	**[+0.016, +0.311]**	25.3
Native + comp.	0.6	10/71	9.2	0.499	[−0.047, +0.022]	101.3
Support + corridor tube	0.6	9/71	7.3	0.469	**[+0.006, +0.030]**	11.2
DGSP	0.6[§]	9/71	7.1	0.485	–	7.3
Held-out sequences						
Cheap (u320)	1.0	18/154	34.5	0.000	**[+0.412, +0.529]**	3.1
u960 + comp.	0.6	20/154	8.0	0.457	**[+0.002, +0.027]**	25.3
Native + comp.	0.6	19/154	9.5	0.483	[−0.023, +0.006]	101.3
Support + corridor tube	0.6	24/154	8.6	0.474	[−0.011, +0.008]	13.4
DGSP	0.6[§]	20/154	7.1	0.474	–	7.9

Table S31. **Held-out sequences** (65 Argoverse-HD training sequences, 39,265 frames, 154 required episodes), in the first of two evaluations under protocols fixed in advance. ΔG: 95% paired interval of (DGSP − row), bold if above zero. Missed: required braking episodes without braking. "+ comp.": with the refiner and the learned looming estimate.

Pipeline	G	ΔG	px (%)	False	Missed
Cheap (u320)	0	**[+0.313, +0.480]**	3.1	34.5	18
u960 + comp.	0.400	[−0.012, +0.021]	25.3	6.4	42
Native + comp.	0.417	[−0.026, +0.008]	101.3	8.2	44
Support + corridor tube	0.410	[−0.011, +0.005]	12.6	7.7	44
Uncertainty crops	0.338	**[+0.046, +0.093]**	15.2	20.7	32
AnA-style, iterated	0.408	[−0.022, +0.012]	12.4	7.5	44
DGSP	0.407	–	7.6	6.5	43
Bounded + temporal recovery	0.409	[−0.011, +0.002]	6.9	7.6	42

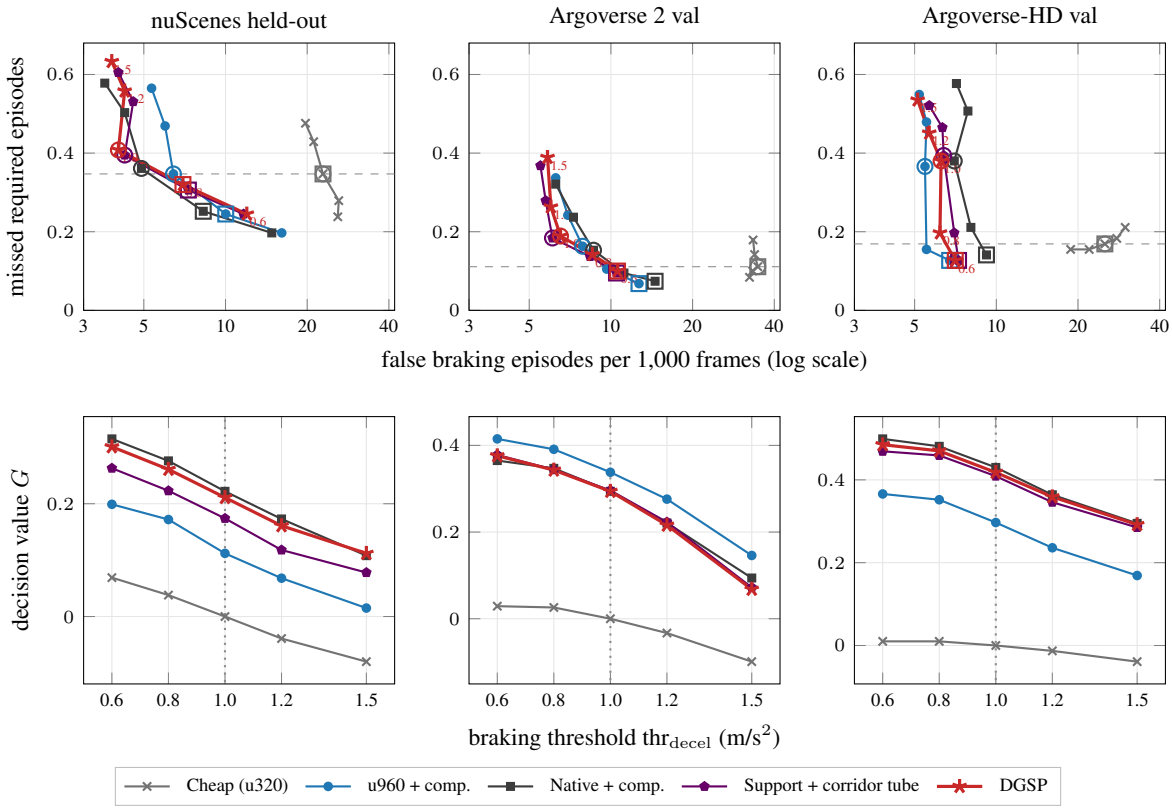


Figure S10. **Is the advantage of DGSP an artifact of its operating point?** Operating curves on the test splits, obtained by varying the braking threshold $\text{thr}_{\text{decel}} \in \{0.6, 0.8, 1.0, 1.2, 1.5\}$ m/s^2 of the planner (labels at the points of DGSP); the required episodes and the loss of G remain those of the deployed planner. Top: share of required braking episodes that are missed *vs*. false braking episodes per 1,000 frames. Circles: deployed threshold (1.0); squares: threshold chosen on development data by the constrained rule of Tab. S30; dashed line: missed share of the cheap detector at the deployed threshold. Bottom: decision value *vs*. threshold. All pipelines except the cheap one use the shared components, and the hybrid receives the per-frame budget of DGSP at each threshold. Values in Tab. S49; false-episode rates from the same runs.

data before the run: bounded-keep (Sec. S6.1), together with the recovery of missing looming observations of verified objects from the retained previous frames. It attains the decision value of DGSP (ΔG in Tab. S31) with 10% fewer pixels and 1.0 more false braking episode per 1,000 frames ($[+0.49, +1.62]$); it is not used elsewhere in this paper.

S5.4. Stratified evaluation

DGSP attains a positive decision value in every stratum of Tab. S32, including the night scenes of nuScenes (0.231, compared with -0.061 for u960) and rain (0.440). In the 14 Argoverse-HD sequences of Pittsburgh, DGSP is significantly better than u960 and the hybrid. The strata are subsets of the evaluated datasets, not an independent test of domain generalization.

S5.5. Robustness of the evaluation

Across 27 weightings of the loss terms (excessive braking, collisions, and command changes each scaled by 0.5, 1, or 2), DGSP ranks second of ten pipelines on nuScenes and Argoverse-HD and third or fourth on Argoverse 2, and its decision value is at least 0.166, 0.159, and 0.335 in every setting, including the one in which excessive braking is halved and collisions are doubled, *i.e.*, in which insufficient braking weighs twice as much relative to excessive braking. With the braking thresholds scaled by 0.8 or 1.2 or a reaction time of 0.3 or 0.6 s, which change the field through the planner, DGSP remains significantly better than uncertainty crops on every dataset and in every tested setting; these checks do not cover every possible loss.

Table S32. **Decision value per stratum** (lighting and rain from the nuScenes scene descriptions; city for Argoverse 2 and Argoverse-HD). All baselines with the shared components. † / *: DGSP significantly better / worse within the stratum.

Stratum	Units	u960	Native	Hybrid	Uncert.	DGSP
nuScenes, day	201	0.233	0.237	0.195	0.163†	0.197
nuScenes, night	27	−0.061	0.201	0.144	−0.379†	0.231
nuScenes, no rain	178	−0.002	0.191	0.091	−0.106†	0.140
nuScenes, rain	50	0.479	0.323	0.440	0.093†	0.440
Argoverse 2, ATX	4	0.272	0.298	0.186	0.199	0.191
Argoverse 2, DTW	21	0.461	0.197	0.473	0.185†	0.491
Argoverse 2, MIA	49	0.303	0.222	0.225	0.160†	0.223
Argoverse 2, PAO	7	0.481	0.504	0.412	0.377	0.442
Argoverse 2, PIT	51	0.292	0.327	0.321	0.237†	0.309
Argoverse 2, WDC	18	0.456	0.468*	0.279	0.357	0.269
Argoverse-HD, MIA	10	0.393	0.442	0.413	0.280†	0.415
Argoverse-HD, PIT	14	0.174†	0.415	0.402†	0.284†	0.421

Table S33. **Decision value when insufficient braking weighs more**, with the insufficient-braking term of the loss multiplied by w and all other terms fixed, computed from the stored actions of each pipeline (uncapped). † / *: DGSP significantly better / worse (95% paired interval); ‡ / **: also after Holm correction over all compared pipelines and weights of a dataset.

	nuScenes held-out				Argoverse 2 val				Argoverse-HD val			
w	1	2	5	10	1	2	5	10	1	2	5	10
u960 + comp.	0.112	0.077	0.052	0.043	0.338	0.239*	0.132*	0.080*	0.297†	0.259†	0.195	0.148
u1280 + comp.	0.292	0.269	0.251*	0.245*	0.348*	0.252*	0.148*	0.097*	0.315†	0.279†	0.218	0.174
Native + comp.	0.222	0.175	0.140	0.128	0.294	0.160	0.015	−0.054	0.430	0.369	0.265	0.189
AdaScale	0.144	0.113	0.091	0.083	0.341	0.244*	0.139*	0.088*	0.298‡	0.261†	0.198	0.152
SAHI	0.303	0.309*	0.314**	0.315**	0.256	0.210	0.160*	0.135**	0.229‡	0.231†	0.234	0.236
AnA-style, iterated	0.158	0.117	0.088	0.077	0.349*	0.241*	0.125*	0.068*	0.315†	0.268†	0.188†	0.130
Support + tube	0.174	0.129	0.096	0.083	0.296	0.170	0.033	−0.033	0.408	0.346	0.242	0.166
Reference $\pi(F)$	0.085	0.049	0.023	0.013	0.290	0.256*	0.219**	0.201**	0.296	0.322	0.367	0.400*
DGSP w/o refiner	0.181	0.131†	0.094†	0.081†	0.241†	0.169	0.091	0.053*	0.422	0.422	0.423*	0.423**
DGSP	**0.211**	**0.166**	**0.134**	**0.122**	**0.294**	**0.166**	**0.027**	**−0.039**	**0.418**	**0.357**	**0.254**	**0.179**

S5.6. Losses that weight insufficient braking more

The weightings of Sec. S5.5 scale the excessive-braking, collision, and command-change terms. Table S33 instead multiplies the insufficient-braking term by 2, 5, and 10. Since most of the gain of DGSP comes from fewer excessive braking actions (Sec. S3.6), its decision value falls with this weight, to 0.122, −0.039, and 0.179 at $w = 10$, where it no longer differs significantly from that of the cheap pipeline on any dataset. On nuScenes, only SAHI (from $w = 2$) and whole-frame inference at 1280 pixels (from $w = 5$) score significantly higher, and on Argoverse-HD, no compared method does at any weight. On Argoverse 2, whole-frame inference at 960 and 1280 pixels, AdaScale, AnA-style refinement, and SAHI score significantly higher from $w = 2$ or 5, after Holm correction only SAHI at $w = 10$. The range refiner accounts for much of this decline on Argoverse 2 and Argoverse-HD: without it, DGSP keeps $G = 0.42$ on Argoverse-HD at every weight and attains 0.053 on Argoverse 2 at $w = 10$, and the reference without the refiner attains 0.201 and 0.400; on nuScenes, the refiner raises the decision value at every weight. The refiner thus trades insufficient for excessive braking, and the decision value under the deployed weights is not a measure of safety.

S6. Embedded latency, deadlines, closed loop, and planning

This section reports the latency and energy of DGSP and of all compared methods on a Jetson AGX Xavier (Sec. S6.1), the decision value under per-frame deadlines run on the device (Sec. S6.2), the controlled closed loop in CARLA (Sec. S6.3), and the planning benchmarks (Sec. S6.4).

S6.1. Latency and energy

The detectors run as TensorRT FP16 engines at maximum clocks; image decoding is excluded, the first 200 frames serve as warm-up, and all pipelines are timed on the same frames. The looming estimate encodes all crops of a lift step

Table S34. **Latency on a Jetson AGX Xavier** (ms). DGSP is 1.7–1.8× faster than native-resolution inference with the same components. G on the timed frames (805 frames of 20 nuScenes held-out scenes; 586 frames of 8 Argoverse 2 logs). Bounded-drop: at most two crop batches, pending truncated detections discarded (Sec. S6.1).

	nuScenes			Argoverse 2		
Pipeline	$G\uparrow$	Mean	p95	$G\uparrow$	Mean	p95
Cheap (u320)	0	7.5	8.2	0	10.4	10.9
u512 + comp.	0.336	13.9	22.1	0.363	24.3	39.7
u960 + comp.	0.386	22.7	30.9	0.323	34.3	49.2
u1280 + comp.	0.395	28.6	36.5	0.311	43.6	59.3
Native + comp.	0.364	37.9	45.5	0.410	75.7	88.7
DGSP	0.380	21.9	59.2	0.419	41.7	93.8
DGSP, bounded-drop	0.382	19.8	46.2	0.423	37.1	73.5

in one batched TensorRT call, caches the feature of each crop for the two following frames, and skips the objects whose demand cannot depend on their time-to-collision, *i.e.*, objects outside the ego corridor or beyond 80 m; the 64×64 crops are shared with the refiner. On the full development splits, these changes leave every action of every pipeline unchanged. The tail of the latency of the unbounded loop stems from the detector calls of the truncation guard, not from the looming estimate. We therefore evaluate two variants that return the action of the merged scene after at most two crop batches after the draft; each batch processes the new crops of the field and the support together with the enlarged crops that the guard has queued in the previous batch, and a batch can be split into several executions of the detector engine. *Bounded-drop* discards the detections whose guard re-detection is still pending at the bound; *bounded-keep*, selected on the development data, keeps them as detected. The deadline experiments (Fig. S12, Tabs. S35 and S51) and the held-out extension (Sec. S5.3) use bounded-keep; Tab. S34, the single run from which main Tab. 6 takes its latencies, reports bounded-drop.

- *Bounded-drop* exceeds 50, 66.7, and 100 ms on 3.1%, 0.2%, and none of the timed nuScenes frames, compared with 9.4%, 2.7%, and 0.2% for DGSP and 0.5% and 1.2% above 50 ms and none above 66.7 ms for u960 and native-resolution inference with the components; on the full test splits, however, it attains $G = 0.091$, 0.282, and 0.419, the loss on nuScenes stemming from one scene.
- *Bounded-keep* attains $G = 0.209$, 0.293, and 0.415 on the full test splits with 6–10% fewer pixels than DGSP, within $[-0.005, +0.002]$, $[-0.007, +0.005]$, and $[-0.008, +0.001]$ of DGSP. Timed in separate runs on the same frames, it runs at 20.3 and 38.5 ms on average on nuScenes and Argoverse 2, with a 95th percentile of 49.1 and 77.4 ms, compared with 21.8 and 40.4 ms and 59.7 and 91.2 ms for DGSP (paired differences of the 95th percentile $[-14.7, -2.5]$ and $[-22.8, -6.4]$ ms); it exceeds 100 ms on none and 1.1% of the frames.

Table S50 times the methods of Tab. 2 in separate runs on the same frames: every published, learned, or object-level method that attains at least the decision value of DGSP takes 25.1–84.8 ms on average, compared with 21.5 and 39.8 ms for DGSP in the same runs. Measured on the module rails and above the cheap detector, DGSP requires 0.29 and 0.54 J per frame on nuScenes and Argoverse 2, compared with 0.54 and 0.76 J for whole-frame inference at the verification resolution and 1.12 and 2.39 J at the native resolution, both with the same components; whole-frame inference at 512 pixels requires 0.26 and 0.42 J, less than DGSP (one run per pipeline).

Where the tail latency comes from. In a separate run with a synchronization after every stage (which adds about 4% to the mean), the mean latency of DGSP over all frames and over the frames at or above its 95th percentile is 21.5 and 70.2 ms on nuScenes and 33.3 and 97.6 ms on Argoverse 2. From all frames to these tail frames, the crop rounds grow from 8.8 to 45.0 ms and from 13.4 to 55.3 ms, and the looming estimate of the crops from 1.4 to 5.3 ms and from 4.0 to 15.8 ms, whereas the draft stays at 7–10 ms, the refiner below 4 ms, and the field, guard, lift, and merge together below 13 ms. The tail therefore scales with the number of sequential crop rounds, which is why bounding the loop to two batches (Tab. S34) shortens it. Two exact optimizations of the implementation, which leave every action unchanged, did not shorten it measurably: omitting an unused reduction of the raw detections changed the 95th percentile from 58.8 to 59.0 ms and from 111.7 to 111.4 ms on 712 development frames per dataset, and replaying the looming encoder as a fixed graph changed it by −1.2% and −0.2%, all intervals containing zero.

KITTI. On the 0.47 MP frames of KITTI, a single native-resolution pass takes 15.4 ms on the Jetson AGX Xavier, so that region-wise verification offers no cost advantage, and native-resolution inference with the refiner is the natural choice on this dataset. Megapixel frames are the intended operating regime.

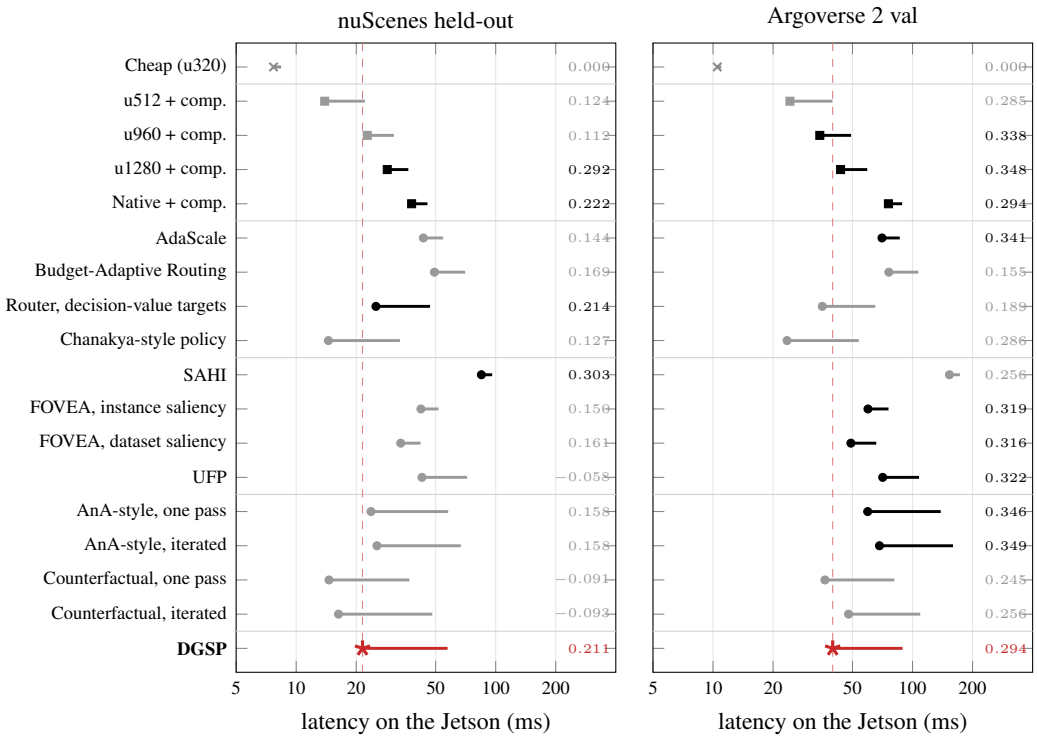


Figure S11. **Which methods reach the decision value of DGSP, and at what latency?** Mean latency (marker) and 95th percentile (end of the bar) on the Jetson AGX Xavier, log scale, with the decision value on the full test split at the right. Black: decision value at least that of DGSP on the dataset; grey: lower; dashed: mean latency of DGSP. Rows, from the top: the cheap detector; whole-frame inference with the components (squares; latency from Tab. S34, separate runs on the same frames; decision value from Tab. S44); the methods of Tab. 2, all with the shared components (Tab. S50).

S6.2. Decision value under deadlines

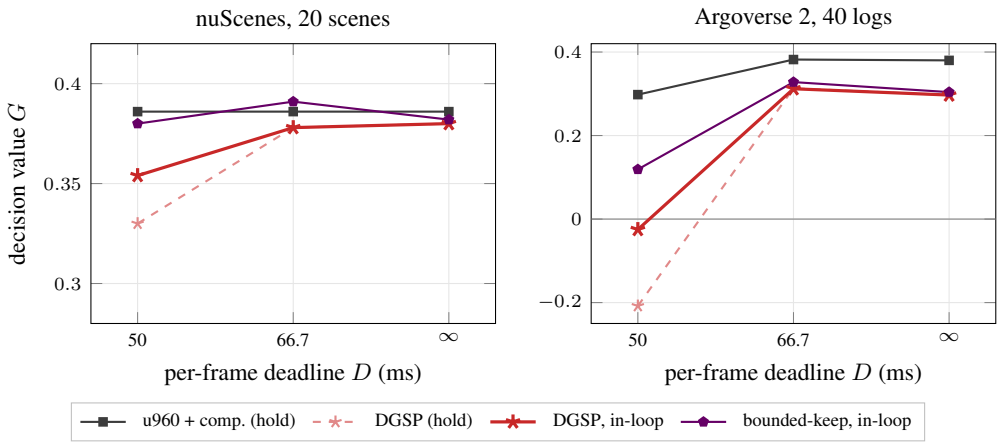


Figure S12. **Decision value under per-frame deadlines**, run in the loop on the Jetson AGX Xavier (nuScenes: 20 test scenes; Argoverse 2: 40 test logs; three repetitions). Hold: a late action is replaced by the last action that was on time. In-loop: the loop stops before a round that would exceed the deadline and returns the action of the last completed round, and a frame that is still late is held. Bounded-keep: at most two crop batches after the draft, keeping pending truncated detections. Decision values on these subsets, not on the full test splits; values and late shares in Tab. S51.

Table S51 runs the pipelines on the Jetson under per-frame deadlines, each with its own history; the rule for stopping the loop and the 40 Argoverse 2 logs were fixed before the run, and without a deadline the actions equal those of Tab. S34 on every common frame. Stopping the loop before a round that would exceed the deadline is better

than holding a late action (Argoverse 2, 50 ms: $+0.18$ $[+0.03, +0.36]$), and bounded-keep retains its decision value from 50 ms on nuScenes and from 66.7 ms on Argoverse 2. Whole-frame inference at 960 pixels with the components runs a single engine pass with a lower tail latency on this device (95th percentile 31 and 47 ms), so that it is rarely late; it matches DGSP on nuScenes and leads on Argoverse 2 (0.30 *vs*. 0.12 at 50 ms and 0.38 *vs*. 0.33 at 66.7 ms, not significant) with 4–5× its pixels. Without a deadline, DGSP attains 0.297 and u960 with the components 0.380 on the 40 logs, the ordering of the full Argoverse 2 split, whereas the 8 logs of Tab. S34 give the opposite ordering; the timed subsets therefore differ in their relative decision values.

Table S35. **Latency–decision frontier on the Jetson AGX Xavier** under per-frame deadlines D, on the frames of Fig. S12, three repetitions in one session per repetition, in rotated order. The fixed-shape engines replay CUDA graphs, which leaves every action, crop, and pixel count unchanged. px: detector-input pixels in % of the native frame; p95: 95th-percentile latency without a deadline (ms). Whole frame and hybrid hold a late action; DGSP stops the loop in time (in-loop). Hybrid: support + ego-corridor tube with the components under the nominal cap at the pixels of DGSP. ‡: on the dynamic engine. The one-batch variant was added after the other results were known.

	nuScenes, 20 test scenes						Argoverse 2, 40 test logs					
			G at D (ms)						G at D (ms)			
Pipeline	px	p95	33.3	50	66.7	∞	px	p95	33.3	50	66.7	∞
Cheap (u320)	4.3	7.7	0.000	0.000	0.000	0.000	2.6	10.1	0.000	0.000	0.000	0.000
u512 + comp.	10.2	21.4	0.351	0.336	0.336	0.336	6.7	36.3	−0.095	0.361	0.361	0.358
u640 + comp.	17.1	23.5	0.345	0.330	0.330	0.330	10.3	38.9	0.139	0.367	0.375	0.373
u800 + comp.‡	26.7	28.6	0.414	0.369	0.369	0.369	15.3	43.3	−0.300	0.383	0.383	0.381
u960 + comp.	38.4	29.1	0.423	0.386	0.386	0.386	23.2	45.2	−0.692	0.352	0.382	0.380
u1280 + comp.	65.4	34.8	0.434	0.410	0.395	0.395	40.0	53.1	−1.649	0.119	0.408	0.400
Native + comp.	103.1	43.7	−8.024	0.364	0.364	0.364	101.2	83.3	−1.649	−1.649	−1.647	0.147
Hybrid, capped	10.3	73.1	−0.280	0.153	0.196	0.373	7.3	96.8	−0.726	−0.279	−0.089	0.296
DGSP	7.4	55.1	−0.204	0.354	0.378	0.380	5.3	74.3	0.000	0.213	0.315	0.297
DGSP, bounded-keep	7.8	46.8	0.290	0.383	0.391	0.382	5.9	63.8	−0.241	0.130	0.336	0.304
DGSP, one batch, keep (post hoc)	6.2	32.8	0.437	0.400	0.391	0.391	4.2	44.6	−0.148	0.277	0.269	0.269

Latency–decision frontier. Table S35 repeats the deadline experiment with more whole-frame resolutions, the capped hybrid, and CUDA-graph replay of the fixed-shape engines, which shortens all pipelines; without a deadline, the actions, crops, and pixels of DGSP and of bounded-keep equal those of the earlier run on every timed frame. Since the deadlines lie close to the latencies, individual values under a deadline differ from Fig. S12; u960 with the components again matches DGSP on nuScenes and leads on Argoverse 2. Whole-frame inference at 512 to 960 pixels is a strong baseline on this device: it runs a single engine pass, is rarely late, and on Argoverse 2 attains a higher decision value than DGSP and bounded-keep at every deadline from 50 ms (0.35–0.38 *vs*. 0.13–0.34), significantly only at 50 ms against u800 (-0.170, $[-0.312, -0.015]$) and against u960 on the dynamic engine. On nuScenes, DGSP and bounded-keep do not differ significantly from whole-frame inference at any resolution from 50 ms, except that both are significantly better than u640 at some deadlines, while at 33.3 ms both are significantly worse than whole-frame inference at 800 to 1280 pixels. With the same nominal pixels, the hybrid has a 95th-percentile latency of 73 and 97 ms, and DGSP is significantly better than it at 50 ms on nuScenes ($+0.201$, $[+0.008, +0.768]$) and at 33.3 to 66.7 ms on Argoverse 2 ($+0.40$ to $+0.73$). A variant bounded to a single crop batch, added after these results were known, attains 0.437 at 33.3 ms and 0.400 at 50 ms on nuScenes with 6.2% of the pixels, but only 0.269 on Argoverse 2; we do not use it elsewhere. The advantage of DGSP on this device therefore lies in the decision value per pixel and in the tail latency relative to other crop selection, not in the latency relative to whole-frame inference.

All compared methods under deadlines. Table S36 repeats the benchmark of Tab. S35 with every method of Tab. 2, on the same frames, in one exclusive session of the device. The actions, pixels, and crops of DGSP and of the bounded variant equal those of the earlier device runs on every frame, and the actions of all methods agree with their evaluated records on 99.2–99.9% of the frames (96.8–100% for the sizes chosen by the routers), the differences coming from the half-precision draft on the device. No method is significantly better than DGSP after Holm correction at any deadline, whereas DGSP is significantly better than 2 to 5 methods on nuScenes from 50 ms and than 1 to 8 on Argoverse 2, mainly those that are too slow for the tighter deadlines. Without correction, three of the four whole-frame resolutions

Table S36. **All compared methods under per-frame deadlines** on the Jetson AGX Xavier, on the frames of Tab. S35 (nuScenes: 20 test scenes; Argoverse 2: 40 test logs), three repetitions. DGSP and the iterated methods stop their loop before a round that would exceed the deadline; every other pipeline holds the last action that was on time. p95: 95th-percentile latency without a deadline (ms). † / *: DGSP significantly better / worse (95% paired interval); ‡: better also after Holm correction over the rows at that deadline; no row is better than DGSP after this correction.

	nuScenes						Argoverse 2					
		G at D (ms)						G at D (ms)				
Pipeline	p95	33.3	50	66.7	100	∞	p95	33.3	50	66.7	100	∞
Cheap (u320)	8	0.00	0.00	0.00	0.00	0.00	10	0.00	0.00	0.00	0.00	0.00
u512 + comp.	21	0.35*	0.34	0.34	0.34	0.34	36	−0.09	0.36*	0.36	0.36	0.36
u640 + comp.	23	0.35	0.33	0.33	0.33†	0.33†	39	0.14	0.37*	0.37	0.37	0.37
u960 + comp.	29	0.42*	0.39	0.39	0.39	0.39	45	−0.78†	0.35	0.38	0.38	0.38
u1280 + comp.	35	0.43*	0.41	0.39	0.39	0.39	53	−1.6‡	0.11	0.41	0.40	0.40
Native + comp.	44	−8.0†	0.36	0.36	0.36	0.36	83	−1.6‡	−1.6‡	−1.6‡	0.15	0.15
Hybrid, capped	74	−0.28	0.14‡	0.19	0.35	0.37	98	−0.84†	−0.34‡	−0.10†	0.33	0.30
Uncertainty crops, capped	109	−0.02	−0.25†	0.05‡	0.11‡	0.20	160	−0.98†	−0.51‡	−0.43‡	0.12†	0.19‡
AdaScale	51	−8.0†	0.39	0.37	0.37	0.37	82	−1.6‡	−1.6‡	−0.92‡	0.41	0.41
Budget-Adaptive Routing	66	−0.48	−0.47‡	0.17	0.16†	0.16†	97	−1.6‡	−0.14†	−0.12‡	0.10†	0.16†
Router, decision-value targets	44	−1.0	0.23	0.23‡	0.23†	0.23†	59	−0.30†	−0.03	0.22‡	0.20‡	0.20‡
Chanakya-style policy	32	0.25	0.23	0.23‡	0.23‡	0.23‡	47	−0.42	0.24	0.24	0.23†	0.23†
FOVEA, instance saliency	47	−8.0†	0.42	0.40	0.40	0.40	67	−1.6‡	−1.3‡	0.23	0.37	0.37
FOVEA, dataset saliency	38	−0.27	−0.18	−0.20	−0.20	−0.20	58	−1.6‡	−0.16	0.31	0.33	0.33
AnA-style, one pass	57	0.32	0.32	0.31†	0.33	0.33	111	−1.1‡	−0.21	0.13	0.34	0.40
AnA-style, iterated	67	0.32	0.32	0.31†	0.33	0.33	128	0.00	−0.21	0.11	0.26	0.40
Counterfactual, one pass	38	−0.02	0.11	0.07‡	0.07‡	0.07‡	68	−1.5‡	0.07	0.28	0.28	0.28
Counterfactual, iterated	48	0.03	0.12	0.07‡	0.07‡	0.07‡	105	0.00	0.12	0.28	0.31	0.32
DGSP	56	**−0.19**	**0.35**	**0.38**	**0.38**	**0.38**	75	**0.00**	**0.15**	**0.31**	**0.30**	**0.30**
DGSP, ≤ 2 batches	48	0.31	0.38	0.39	0.38	0.38	64	−0.35	0.10	0.34	0.31	0.30
DGSP, one batch (post hoc)	34	0.43	0.40	0.39	0.39	0.39	45	−0.35	0.28	0.27	0.27	0.27

score higher on nuScenes at 33.3 ms, where DGSP is late on 16% of the frames and stops its loop early on 55%, and u512 and u640 score higher on Argoverse 2 at 50 ms, where DGSP is late on 11.5% (-0.220, $[-0.404, -0.011]$ against u640). On Argoverse 2, whole-frame inference leads DGSP by 0.05–0.10 from 66.7 ms, and AdaScale and AnA-style refinement by up to 0.11 without a deadline, not significantly. The variant bounded to one crop batch, added post hoc, is the best variant of DGSP at 33.3 ms on nuScenes (0.43) but is worse on Argoverse 2. Values under a deadline differ slightly from Tab. S35 (*e.g.*, 0.15 *vs.* 0.21 for DGSP on Argoverse 2 at 50 ms), since the deadlines lie close to the latencies and the runs are separate. The mean power over the session was 18.6 W; since the rows alternate frame by frame, the energy of each row cannot be separated in this run.

S6.3. Controlled closed loop with estimated states

Bench2Drive keeps the annotated states of matched objects, so that errors in range and time-to-collision never reach its planner. We therefore run a controlled closed loop in CARLA 0.9.15 on the 155 Bench2Drive evaluation routes whose scenarios require braking for vehicles, pedestrians, or bicycles and need no lateral maneuver around an actor, with three traffic seeds. The PDM-Lite route follower controls the steering and the target speed without seeing any actor, and the braking controller of Sec. S2.1 acts on the scene of each perception pipeline, with a hold at standstill. All pipelines use the same YOLOv8s and the geometric lift with the hand-crafted looming estimate, with the time-to-collision taken over 0.5 s; the refiner and the learned looming estimate, which were trained on real images, are not used in simulation. In setting L1, the action of a frame is applied after its latency, drawn from the Jetson measurements of Tab. S34, for DGSP conditional on its number of detector calls; uncertainty crops, which were not timed on the device, use a latency model fitted to these measurements (root-mean-square error 6.6 ms on the frames of DGSP). As an operational definition for this benchmark, a collision is avoidable unless it is a side or rear impact or the braking demand on the annotated states arises less than $v/6 + 0.4$ s before the contact, where v is the ego speed; the latency of L1 is simulated from the measured Jetson latency, not run live.

In Tab. S37, most collisions cannot be avoided by longitudinal control, *i.e.*, they are side or rear impacts or late

Table S37. **Controlled closed loop with estimated states** (CARLA, 155 Bench2Drive routes with braking scenarios, three seeds). Missed: required braking episodes without braking, per route (registered metric); Miss/ev.: share of avoidable required-braking events without braking (%), the miss rate reported with the onset-delay analysis; Coll. / Avoid.: share of routes (%) with a collision / an avoidable collision; Delay: mean braking-onset delay (s) relative to privileged perception on avoidable events; Lat.: mean applied latency (ms), measured on the Jetson for the pipeline with its components. L0: no latency; L1: Jetson latency. All pipelines use the geometric lift with the hand-crafted looming estimate and no refiner. † / *: DGSP significantly better / worse (paired bootstrap over routes for Missed, over route × seed for Miss/ev. and Delay).

	L0					L1					
Pipeline	Missed	Miss/ev.	Coll.	Avoid.	Delay	Missed	Miss/ev.	Coll.	Avoid.	Delay	Lat.
Privileged	0	–	31.2	0.2	–	–	–	–	–	–	–
Cheap (u320)	0.44†	6.5†	30.3	1.5	0.091†	0.64†	10.4†	33.1	3.2†	0.124†	7.5
u960	0.33	2.9	30.3	1.1	0.029*	0.43	4.8*	32.3	1.3	0.059*	22.7
Native	0.32	3.0	31.6	1.3	0.029*	0.45	4.9*	32.5	2.8	0.059*	37.9
Uncertainty crops	0.43†	5.8†	31.4	1.9	0.079†	0.62†	9.8†	33.1	2.6	0.130†	16.1
Support + corridor tube	0.33	3.6	32.0	1.7	0.042	0.47	6.7	34.4	2.2	0.081	24.1
DGSP, $\leq$ 2 batches	0.32	3.8	31.0	1.9	0.041	0.45	6.3	34.8	1.9	0.078	18.3
DGSP	0.29	3.7	30.1	0.6	0.040	0.48	6.8	33.8	1.3	0.084	21.8

entries into the corridor, as the 31% of the routes with a collision under privileged perception indicate, and no pipeline differs significantly from DGSP in the collision rate. DGSP has the fewest avoidable collisions without latency and as few as u960 with latency, significantly fewer than the cheap detector in this setting. On avoidable events, DGSP starts braking significantly earlier than the cheap detector and uncertainty crops, as early as the hybrid, and within 30 ms of, but significantly later than, whole-frame inference (+11 and +26 ms against u960 in L0 and L1).

Two measures of missed braking. The registered measure counts required braking episodes without braking per route; the miss rate counts the share of avoidable required-braking events without braking, paired over route × seed, and accompanies the onset-delay analysis. Both rank DGSP well ahead of the cheap detector and uncertainty crops in L0 and L1. Against whole-frame inference they agree in L0, where neither difference is significant, but not in L1: per route, DGSP misses +0.04 [−0.10, +0.17] episodes more than u960 and +0.02 [−0.09, +0.13] more than native-resolution inference, whereas per event its miss rate is higher by 2.0 percentage points, [+0.9, +3.2] against u960 and [+0.8, +3.3] against native-resolution inference. With latency, DGSP therefore misses a larger share of avoidable braking events than whole-frame inference, although the per-route count does not resolve the difference; the onset delay of the same events (Delay) points the same way.

S6.4. Planning benchmarks

Table S38. **Closed-loop evaluation on Bench2Drive** (219 routes, front-camera perception in the loop). DS, SR: driving score and success rate, mean and standard deviation over three runs; Abil.: mean of the five Multi-Ability success rates; Rec.: recall of the actors within 3 s of collision, from a fourth run that logs the perception.

Perception	DS	SR	Abil.	Rec. ↑
Privileged	96.6 ± 0.2	0.915 ± 0.003	0.911	1
Cheap (u320)	92.3 ± 0.9	0.834 ± 0.018	0.830	0.889
u960	93.2 ± 0.7	0.857 ± 0.010	0.855	0.845
Native (u1600)	92.0 ± 0.7	0.837 ± 0.013	0.839	0.897
DGSP	92.5 ± 0.8	0.839 ± 0.011	0.833	0.906

Table S39. **NAVSIM v2** with PDM-Closed and front-camera perception in the loop: EPDMS on navtest (12,146 scenes) and on the two-stage navhard split.

Perception	navtest	navhard
Privileged (ground truth)	0.870	0.528
Cheap (u320)	0.862	0.508
u960	0.864	0.508
Native	0.864	0.510
Support + route tube	0.865	0.510
DGSP	0.865	0.509

On NAVSIM, an annotated object that projects into the front camera is retained only if it is matched to a detection, and unmatched detections are converted to static objects (Sec. S2.1). Our reference PDM-Closed with ground-truth objects attains 0.900 on navtest, compared with the published value of 0.881; its ego-progress sub-score (0.936) accounts for most of the difference, but we did not isolate every difference of the setup, which all pipelines share. On navtest, the scores with the cheap detector and with ground-truth objects differ by 0.007 PDMS. The field probed from PDM-Closed attains a PDMS of 0.8945, compared with 0.8926 for the closed-form braking field ([+0.0002,

$+0.0040]$). The two fields differ in shape on 99.8% of the $12{,}146$ scenes (intersection over union below 0.9; median 0.33), and 67% of the footprint of the probed field lies outside the braking field, but the selected trajectory differs by more than 0.5 m on only 4.3% $[3.6,\ 4.9]$ of the scenes (517). On these scenes, the probed field improves the PDMS by 0.045 $[+0.005,\ +0.091]$; it scores better on 186, worse on 265, and equally on 66, and the mean gain stems from 35 scenes whose score changes by 1, 31 of them in its favor. On nuPlan-mini ($1{,}440$ states), whose logs coincide with the NAVSIM development split, DGSP attains the highest PDMS (0.940, compared with 0.937 for FOVEA and u960, 0.935 for u640, and 0.933 for the cheap detector). Table S38 reports the closed-loop results on Bench2Drive with CARLA [4], where matched objects retain their true state. One of the 220 routes is excluded for all pipelines, as the simulator repeatedly fails on it. Paired over the routes and pooled over the three runs, the driving score of DGSP does not differ significantly from that of whole-frame inference at the verification resolution ($[-2.20,\ +1.03]$) or at the native resolution ($[-1.37,\ +2.42]$). In the logged run, the recall of the actors within 3 s of collision is 0.906 for DGSP, compared with 0.889 for the cheap detector and 0.845 and 0.897 for whole-frame inference at the verification and the native resolution. On NAVSIM v2 (Tab. S39), all perception pipelines lie within 0.003 extended PDMS of each other on navtest and within 0.002 on navhard. The scenarios in which all perception pipelines fail are accident and construction scenarios, which contain cones and warning signs outside the classes of the COCO-pretrained detector.

Cost of probing the planner. On NAVSIM, the field and the support require 50.2 calls of PDM-Closed per frame on average (45.1 probe queries, 4.0 removal queries, and the base call). Since the planner input is identical for all calls of a frame, it is constructed once, and the simulated ego roll-outs are reused whenever a query leaves the proposals unchanged, which is the case for 41 of the 50 calls; the outputs are identical to those of the reference implementation. This reduces the cost of a frame by a factor of 11, to 1.3 s at the median and 2.0 s at the 95th percentile with eight CPU worker processes (6.0 and 10.1 s with one). Coarser fields reduce the number of calls further without a measurable change of the score: a 5×5 grid (26 calls), a 3×3 grid (14 calls), and the cells along the current plan (15 calls) change the PDMS on navtest by at most 0.0002, and all intervals of the difference contain zero; measured on the same $1{,}000$ frames, they take 51%, 32%, and 33% of the time of the 7×7 grid. For the bottleneck selector of Corollary 2, an output-identical implementation that caches the base score of each candidate and evaluates only the probe columns reduces the cost of the field and the support from 101.0 to 11.0 ms per frame on one CPU thread for the 7×7 grid and from 24.1 to 6.4 ms for the 3×3 grid, with identical actions, fields, and supports on all $1{,}000$ frames. The measured violations of certification by single-object queries are reported in Sec. S1.7. On Bench2Drive, the field is that of the braking controller and requires no planner query.

S6.5. Planner-anchored crops, PDMS, and interactions on NAVSIM

Table S40. **Bottleneck selector on NAVSIM navtest** ($12{,}146$ scenes, real crop detections, official PDMS of NAVSIM v1.1). Agr.: agreement with $\pi(F)$; Repair: agreement on the 724 scenes on which the cheap draft differs from $\pi(F)$, with the interval of DGSP minus the row; px: crop pixels in the unit of the field. VeteranAD-style rows as in Tab. S15, with the 15 candidates of the selector as anchors. Every interval of the PDMS difference between rows contains zero.

Row	PDMS	Agr.	Repair	Δ repair	px
Cheap draft	0.512	0.940	0	–	–
Reference $\pi(F)$	0.510	1	1	–	–
(a) all anchors	0.510	0.965	0.638	**[+0.019, +0.083]**	1.26
(b) selected anchors	0.511	0.965	0.601	**[+0.057, +0.124]**	0.98
(c) selected + support	0.511	0.967	0.663	[-0.004, +0.058]	0.84
Support first, then geometric field	0.511	0.968	0.674	[-0.012, +0.044]	0.91
Support + route tube	0.511	**0.973**	0.671	–	–
DGSP	0.511	0.968	**0.689**	–	0.83

Agreement is not PDMS. For the bottleneck selector of Sec. S1.7, the PDMS of DGSP (0.511, $[0.476,\ 0.546]$) does not differ significantly from that of the cheap draft (0.512) or of the reference (0.510): the selector with its 15 fixed candidates is a weak planner, and its choices with the cheap and the whole-frame detections score alike, so this benchmark tests fidelity, not driving quality. Table S40 adds VeteranAD-style crops along the candidates. DGSP repairs significantly more disagreements than crops along all candidates or along the selected ones, with fewer pixels, while adding its support to the selected candidates or verifying the support before a geometric field makes the difference not significant. The hybrid of the support with a fixed tube along the route agrees with $\pi(F)$ more often than

DGSP (97.3% *vs*. 96.8%; DGSP minus hybrid -0.52 points, $[-0.76, -0.27]$) but repairs fewer of the disagreements of the draft (0.671 *vs*. 0.689). A field built from pairs of probes ($k = 2$) equals the single-probe field of PDM-Closed on all 1,000 scenes of Sec. S1.7, since no pair of adjacent cells changes the plan jointly.

Table S41. **Joint effects of objects on PDM-Closed** that no single object explains, on the 11,146 navtest scenes not in Sec. S1.7 (sampled combinations; a joint effect moves the plan by more than 0.5 m within 4 s). Rate: share of the tested combinations, with a 95% Wilson interval in the same power of ten; Scenes: share of the scenes with a tested combination that have at least one effect. ΔPDMS: mean change of the PDMS from the base plan to the jointly changed plan, with the numbers of effects that lower / raise it.

Family	Effects / tested	Rate	Scenes	ΔPDMS (lower / higher)
Removal of three objects	528 / 166,866	$3.2 \cdot 10^{-3}$ [2.9, 3.5]	2.2%	−0.33 (252 / 175)
Insertion of two probes	20 / 1,114,600	$1.8 \cdot 10^{-5}$ [1.2, 2.8]	0.16%	−0.26 (19 / 1)
Insertion of three probes	55 / 1,114,600	$4.9 \cdot 10^{-5}$ [3.8, 6.4]	0.32%	−0.23 (50 / 5)
Adjacent pair of cells	0 / 651,247	0 $[0, 5.9] \cdot 10^{-6}$	0	–

Interactions in a general planner. Table S41 extends the measurement of Sec. S1.7 to combinations of up to three objects on the remaining navtest scenes, with a fresh instance of PDM-Closed for every scene, at a cost of 337 CPU hours for 3.1 million planner calls. Pairs of adjacent cells never change the plan jointly (0 of 709,645 on all scenes), and insertions of two or three probes do so for a few combinations in 10^5. Removing three objects together changes the plan more often, on 2.2% of the scenes with such triples; this rate per combination is higher than on the first 1,000 scenes ($1.9 \cdot 10^{-3}$, $[1.3, 2.7] \cdot 10^{-3}$), so the smaller sample underestimated it. When a joint effect occurs, it lowers the PDMS more often than it raises it (mean -0.32; on 114 of the 182 affected new scenes at least one effect lowers it). A support from single-object queries therefore misses a small but consequential set of multi-object interactions, and these diagnostics of finite order do not certify the absence of larger ones.

S7. Qualitative results, failure cases, and reproducibility

S7.1. Qualitative results

Fig. 4 and Figs. S13 to S15 show frames selected by a fixed rule on the stored test results of the frozen configuration, with categories defined by the actions: unnecessary braking removed (the required action is KEEP, the cheap pipeline brakes, and DGSP and u960 with the components keep), range corrected (the cheap detector finds a decision-relevant object, the cheap pipeline and DGSP without the refiner take an action other than the required one, and DGSP takes it), missed object recovered by the field (a decision-relevant object is missed by the cheap detector and by the support-only ablation and found by u960 and by DGSP, which takes the required action), and failure (DGSP takes an action other than the required one and u960 with the shared components takes it). Per category and dataset, the first qualifying frame in time order is shown; none is chosen by inspection.

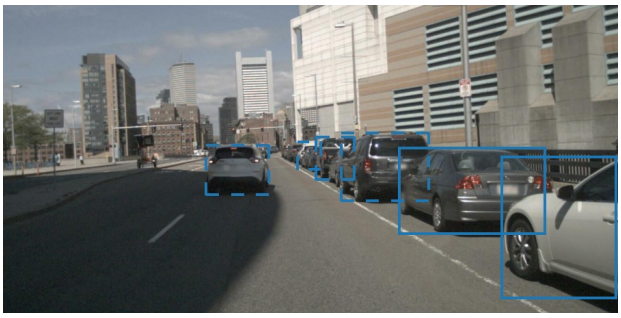
S7.2. Failure modes

Table S42 summarizes the principal failure modes and points to the analyzes that support them.

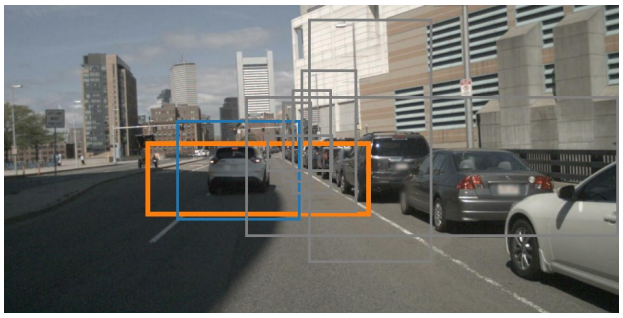
S7.3. Reproducibility

The frozen configuration, the probe grid, the verification loop, the truncation guard, and the training data of the range refiner and the looming estimate are given in Sec. S2.2 and Tab. S7; the planner, the loss, and all constants in Sec. S2.1, and the second planner and the curved corridor in Sec. S4.3. The splits, the selection rule, the configurations evaluated on the test splits, and the two held-out evaluations under protocols fixed in advance are documented in Secs. S2.2, S5.2 and S5.3; the baselines, their candidate grids and selected settings, and the pixel accounting in Secs. S2.3 and S2.5. All intervals are paired bootstrap intervals over the units (scenes, logs, sequences, or routes), with 2,000 resamples, a fixed seed, and 95% percentiles; DGSP is *non-inferior* to a baseline if the lower bound of the difference exceeds -0.02, a margin of 2% of the decision loss of the cheap pipeline fixed in advance. The Jetson timing and deadline protocols are given in Secs. S6.1 and S6.2, the qualitative frames are selected by the rule of Sec. S7.1, and the plots are generated from the values of Sec. S8.

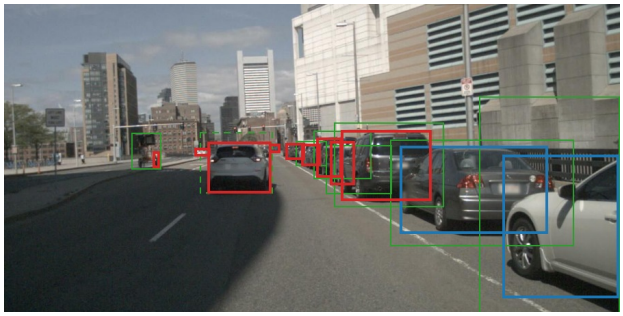
(1) Range corrected (nuScenes held-out)



(a) draft



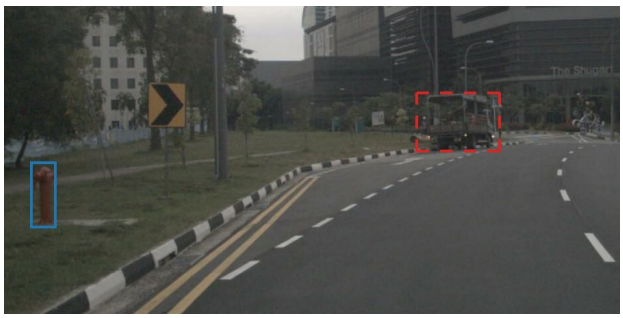
(b) verified crops



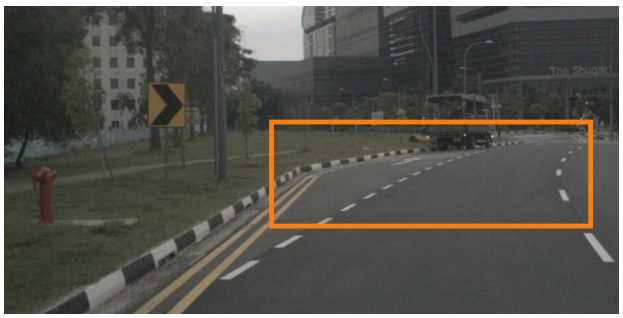
(c) final scene

The draft decelerates for the vehicle ahead. After its verification, the refiner corrects its range, and DGSP takes the required HARD-BRAKE; without the refiner, it decelerates.

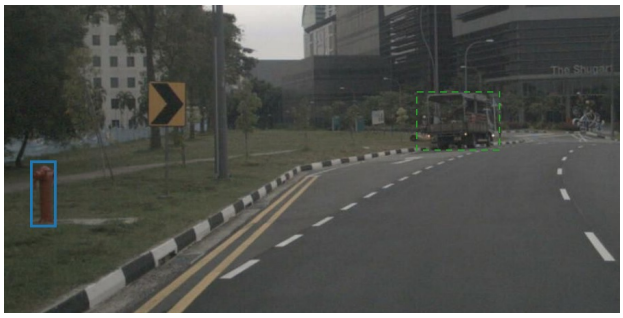
(2) Failure (nuScenes held-out)



(a) draft



(b) verified crops

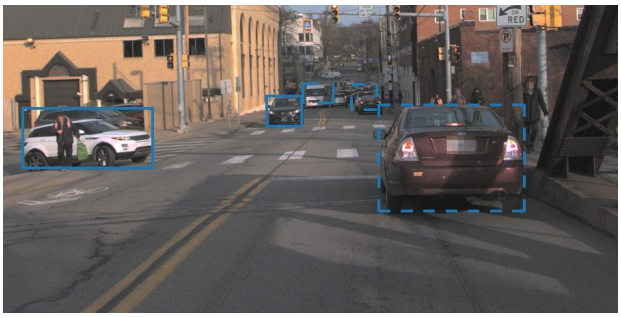


(c) final scene

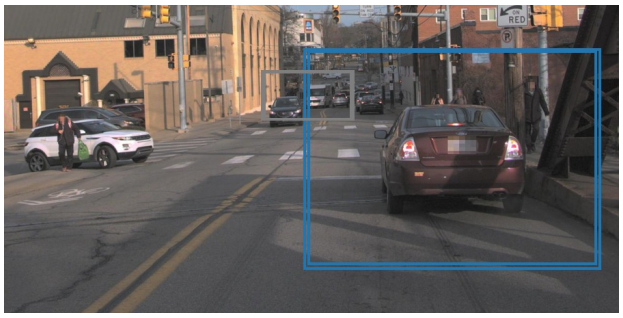
The decision-relevant truck lies above the upper edge of the field footprint and is detected neither in the draft nor in the crop. DGSP keeps KEEP, whereas u960 with the components decelerates as required.

Figure S13. **Qualitative results selected by a fixed rule** (nuScenes held-out, the two categories of Fig. 4 not shown there). Blue: draft detections (dashed if replaced) and support crop; dashed red: decision-relevant object missed by the draft; orange: field; grey: guard crops; red: verified detections; green: annotation. Images are cropped to the union of the crops and the decision-relevant objects.

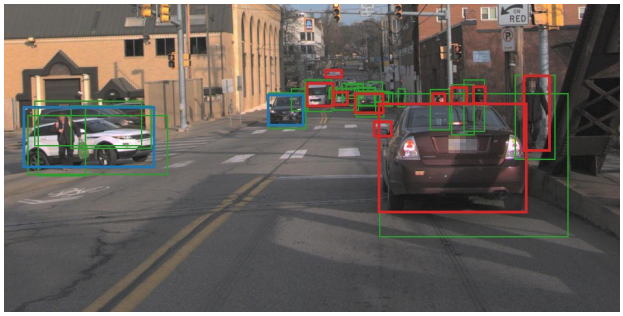
(3) Unnecessary braking removed (Argoverse 2 val)



(a) draft



(b) verified crops



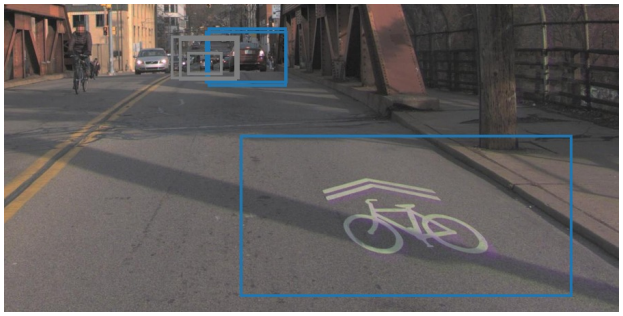
(c) final scene

The draft decelerates for the car on the right. The support crop verifies it, and with the refined range, DGSP takes the required KEEP.

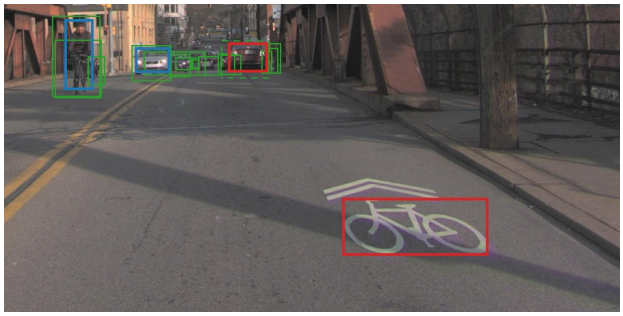
(4) Range corrected (Argoverse 2 val)



(a) draft



(b) verified crops

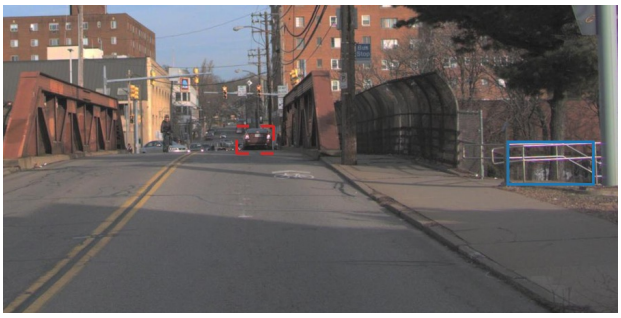


(c) final scene

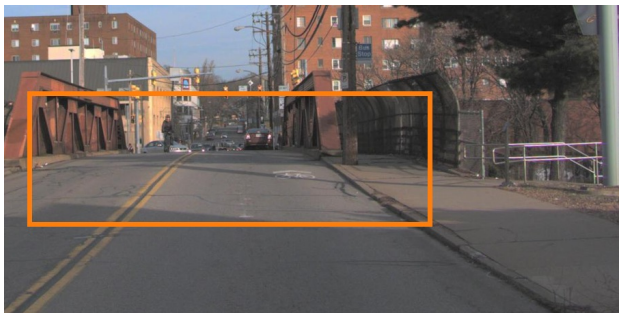
The draft detects the painted bicycle symbol as an object and brakes hard. The verification re-detects it, the refiner corrects its range, and DGSP takes the required DECELERATE.

Figure S14. **Qualitative results selected by a fixed rule** (Argoverse 2 val). Blue: draft detections (dashed if replaced) and support crop; dashed red: decision-relevant object missed by the draft; orange: field; grey: guard crops; red: verified detections; green: annotation. Images are cropped to the union of the crops and the decision-relevant objects.

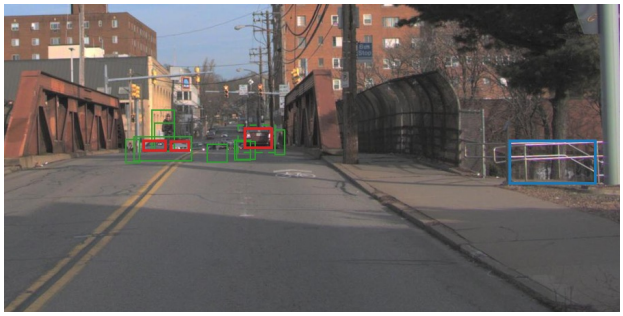
(5) Missed object recovered (Argoverse 2 val)



(a) draft



(b) verified crops



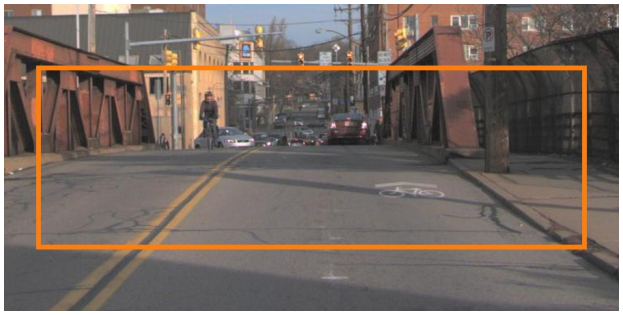
(c) final scene

Verifying the field footprint recovers the distant vehicle that the draft misses, and DGSP takes the required DECELERATE.

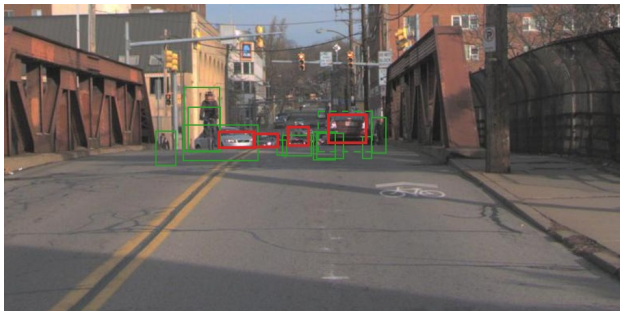
(6) Failure (Argoverse 2 val)



(a) draft



(b) verified crops



(c) final scene

Verifying the field footprint recovers the vehicle, but the refined range keeps the action at KEEP, where DECELERATE is required; without the refiner, DGSP decelerates.

Figure S15. **Qualitative results selected by a fixed rule** (Argoverse 2 val, continued). Blue: draft detections (dashed if replaced) and support crop; dashed red: decision-relevant object missed by the draft; orange: field; grey: guard crops; red: verified detections; green: annotation. Images are cropped to the union of the crops and the decision-relevant objects.

Table S42. **Failure modes of DGSP** and where they are measured.

Failure mode	Evidence	Section
Decisive object outside the verified region	Coverage 0.92–0.96; lowest for objects more than 3 m from the ego lane, for objects that the draft misses, and for the fastest-closing third on nuScenes (0.80), which the static probe does not anticipate; example (2)	S4.2, S7.1
Decisive object verified, but its state estimated inaccurately	Of the 60 costliest frames in which DGSP brakes less than required, 46 have a detected decisive object with an inaccurate time-to-collision, lateral position, or range, 11 an object in a crop that is missed, and 3 an object outside every crop; example (6)	S1.8, S7.1
Intermittently verified objects lack looming history	An object that is verified only intermittently lacks the three same-scale observations of the looming estimate and carries no time-to-collision; iterated AnA-style refinement, which verifies every corridor detection on every frame, gives it a continuous history	S2.2, S3.1
Low-confidence draft boxes	78 of 117 natural misses on nuScenes have a draft box with a confidence between 0.05 and 0.25, which DGSP does not verify	S4.1
Missed braking below the hard-braking threshold	At the deployed threshold, more missed DECELERATE episodes than the cheap detector, as for whole-frame inference with the same components, in part through the refiner that all pipelines with the components share; not significant at the constrained threshold	S5.1, S5.2
Classes outside the detector	All perception pipelines fail in the accident and construction scenarios of NAVSIM, which contain cones and warning signs outside the classes of the COCO-pretrained detector	S6.4
Tail latency	The re-detections of the truncation guard set the tail; whole-frame inference at 960 pixels with the components has a lower 95th percentile and leads under tight deadlines on Argoverse 2 (not significant)	S6.1, S6.2

S8. Reference tables

The plots of this document are generated from the following tables, which list the exact values and intervals. Unless stated otherwise, an interval is the 95% paired bootstrap interval of (DGSP − method), and an interval in bold lies above zero, *i.e.*, favors DGSP.

Table S43. **Intervals for Tab. 2**: decision value G and 95% paired bootstrap intervals of (DGSP − method) without a cap (Δ) and with the per-frame cap at the pixels of DGSP (Δ_{cap}); bold: above zero; *: below zero. The table includes the rows that Tab. 2 omits (u640, FOVEA with dataset saliency, and the one-pass variants) and the variants u768, AnA-style refinement with the draft threshold of the other pipelines ($t = 0.25$), and the development-best Chanakya-style policy (§: its development pixels exceed those of DGSP by 17%, outside the matching rule).

	nuScenes held-out			Argoverse 2 val			Argoverse-HD val		
Method	G	Δ	Δ_{cap}	G	Δ	Δ_{cap}	G	Δ	Δ_{cap}
Whole frame, u640	−0.036	[−0.070, +0.666]	–	0.311	[−0.076, +0.035]	–	0.279	**[+0.026, +0.329]**	–
Whole frame, u768	0.046	[−0.061, +0.459]	–	0.308	[−0.080, +0.043]	–	0.273	**[+0.032, +0.336]**	–
Whole frame, u960	0.112	[−0.078, +0.308]	–	0.338	[−0.102, +0.004]	–	0.297	**[+0.017, +0.320]**	–
Whole frame, u1280	0.292	[−0.165, +0.017]	–	0.348	[−0.113, −0.005]*	–	0.315	**[+0.013, +0.278]**	–
Whole frame, native	0.222	[−0.087, +0.122]	–	0.294	[−0.077, +0.080]	–	0.430	[−0.048, +0.028]	–
AdaScale	0.144	[−0.089, +0.228]	[−0.065, +0.112]	0.341	[−0.107, +0.002]	[−0.040, +0.099]	0.298	**[+0.017, +0.319]**	**[+0.050, +0.414]**
Budget-Adaptive Routing	0.169	[−0.032, +0.125]	**[+0.096, +0.345]**	0.155	**[+0.078, +0.201]**	**[+0.214, +0.344]**	0.207	**[+0.124, +0.297]**	**[+0.304, +0.508]**
Router, decision-value targets	0.214	[−0.088, +0.095]	**[+0.094, +0.339]**	0.189	**[+0.053, +0.157]**	**[+0.216, +0.328]**	0.011	**[+0.306, +0.505]**	**[+0.313, +0.516]**
Chanakya-style policy	0.127	[−0.032, +0.207]	**[+0.064, +0.316]**	0.286	[−0.039, +0.051]	**[+0.193, +0.309]**	0.297	**[+0.025, +0.237]**	**[+0.290, +0.497]**
Chanakya-style, dev.-best§	0.165	[−0.009, +0.095]	**[+0.071, +0.324]**	0.272	[−0.028, +0.065]	**[+0.198, +0.311]**	0.310	**[+0.020, +0.209]**	**[+0.285, +0.490]**
SAHI	0.303	[−0.207, +0.029]	**[+0.090, +0.289]**	0.256	[−0.031, +0.105]	**[+0.246, +0.424]**	0.229	**[+0.062, +0.388]**	**[+0.166, +0.430]**
FOVEA, instance saliency	0.150	[−0.049, +0.179]	**[+0.148, +1.020]**	0.319	[−0.086, +0.027]	**[+0.026, +0.189]**	0.313	**[+0.016, +0.278]**	**[+0.202, +0.602]**
FOVEA, dataset saliency	0.161	[−0.061, +0.197]	**[+0.206, +1.374]**	0.316	[−0.074, +0.023]	[−0.047, +0.071]	0.312	**[+0.019, +0.276]**	**[+0.222, +0.677]**
UFP	−0.058	[−0.030, +0.670]	[−0.065, +0.243]	0.322	[−0.089, +0.026]	[−0.038, +0.119]	0.282	**[+0.021, +0.328]**	**[+0.032, +0.346]**
AnA-style, one pass	0.158	[−0.074, +0.181]	**[+0.003, +0.273]**	0.346	[−0.111, +0.002]	[−0.011, +0.047]	0.314	**[+0.007, +0.297]**	**[+0.008, +0.034]**
AnA-style, iterated	0.158	[−0.074, +0.181]	**[+0.001, +0.270]**	0.349	[−0.113, −0.002]*	[−0.013, +0.045]	0.315	**[+0.006, +0.296]**	**[+0.008, +0.034]**
AnA-style, $t = 0.25$	−0.037	[−0.014, +0.470]	**[+0.039, +0.485]**	0.304	[−0.081, +0.063]	**[+0.014, +0.107]**	0.225	**[+0.036, +0.396]**	**[+0.020, +0.232]**
Counterfactual, one pass	−0.091	**[+0.054, +0.527]**	**[+0.049, +0.524]**	0.245	**[+0.011, +0.098]**	**[+0.016, +0.101]**	0.307	**[+0.019, +0.233]**	**[+0.008, +0.222]**
Counterfactual, iterated	−0.093	**[+0.056, +0.530]**	**[+0.051, +0.526]**	0.256	[−0.005, +0.090]	**[+0.014, +0.102]**	0.285	**[+0.033, +0.253]**	**[+0.020, +0.228]**
DGSP	0.211	–	–	0.294	–	–	0.418	–	–

Table S44. **Decision value G and detector-input pixels (px) of whole-frame inference** (braking controller, identical COCO-pretrained YOLOv8s). "Plain": the pipeline as plotted in Fig. S5. "Ref.": with the range refiner of DGSP. "Hand." and "Learn.": with the refiner and the hand-crafted or the learned looming estimate ("Learn." is the shared lift).

	Argoverse 2					nuScenes					Argoverse-HD				
Long side	Plain	Ref.	Hand.	Learn.	px (%)	Plain	Ref.	Hand.	Learn.	px (%)	Plain	Ref.	Hand.	Learn.	px (%)
448	–	–	–	–	–	−0.026	0.011	0.073	0.086	8.0	−0.015	0.055	0.240	0.247	5.6
512	−0.025	0.105	0.185	0.285	6.7	−0.003	0.039	0.093	0.124	10.2	−0.015	0.068	0.266	0.263	7.1
640	−0.019	0.131	0.219	0.311	10.3	−0.166	−0.112	−0.069	−0.036	17.1	−0.034	0.099	0.278	0.279	11.6
960	−0.021	0.182	0.255	0.338	23.2	0.011	0.055	0.096	0.112	38.4	−0.032	0.124	0.287	0.297	25.3
1280	−0.008	0.189	0.272	0.348	40.0	0.201	0.251	0.269	0.292	65.4	−0.067	0.130	0.262	0.315	44.4
1536	−0.031	0.168	0.233	0.328	57.3	0.191	0.257	0.280	0.302	92.2	−0.075	0.130	0.268	0.357	64.0
Native	−0.058	0.155	0.224	0.294	101.2	0.121	0.176	0.197	0.222	103.1	−0.092	0.147	0.283	0.430	101.3
DGSP w/o refiner		0.241			5.4		0.181			7.8		0.422			7.0
DGSP		**0.294**			5.4		**0.211**			7.8		**0.418**			7.0

Table S45. **Intervals for Tab. S44**: 95% paired bootstrap intervals of (DGSP − whole-frame inference with the refiner); bold: above zero; *: below zero.

Long side	Argoverse 2	nuScenes	Argoverse-HD
448	–	**[+0.029, +0.382]**	**[+0.275, +0.463]**
512	**[+0.130, +0.249]**	**[+0.020, +0.340]**	**[+0.274, +0.445]**
640	**[+0.107, +0.218]**	**[+0.026, +0.725]**	**[+0.252, +0.414]**
960	**[+0.056, +0.167]**	[-0.016, +0.354]	**[+0.198, +0.418]**
1280	**[+0.053, +0.156]**	[-0.144, +0.079]	**[+0.166, +0.450]**
1536	**[+0.069, +0.181]**	[-0.188, +0.128]	**[+0.169, +0.440]**
Native	**[+0.060, +0.222]**	[-0.057, +0.179]	**[+0.143, +0.429]**

Table S46. **Decomposition of the decision loss** into insufficient braking (Insuff.), excessive braking (Exc.), and command changes and collisions (Other), each as a share of the total loss of the cheap pipeline; $1 - G$ is the sum of the three. All pipelines except the cheap one use the refiner and the learned looming estimate.

	nuScenes held-out			Argoverse 2 val			Argoverse-HD val		
Pipeline	Insuff.	Exc.	Other	Insuff.	Exc.	Other	Insuff.	Exc.	Other
Cheap (u320)	0.782	0.191	0.027	0.443	0.534	0.023	0.192	0.793	0.015
u960 + comp.	0.756	0.102	0.030	0.437	0.205	0.020	0.180	0.503	0.021
Native + comp.	0.692	0.067	0.019	0.506	0.177	0.023	0.182	0.367	0.022
Uncertainty crops	0.943	0.079	0.037	0.527	0.233	0.021	0.279	0.421	0.018
Support + corridor tube	0.727	0.075	0.024	0.495	0.187	0.022	0.187	0.386	0.019
DGSP	0.696	0.071	0.022	0.498	0.187	0.022	0.184	0.379	0.019

Table S47. **Attribution of the disagreement with $\pi(F)$** on the test splits (DGSP without refiner; F: whole-frame inference at the verification resolution with the learned looming estimate). Each row adds one intervention to the row above.

Setting	nuScenes	Argoverse 2	Argov.-HD
DGSP, crop detections	0.976	0.968	0.940
+ reference detections in the crops	0.984	0.980	0.953
+ looming estimate of the reference	0.988	0.986	0.956
+ closing probe (10 m/s)	0.991	0.989	0.963
+ extended grid	0.992	0.993	0.968
+ three probe sizes	0.995	0.996	0.971
+ limit of ten rounds	0.995	0.996	0.971

Table S48. **Coverage of the decisive object** by the verified region of DGSP (field footprint, support and guard crops accumulated up to the final round) on the frames that require braking, by stratum (n: frames). Coverage is box-center geometry, not detection recall. On Argoverse-HD, the annotations do not separate vehicle classes.

	nuScenes		Argoverse 2		Argoverse-HD	
Stratum	Cov.	n	Cov.	n	Cov.	n
All	0.92	928	0.95	2,854	0.96	3,593
Car	0.94	779	0.97	2,381	–	–
Truck	0.83	59	0.92	152	–	–
Bus	0.74	27	0.81	69	–	–
Pedestrian	0.93	29	1.00	66	0.98	177
Cyclist	0.65	17	0.87	62	–	–
Other (AHD: vehicles)	0.88	17	0.89	124	0.96	3,416
Range < 20 m	0.91	439	0.95	1,108	0.93	716
Range 20–40 m	0.93	345	0.97	1,066	0.96	1,789
Range 40–60 m	0.92	127	0.94	511	0.98	783
Range > 60 m	0.88	17	0.96	169	0.94	305
Lateral < 1.2 m	0.95	560	0.97	1,588	0.95	2,505
Lateral 1.2–3 m	0.87	342	0.95	1,149	0.97	1,015
Lateral > 3 m	0.85	26	0.83	117	0.95	73
Closing speed, low	0.99	309	0.97	952	0.96	1,198
Closing speed, middle	0.96	309	0.95	951	0.95	1,197
Closing speed, high	0.80	309	0.94	951	0.96	1,198
Detected by the draft	0.93	784	0.96	2,565	0.96	2,524
Missed by the draft	0.85	144	0.90	289	0.97	1,069

Table S49. **Operating curves with identical components** on the test splits: missed required episodes and G at each braking threshold $\text{thr}_{\text{decel}}$ (m/s^2; deployed: 1.0). The support + corridor tube receives the per-frame budget of DGSP at the same threshold.

	nuScenes held-out (147 episodes)				Argoverse 2 val (190)				Argoverse-HD val (71)			
$\text{thr}_{\text{decel}}$	u960	Native	Tube	DGSP	u960	Native	Tube	DGSP	u960	Native	Tube	DGSP
Missed episodes (cheap at 1.0: 51, 21, 12)												
0.6	29	29	36	36	13	14	18	19	9	10	9	9
0.8	36	37	45	47	20	18	26	27	11	15	14	14
1.0	51	53	58	60	31	29	35	36	26	27	28	27
1.2	69	74	78	82	46	45	53	50	34	36	33	32
1.5	83	85	89	93	64	61	70	74	39	41	37	38
Decision value G												
0.6	0.199	0.315	0.263	0.301	0.415	0.365	0.378	0.376	0.366	0.499	0.469	0.485
0.8	0.172	0.276	0.223	0.261	0.391	0.347	0.344	0.343	0.352	0.481	0.459	0.470
1.0	0.112	0.222	0.174	0.211	0.338	0.294	0.296	0.294	0.297	0.430	0.409	0.418
1.2	0.068	0.173	0.118	0.161	0.276	0.221	0.223	0.216	0.236	0.364	0.346	0.359
1.5	0.015	0.108	0.078	0.112	0.146	0.094	0.072	0.068	0.169	0.295	0.285	0.292

Table S50. **Latency of the methods of Tab. 2** on the Jetson AGX Xavier (ms, protocol of Tab. S34, separate runs on the same frames). G on the full test splits; all methods with the shared components.

	nuScenes			Argoverse 2		
Method	$G\uparrow$	Mean	p95	$G\uparrow$	Mean	p95
Cheap (u320)	0.000	7.7	8.4	0.000	10.5	11.0
AdaScale	0.144	43.4	54.5	0.341	70.3	86.3
Budget-Adaptive Routing	0.169	49.3	70.3	0.155	76.1	106.9
Router, decision-value targets	0.214	25.1	46.9	0.189	35.3	65.0
Chanakya-style policy	0.127	14.5	33.1	0.286	23.5	53.8
SAHI	0.303	84.8	96.0	0.256	153.2	172.8
FOVEA, instance saliency	0.150	42.2	51.7	0.319	59.8	75.8
FOVEA, dataset saliency	0.161	33.4	42.0	0.316	49.1	65.8
UFP	−0.058	42.7	71.9	0.322	70.9	107.8
AnA-style, one pass	0.158	23.7	57.8	0.346	59.6	138.5
AnA-style, iterated	0.158	25.4	66.9	0.349	68.3	159.6
Counterfactual, one pass	−0.091	14.6	36.9	0.245	36.4	81.1
Counterfactual, iterated	−0.093	16.3	48.1	0.256	47.8	109.3
DGSP	0.211	21.5	57.4	0.294	39.8	89.1

Table S51. **Decision value under per-frame deadlines** D, run on the Jetson AGX Xavier (nuScenes: 20 test scenes; Argoverse 2: 40 test logs; three repetitions). In-loop: the loop stops before a round that would exceed D and returns the action of the last completed round; a frame that is still late keeps the last action that was on time (hold), as for u960. At 50 ms, u960, DGSP in-loop, and bounded-keep are late on 0.5, 6.4, and 3.9% of the nuScenes and 3.0, 13.4, and 19.4% of the Argoverse 2 frames.

	nuScenes			Argoverse 2		
Pipeline	D=50	66.7	∞	50	66.7	∞
u960 + comp. (hold)	0.386	0.386	0.386	0.298	0.382	0.380
DGSP (hold)	0.330	0.378	0.380	−0.208	0.313	0.297
DGSP, in-loop	0.354	0.378	0.380	−0.025	0.312	0.297
Bounded-keep, in-loop	0.380	0.391	0.382	0.119	0.328	0.304

References

[1] Fatih Cagatay Akyon, Sinan Onur Altinuc, and Alptekin Temizel. Slicing aided hyper inference and fine-tuning for small object detection. In *ICIP*, 2022. 16, 17

[2] Ting-Wu Chin, Ruizhou Ding, and Diana Marculescu. AdaScale: Towards real-time video object detection using adaptive scaling. In *MLSys*, 2019. 16, 17

[3] Wonkyo Choe, Rongxiang Wang, and Felix Xiaozhu Lin. AnA: An attentive autonomous driving system. In *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 33–46, 2025. 16

[4] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. CARLA: An open urban driving simulator. In *CoRL*, 2017. 40

[5] Wei Geng, Nitinder Mohan, and Jörg Ott. Budget-adaptive routing: Skipping the weak when the strong answers anyway. In *Proceedings of the ACM SIGCOMM 2026 Conference*, pages 2001–2009, 2026. 16, 17

[6] Anurag Ghosh, Vaibhav Balloli, Akshay Nambi, Aditya Singh, and Tanuja Ganu. Chanakya: Learning runtime decisions for adaptive real-time perception. In *NeurIPS*, 2023. 16

[7] Pranay Gupta, Abhijat Biswas, Henny Admoni, and David Held. Object importance estimation using counterfactual reasoning for intelligent driving. *IEEE Robotics and Automation Letters*, 9(4):3648–3655, 2024. 16

[8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 13

[9] Cong Hoang Pham and Ho Viet Duc Luong. The decision value of perception compute. *arXiv preprint arXiv:2609.35910*, 2026. 12, 17, 25

[10] Yecheng Huang, Jiaxin Chen, and Di Huang. UFPMP-Det: Toward accurate and efficient object detection on drone imagery. In *AAAI*, 2022. 16, 25

[11] Glenn Jocher and Jing Qiu. Ultralytics YOLO11. `https://github.com/ultralytics/ultralytics`, 2024. 23

[12] Kai Liu, Zhihang Fu, Sheng Jin, Ze Chen, Fan Zhou, Rongxin Jiang, Yaowu Chen, and Jieping Ye. ESOD: Efficient small object detection on high-resolution images. *IEEE TIP*, 34:183–195, 2025. 16, 25

[13] Chittesh Thavamani, Mengtian Li, Nicolas Cebron, and Deva Ramanan. FOVEA: Foveated image magnification for autonomous navigation. In *ICCV*, 2021. 16, 17, 25

[14] Chittesh Thavamani, Mengtian Li, Francesco Ferroni, and Deva Ramanan. Learning to zoom and unzoom. In *CVPR*, 2023. 16, 25

[15] Tao Wang, Chenyu Lin, Chenwei Tang, Jizhe Zhou, Deng Xiong, Jianan Li, Jian Zhao, and Jiancheng Lv. Adaptive image zoom-in with bounding box transformation for UAV object detection. *ISPRS Journal of Photogrammetry and Remote Sensing*, 233:452–466, 2026. 16, 25

[16] Bozhou Zhang, Jingyu Li, Nan Song, and Li Zhang. Perception in plan: Coupled perception and planning for end-to-end autonomous driving. In *AAAI*, pages 12376–12384, 2026. 22